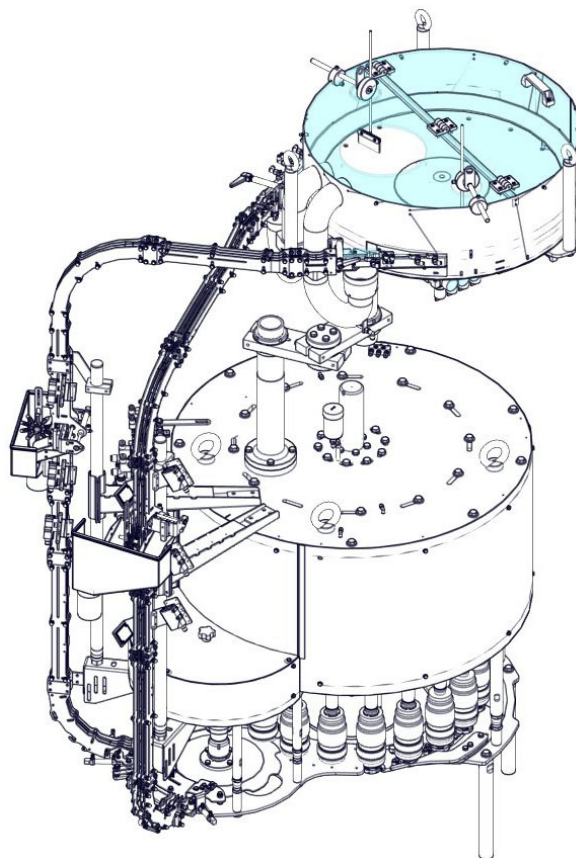




Model
TS - EURO PK TWIN CHUTE D

Serial number
15610

Pitch diameter
720 mm

Nr of heads
20

Manufacturing year
2016

USE AND MAINTENANCE

TRANSLATION OF THE
ORIGINAL INSTRUCTIONS

Version
1

Edition
01/2016

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WARNING

BEFORE START UP THE PARTLY COMPLETED MACHINERY, READ CAREFULLY THE TECHNICAL INSTRUCTIONS IN THIS PUBLICATION AND SCRUPULOUSLY FOLLOW THE INDICATIONS IN THIS MANUAL.

KEEP THE MANUAL IN A PLACE EASY TO GET AND WEEL KNOWN BY THE USER (OPERATORS AND PERSONNEL CHARGED WITH THE MAINTENANCE).



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1 PURPOSE OF THE MANUAL - EDITION - UPDATING METHOD

The manual is integral to the partly completed machinery TS - EURO PK TWIN CHUTE D, matricola 15610 model; it should be kept all the life of partly completed machinery; it must follow the same machinery when it is moved to other premises or if it sold.

It is in Italian and translated in target language whenever required.

In case of doubts about the interpretation of the text, always refer to the original language and contact Customer Service in case of need:

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The text of the manual will refer to the "closing machine" as an abbreviation of the standard name **"Closing machine which uses a closure"** as stated in the UNI EN 415-1 standard (Terminology and classification of packaging machines).

The manual contains information and instruction about all the operating cycle of the closing machine:

- Installation.
- Start up.
- Use.
- Size change.
- Maintenance.
- Dismantling and disposal.

The manual should be kept so that:

- avoid any damage.
- Arrange for its availability to the professional operators working on the closing machine.
- Prevent it from being modified or damaged during the use.
- Make it available in case of need (training, changed ownership of the closing machine, etc.)



Ask for a replacing copy if damaged.

This manual was written with care; if some of the sections needs to be reviewed to add information or instructions, or simply for clarity reasons, AROL S.p.A. will proceed as follows:

- *should the modification involve only a few pages*: the replacement pages will be sent.
- *should the modifications involve more pages of different chapters*: all the chapters involving modification will be sent again.
- *should the modification involve many chapters* : the whole replacement manual will be sent.

Should the manual be supplied in electronic form, the supplied manual will always be the complete and updated one.

The consultation of this manual is made easier by the addition of the initial pages of the general table of contents, for an immediate tracking of the topic concerned.

The chapters are organised a hierarchy structure for a fast tracking of the info required.



The manual shows graphic symbols that are used to distinguish:

- Informations and prescriptions.
- Obligations.
- Warning about residual risks.

The following documents are attached to this manual:

- Pneumatic diagram.
- Closing machine installation diagram.
- Identification tables for size change.
- Instructions manuals for accessories and/or equipment installed on the closing machine and not integrated in this manual..



Before any operation with or on the closing machine read the procedures and warnings herein carefully.



IMAGES OF ANY GENERAL KIND ARE NOT USED IF THE DETECTION OF SPECIFIC STRUCTURE OF CLOSING MACHINE IS ESSENTIAL IN THE CONTEXT TO THE SUBJECT MATTER. The reported instructions and use modes for all described components anyway remain valid.



2 IDENTIFICATION OF THE CLOSING MACHINE

The upper part of the closing machine is provided with an identification plate. (See Figure).

On the plate are reported:

- Closing machine model (**MACHINE**)
- Type of closing machine (**TYPE**)
- Pitch diameter (**P.D. in mm**)
- Number of closure elements (**HEADS**)
- Serial number (**SERIAL N.**)
- Manufacturing year (**YEAR**)

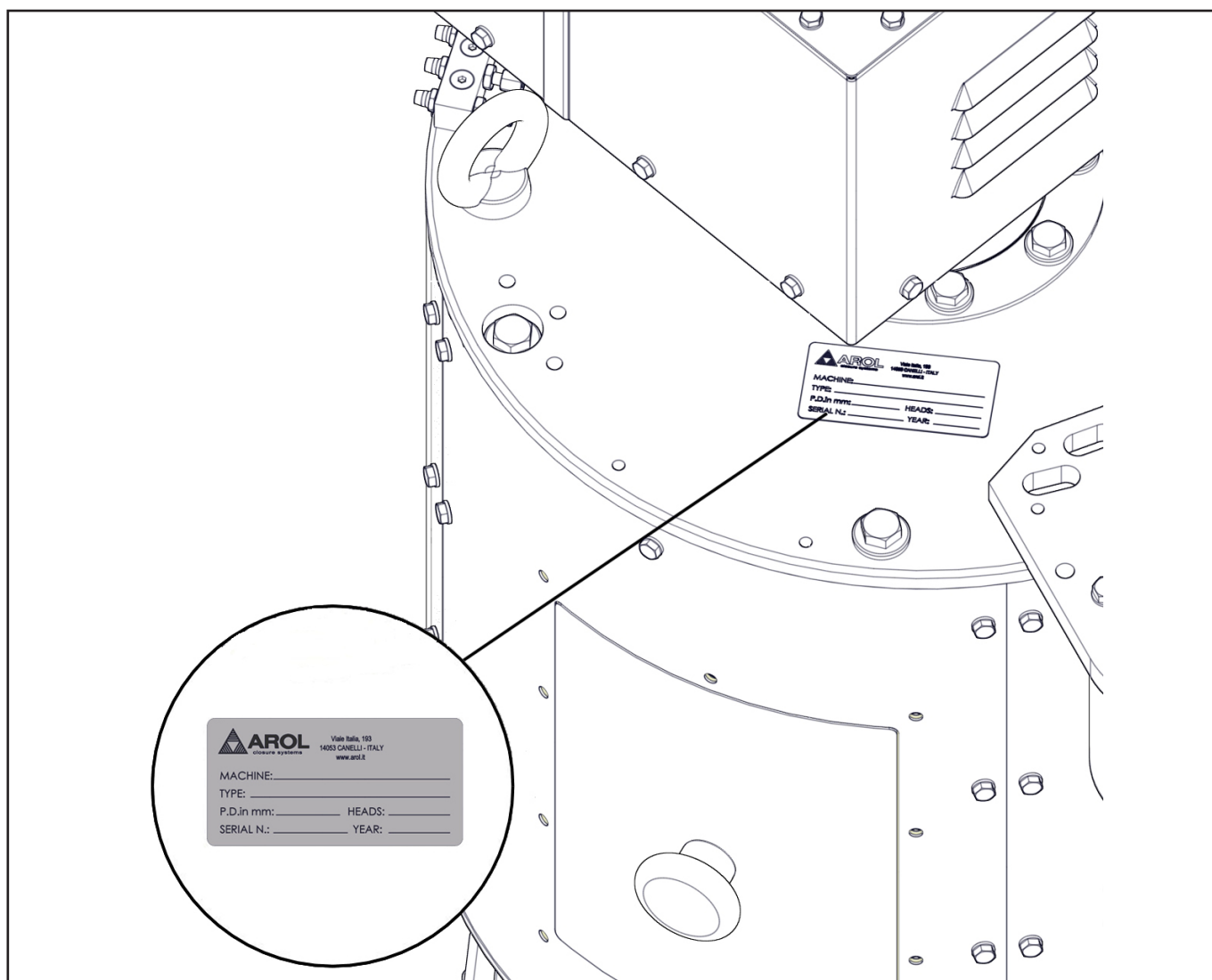


Fig. 1

The data plate is applied using 2 rivets at the opposite corners and should not be removed for whatever reason.

In case of damage, ask for a spare copy to *AROL S.p.A.*.

Clean with neutral detergents paying attention in order not to damage the warning instructions.



3 DECLARATION OF INCORPORATION

The closing machine of this manual is defined as "partly completed machinery" (according to the European Directive 2006/42/EC art. 2 point g), and comes with the *DECLARATION OF INCORPORATION*.

The original copy of the *DECLARATION OF INCORPORATION* is kept at AROL S.p.A..

A compliant copy of the *DECLARATION OF INCORPORATION* is delivered together with the partly completed machinery documents and must be kept till the machine is scrapped.



AROL S.p.A. points out that this statement only and **exclusively** concerns the above-described partly completed machinery, in the conditions and utilisation processes it was placed on the market, thus excluding any added components and/or operations carried out later on without any authorisation by the manufacturer.

In particular, any possible modification or replacement of parts of the closing machine, like as, for example modification to the software of programmable components, made without asking the permission of AROL S.p.A., releases the manufacturer from any responsibility and may cancel the validity of the Declaration of Conformity for the machine where this partly completed machinery is integrated.



Here is a copy of the *DECLARATION OF INCORPORATION*.



DICHIARAZIONE DI INCORPORAZIONE PER UNA QUASI-MACCHINA
(allegato II B della direttiva 2006/42/CE)

Il fabbricante:

AROL S.p.A.
VIALE ITALIA 193
14053 CANELLI (AT) - ITALY

DICHIARA

sotto la propria responsabilità che la nuova quasi-macchina descritta in appresso:

Modello: CAPPING TURRET

Tipo: EURO PK TWIN CHUTE D

Matricola: 15610

Anno di costruzione: 2016

adatta ad applicare tappi su contenitori preformati rigidi, è destinata ad essere incorporata in una macchina a cui si applica la direttiva 2006/42/CE

Dichiara inoltre che la quasi-macchina in oggetto rispetta completamente, entro i suoi limiti, i seguenti Requisiti Essenziali di Sicurezza (RES):

1.1.2, 1.1.3, 1.1.5, 1.3.1, 1.3.2, 1.3.4, 1.4.2.1, 1.7.4, 1.7.4.1

I seguenti RES vengono rispettati solo per la parte specificata:

1.3.3, 1.3.7, 1.4.1 ai componenti della zona di trasmissione del moto alle teste di chiusura

1.3.7 limitatamente al dispositivo di orientamento tappi

2.1.1 limitatamente ai punti a), b) ed e).

2.1.2 limitatamente alle parti oggetto della fornitura

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Canelli, 13/01/2016

CIRIO Sergio

Legale rappresentante

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4 CLOSING MACHINE DESCRIPTION

The closure machine with pre-threaded cap EURO PK TWIN CHUTE D has been designed and built for the application of pre-threaded caps (namely screw caps) on bottles/containers.

4.1 MAIN UNITS OF THE CLOSING MACHINE

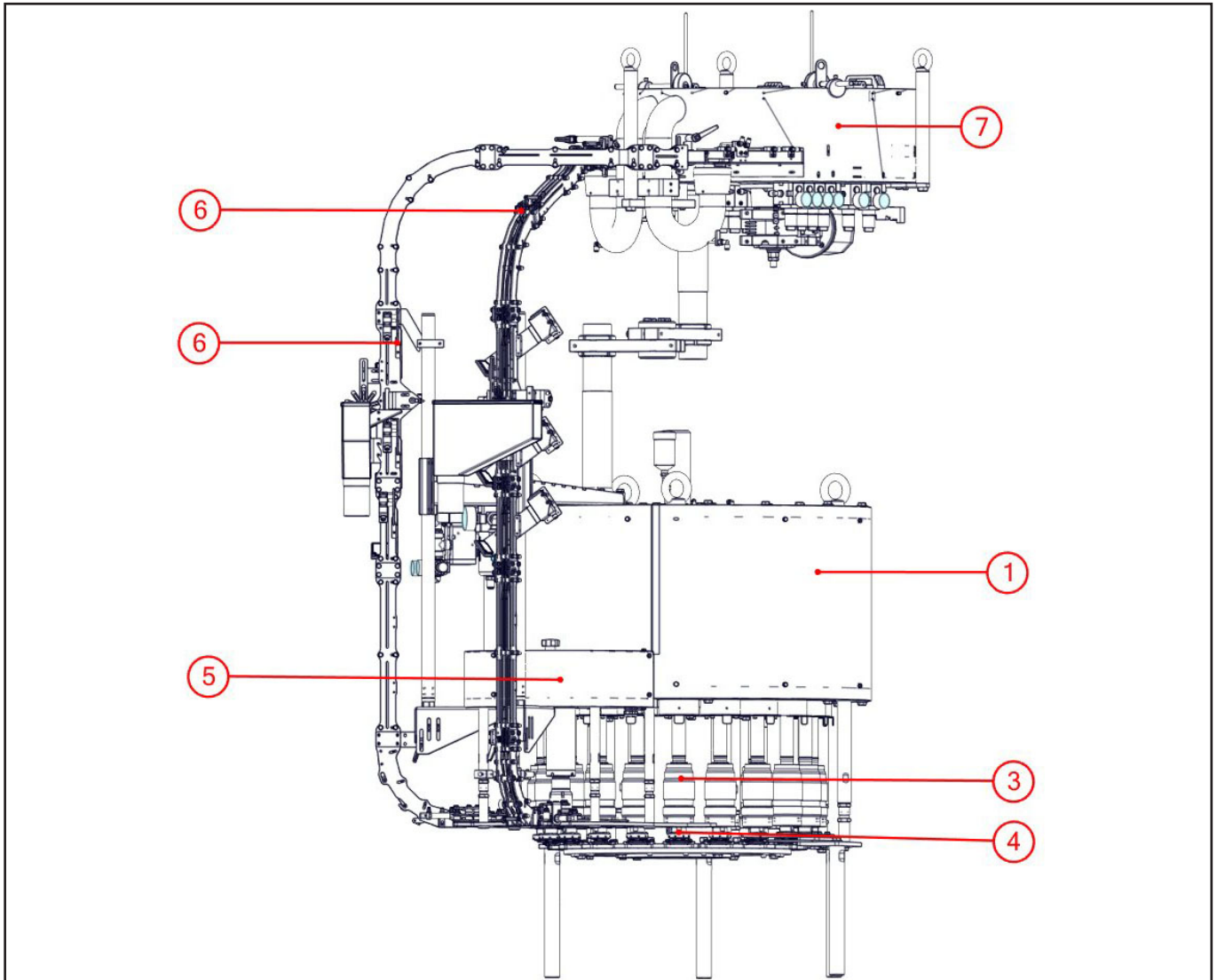


Fig. 2

Legend:

- 1 Upper closure carousel.
- 3 Closure heads.
- 4 Closure chucks.
- 5 Transfer disk for closure systems.
- 6 Chute for closure systems.
- 7 Closure system sorter.



The components forming the closing machine are located inside the main assemblies:

- Closure heads.

Legend:

- 1 Magnet adjustment ring nut
- 2 Graduated scale
- 3 Head upper bush
- 4 Ejector assembly



Fig. 3

- Closure chucks.

Legend:

- 1 Spheres protecting lid
- 2 Cone

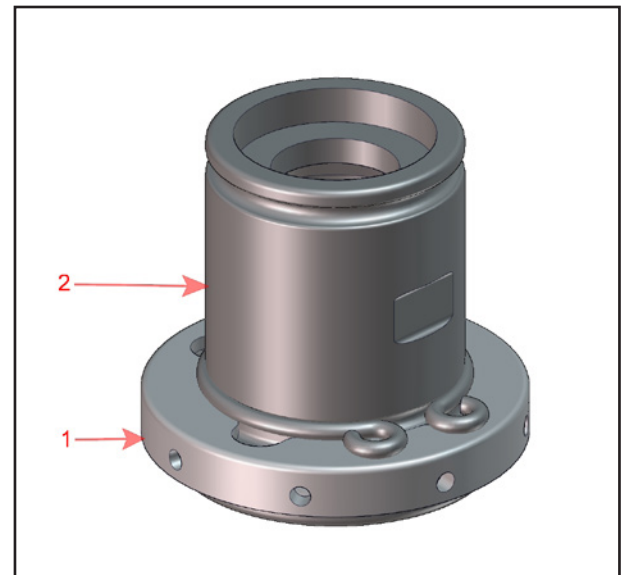


Fig. 4



The "TWIN HOPPER" feeding system is based on the use of two independent caps chutes to obtain a necessary number of caps in order to avoid the stop of the production due to a lack of caps into the caps chute or an error of the first transfer disk release check photocell

The feeding system is made of:

- Centrifugal caps sorter with reversible rotation direction and double exit mouth
- Main caps chute with relevant entrance on the transfer disk

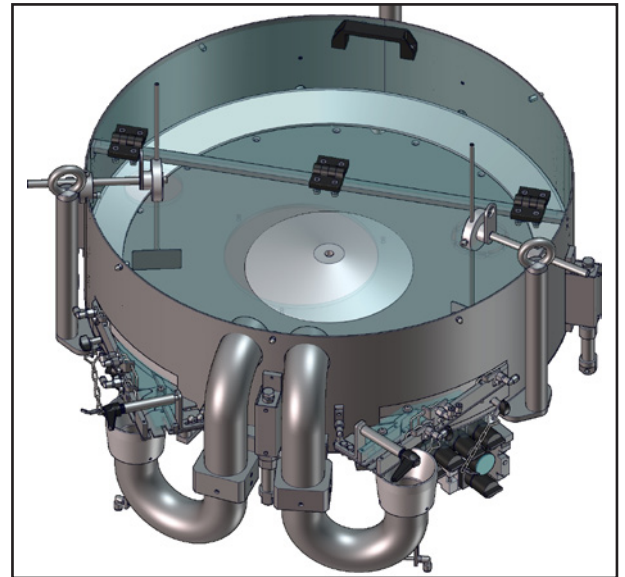


Fig. 5

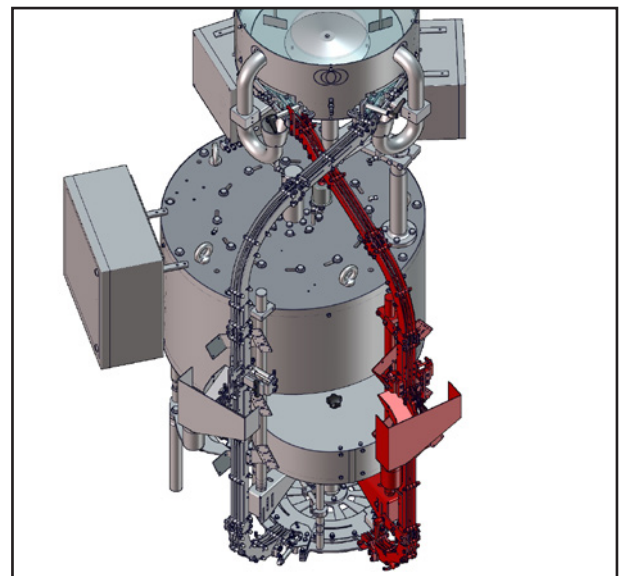


Fig. 6 Example for clockwise direction rotation

- Secondary caps chute with relevant entrance on the transfer disk

To get a good operating of the closing machine, it would be necessary:

- keep both caps chutes full,
- keep the caps sorter full
- operate with the so called first caps chute and , in case of error, use the second one.



The caps chute that must have the priority of operating is the one having the farthest introduction point on the disk in comparison to the cap take position; in this way it is possible to manage the recovering of the jam using the secondary caps chute

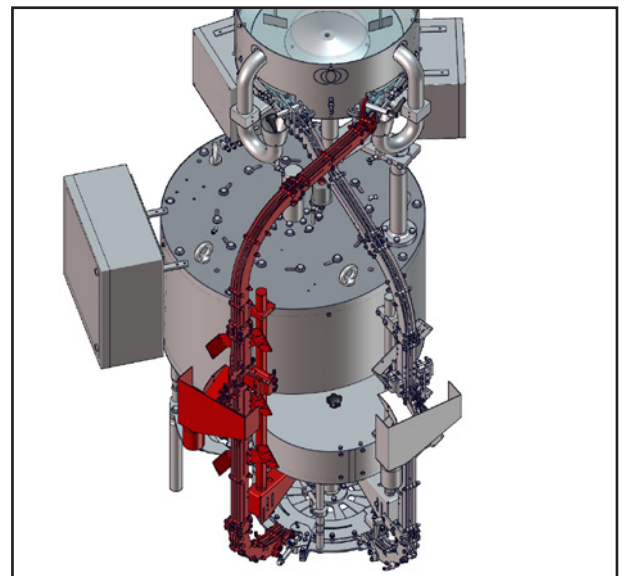


Fig. 7 Example for clockwise direction rotation



4.2 DESCRIPTION OF THE PRODUCTION CYCLE

The application and sealing of closure systems on the containers is the following:

- An orientation device selects the caps inside the hopper.

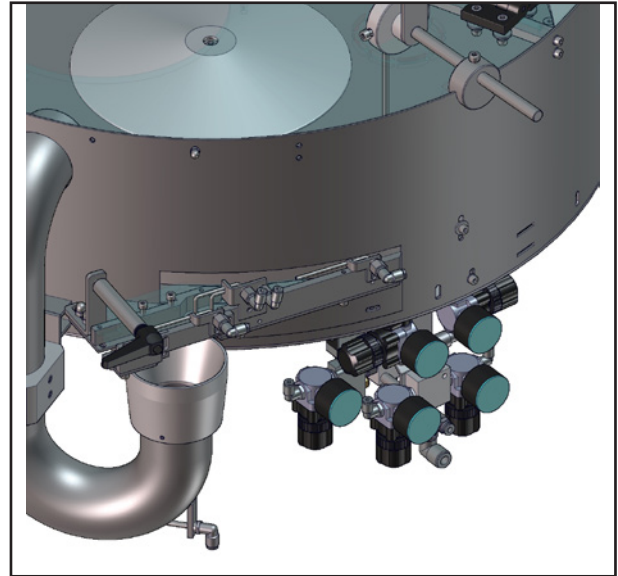


Fig. 8

- A transport system, conducts the caps from caps sorter to transfer disk.

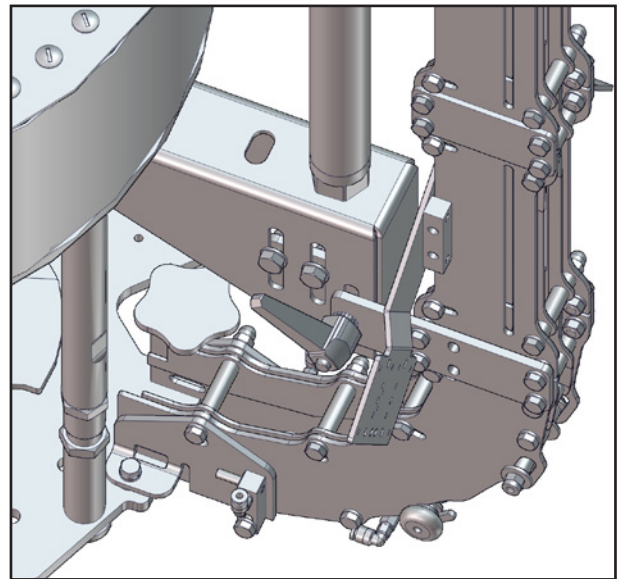


Fig. 9



- A transfer disk, called "pick & place", carries the caps under the closure head.

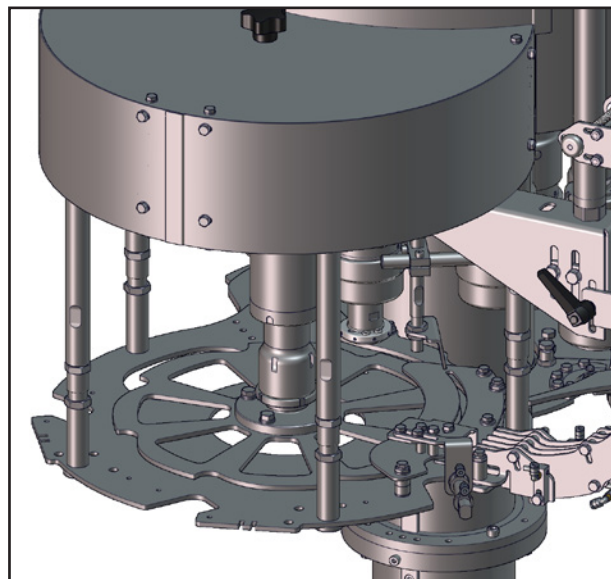


Fig. 10

- The closure head follows the profile of a cam and picks the cap from the conveying device by a chuck with balls.
- After the closure, the exiting container is released, then there is a reset for the following closure.

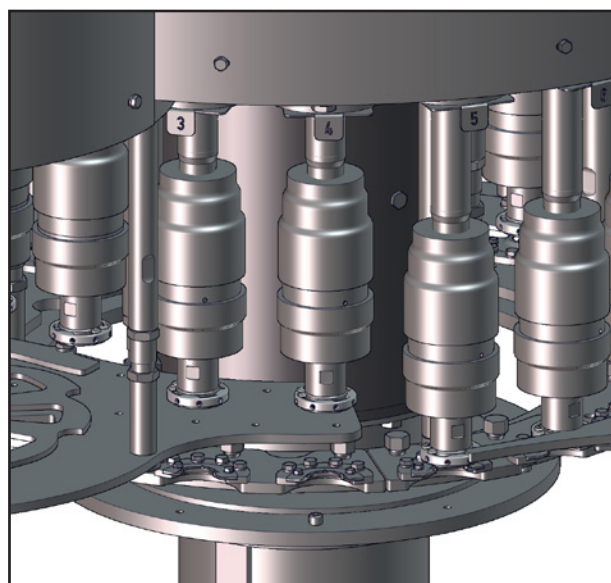


Fig. 11



4.3 CLOSING MACHINE TECHNICAL DATA

TECHNICAL SPECIFICATIONS		
Model	EURO PK TWIN CHUTE D	
Serial number	15610	
Pitch diameter	720 mm	
Nr of heads	20	
Direction of rotation	counter-clockwise	
Manufacturing year	2016	
NOMINAL PRODUCTION		
Pieces/hour	40000	bph
CLOSING MACHINE WEIGHT		
Weight	1600	kg.
ELECTRICAL DATA		
POWER SUPPLY VOLTAGE:		
main	400	V ~
	60	Hz
auxiliary	24	V =
Installed power:		
heads rotation motor	3,00	kW
caps sorter motor	0,45	kW
Total installed power:	3,45	kW
PNEUMATIC DATA		
Total installed power	400 ca	l/min (1 bar 20°C)
Sterile air min. pressure	5	bar
Sterile air max. pressure	8	bar



4.4 USE RESTRICTIONS

4.4.1 PRESCRIBED USE AND FORBIDDEN USES

The closing machine has been designed and made to apply closure systems (caps/corks) on preformed rigid containers with production limits indicated in chapter 4.3 CLOSING MACHINE TECHNICAL DATA page 14.

The closing machine must be used to apply closure systems (caps/corks) on preshaped rigid container already produced according to the order, planning and building stages, as described in the tables attached to this manual.

The closing machine receives the closure systems correctly orientated by its orientation system.

Besides the conduction of the closing machine out of what underlined in the chapter 4.3 CLOSING MACHINE TECHNICAL DATA page 14, just as an example and not limited to, the following uses are forbidden and not expected:

- use of closure systems (caps/corks) and/or preshaped rigid containers whose specifications are different to the ones described in the expected use.
- supply to the closure systems (caps/corks) and/or preshaped rigid hand containers or other means.
- revert the rotation direction of motors.
- tamper guards and/or protection devices.
- overlook the protection means using means to rise on the top or climb on parts of the closing machine.
- perform maintenance operations without insulating and locking all the energy sources and dissipating the residual energies.
- do not comply with the specifications of power sources listed in the table page 14.
- Treat products with characteristics other than those communicated at the order, changing the environmental conditions here the closing machine must be fitted (example: temperature of the treated product; with environmental characteristics other than those communicated at the order).

4.4.2 OPERATING MODES

The closing machine cannot work autonomously.

The installation technician must thus install the closing machine under his own responsibility and according to the European Community directives and take care that all requirements and specifications of the laws in force are complied with.



4.4.3 NUMBER OF OPERATORS

The operations described in this manual, referring the cycle of life of the closing machine, have been analysed by AROL S.p.A.. Therefore, the number of phases and operators indicated is the one most suitable to carry out the functions in a optimal way.

4.4.4 PROFESSIONAL ROLES OF USERS

The closing machine User shall check that the operators have the following requirements:

- had read and understood the indications that are in the manual.
- had received job instruction and training appropriate to their tasks;
- have been properly trained about the correct use of the closing machine, about the residual risks, on the closing machine controls and on the PPM.

Only the operators and the person charged with the maintenance with a suitable training can use the closing machine.

The professional terms used are specified here below to understand the manual fully:

TERM	DESCRIPTION
EMPLOYER	Person legally responsible for the use of the closing machine described herein.
OPERATOR	Person charged with the standard closing machine operation.
MECHANICAL MAINTENANCE OPERATORS	Person charged with the routine and extraordinary mechanical maintenance of the machine.
ELECTRICAL MAINTENANCE OPERATORS	Person charged with the routine and extraordinary electrical maintenance of the closing machine.

4.4.4.1 OPERATOR

The Operator must have:

- familiarity with the use of the check system (operator interface), with capacity of checking the correlation between the set data and the effect of these on the production;
- knowledge of the technology and specific experience in operating the closing machine acquired through the basic training of this manual;
- base general knowledge and sufficient technical knowledge to read and understand the manual content and the annexes, included the correct interpretation of the drawings.
- Sufficient technical knowledge to carry out intervention specified in the manual, in safe operation condition.
- Knowledge of the accident-preventing norms.
 - General Health and safety, prevention of the accidents on the work place.
 - specification (for the type of product processed by the closing machine, according the laws in force of the country in which the machine is installed).



Besides the conduction of the closing machine from the productive point of view the operator must:

- Know the operation of all the safety devices (for the operator and for the closing machine) so as to be able to check their efficiency and signal any possible fault to the personnel in charge according to the procedures described herein.

When the operator locates a failure that might compromise safety, he must stop the closing machine using the emergency stop controls.



it is forbidden to execute any intervention on the electric and/or pneumatic equipment.

4.4.4.2 MAINTENANCE MAN

The Maintenance Man (mechanical/electrical), besides the above characteristics, must also have a good technical preparation and a suitable working experience in the relevant field. Further, he must have specific and special (mechanical and electrical) technical knowledge needed for the operations described in this manual.

4.4.5 EMPLOYER'S OBLIGATIONS

the employer shall guarantee appropriate training and information of all the professional roles of the closing machine on the following topics:

- Instruction and specific procedures for use and maintenance of the closing machine.
- The user of the closing machine should comply with the safety and health prescriptions on the workplace, in force in the premises where the closing machine is started; all the specific risk conditions should be taken into account.

He must have acquired a good knowledge of the contents of this manual so that he can correctly process the training procedures of the professional roles of the closing machine.

At the end of the training, the employer must make sure that operators and maintenance technicians have understood the above points.

Assess the risks resulting from installing the closing machine in the workplace.

The employer must supply the PP (Personal Protection Means) required for all operations linked to the useful life of the closing machine.



4.4.6 INFORMATION - JOB INSTRUCTIONS - TRAINING

4.4.6.1 INFORMATION - JOB INSTRUCTIONS

The customer shall inform and train the personnel charged to use the machine and make sure that:

- On the work place there are suitable information concerning dangers, risks and prevention and protection solutions including the use of P.P.M., if provided.
- the workers the task of using of work equipment receive a suitable training accordingly.
- The workers the task of using of equipment requiring special knowledge and responsibility receive a suitable and specific training, so that they can use such equipment safely and suitably, also as far as risks caused to other people are concerned.

4.4.6.2 TRAINING

Training serves to explain the correct use of the machine, in compliance with the obligations prescribed by national laws transposing 2009/104/EC directive concerning minimum safety and health requirements for the use of working tools by the operators while working.

4.4.7 ARRANGEMENTS BY THE CUSTOMER

Excluding special contract conditions, the user must arrange the following:

- The closing machine must be electrically supplied by the electric board of the machine to which the partly completed machinery is to be installed. The protection against indirect contact must be envisaged during installation, according to point 6.3 of EN 60204-1.
- Compressed air mains connection.
- Generic tools for the closing machine maintenance and consumables.
- Lifting means suitable for the closing machine handling.



4.4.8 SIGNS FOR WARNING, OBLIGATION, PROHIBITION AND DANGER

SYMBOLS CONTAINED IN THE MANUAL

The graphic symbols inside the manual are described and shown here below.

GRAPHIC SYMBOL	DESCRIPTION
	It indicates a generic warning. Pay utmost care to the text blocks marked by this symbol.
	It refers to a warning about key function or useful information. Pay utmost care to the text blocks marked by this symbol.
	Notes, usage tips, or additional information
	Part of the manual dedicated to the OPERATOR.
	Part of the manual dedicated to the MECHANICAL MAINTENANCE OPERATOR.
	Part of the manual dedicated to the ELECTRICAL MAINTENANCE OPERATOR.
	Obligation to insulate all energy sources. The symbol placed close to this description means that all the energy sources must be insulated, dissipating the residual energies by the user, because this involves the risk of an accident.
	Mandatory use of <i>PROTECTIVE GLOVES</i> . The presence of the symbol close to the descriptions informs that the operator must wear safety gloves due to the risk of injury.
	INSPECTION AND OPERATION Visual inspection.
	INSPECTION AND OPERATION Inspection hearing.
	INSPECTION AND OPERATION Intervention with tools
	INSPECTION AND OPERATION Manual intervention.
	INSPECTION AND OPERATION Intervention of lubrication.



SYMBOLS ON THE CLOSING MACHINE

A set of pictograms describing procedures, obligations, cautions and prohibitions are arranged on the closing machine to attract the operator's attention.



The warning and/or prohibition pictograms on the machine have an extremely important safety function; the operators must follow the warning indications.



The pictograms of the closing machine cannot be removed.

Should the pictograms wear out occasionally and become illegible, or if they are removed, they must be restored with new ones that must be asked to AROL S.p.A..

The pictograms on the closing machine are described and shown here below.

PICTOGRAM	DESCRIPTION
	Indicates the duty to see instruction manual before to operate on the machine.
	It indicates a generic warning. Pay attention to the specific pictogram that accompanies it.
 	Means that the lubricant cartridge must be activated before starting the closing machine.



The following figures indicate the pictograms on the closing machine and their position.

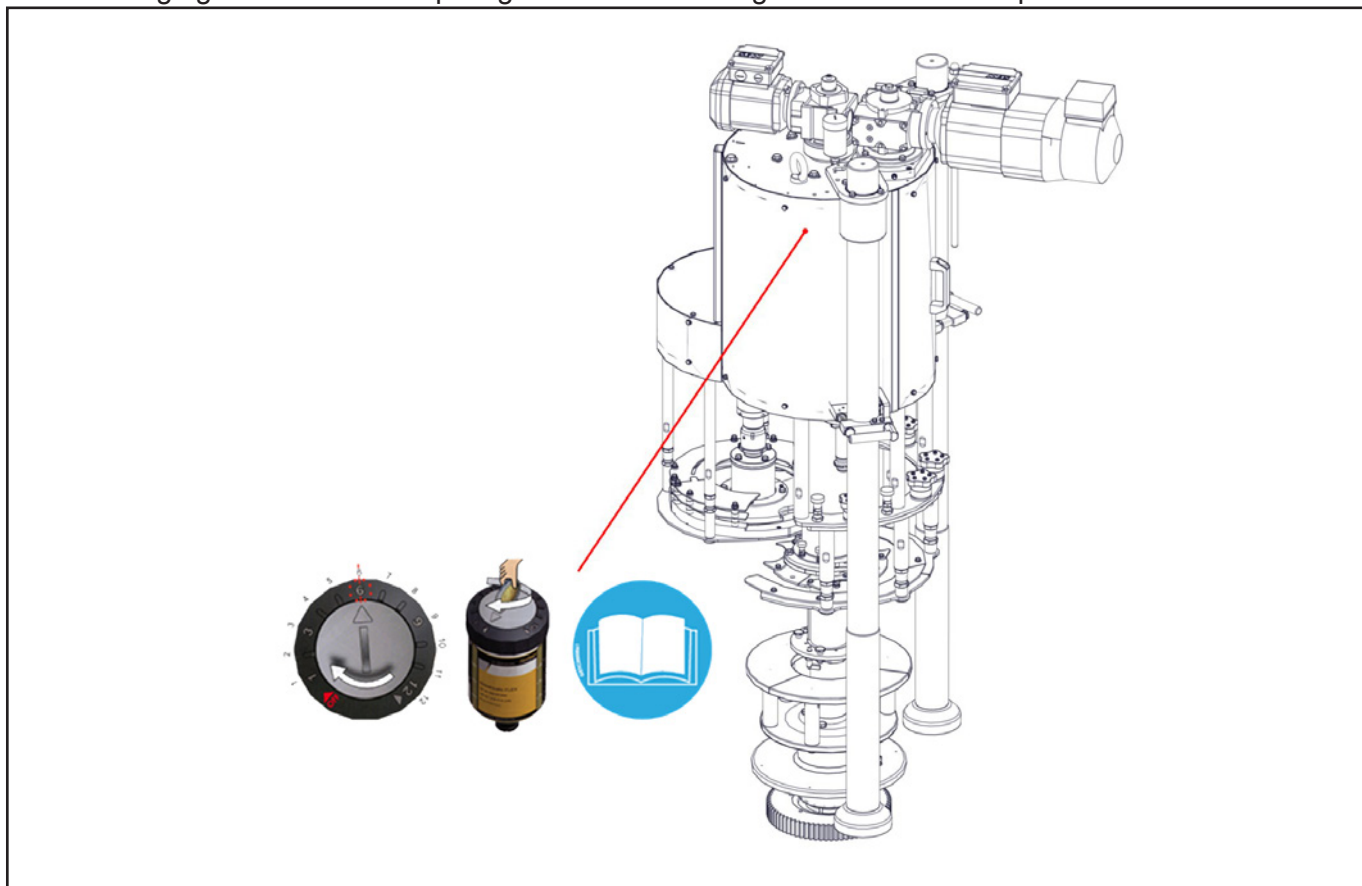


Fig. 12

Re-ordering code: GETIC0000061



4.5 ENVIRONMENTAL LIMITS

4.5.1 ENVIRONMENTAL OPERATING RISKS

USE ENVIRONMENTAL CONDITIONS

The closing machine must work in premises protected by weather agents, with a temperature ranging between 5°C and 40°C, a maximum relative humidity of 80% and height above sea level not over 2000 m.

LIGHTING

The closing machine can work with the standard lighting of the workplace (about 250 lux); local lighting of areas concerned with maintenance is not required very often and must be performed with low voltage lights.

EXPLOSIVE ATMOSPHERE

It is strictly forbidden to operate the closing machine in an area with potentially explosive atmosphere, even if only momentarily.

4.5.2 RESIDUES AND ENVIRONMENTAL CONTAMINATION

No toxic substance is released if the closing machine is used and maintained according to the instructions herein.

The disposal of spent lubricants must respect the norms in force for environment protection in the country of use.



5 SECURITY AND ACCIDENT PREVENTION

5.1 GENERAL INFORMATION

The employer will must provide for the training of the staff about dangers caused by accidents, about equipments arranged for the operator safety, about dangers of noise emission and about the general accident rules provided by international directives and by legislation of the country of designation of the closing machine.

Anyway the behaviour of staff, maintenance operator, cleaning, control, etc. will must scrupulously respect the accident norms of the country of designation of the closing machine.



Before starting the work, the operator must be thoroughly familiar with the location and operation of all controls and characteristics of the closing machine must also have fully read this manual.



The closing machine must be used only from operators that had received a training and understand totally the instructions of this manual.



It is necessary to respect totally instructions, cautions, general accident rules that are in this manual.



Before any maintenance and/or size change operation, the closing machine must be safety conditions:

- **insulate the closing machine from all energy sources dissipating residual energies.**
- **divide and lock all the cutting off devices;**
- **Prevent access to the closing machine to unskilled staff, not charged with maintenance operations, using suitable signs.**



5.2 RISKS - PROTECTIONS - CAUTIONS AND PRECAUTIONS

5.2.1 SAFETY DEVICES OF THE CLOSURE MACHINE

To guarantee health and safety of the people exposed, the closing machine is provided with this safeties:

FIXED GUARDS

All the fixed guards installed on the closing machine are connected to the structure by means of fastening devices which do not allow manual removal, since they necessarily require the use of specific tools.

It is forbidden to use the closing machine after having disassembled the fixed guards; operation can resume after the reassembly of guards.

The housing of the fixed guards in vertical position in the structure of the closing machine has been designed so that the guards, once the fastening devices have been removed, do not remain in closed position, falling by gravity.



To access the area protected by the guard, the closing machine must be insulated from all power sources. (See sub-chapter 5.4 SAFETY PROCEDURES FOR THE MAINTENANCE page 26).

EMERGENCY BUTTONS

Pressing one of the emergency red mushroom buttons placed on the control panel, the following occurs:

- Immediate stop of the closing machine
- Consent inhibition of the auxiliary organs.
- Cutting power off at the actuators.

5.2.1.1 POSITION OF THE SAFETIES

FIXED GUARDS

- Head protection guard.

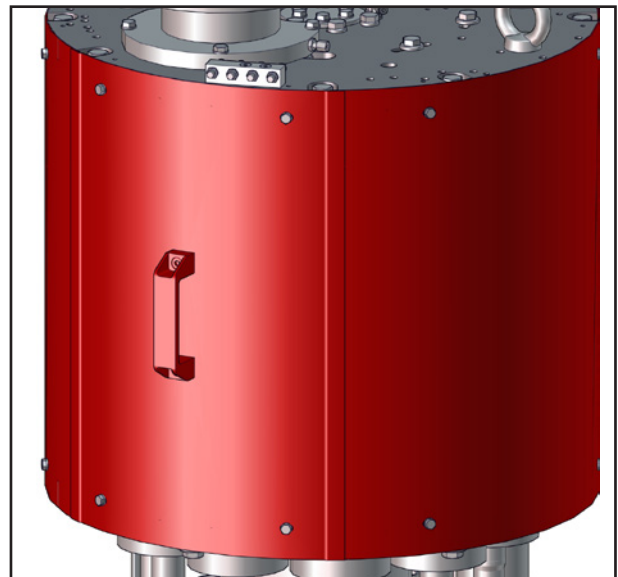


Fig. 13



5.3 SAFETY PROCEDURES FOR THE OPERATION



Wrenches and tools used for the maintenance or the access to closing machine hazardous sections must not be left at the disposal of operators not suitably trained for the size change or the maintenance.

All check, adjustment, maintenance and lubrication steps must be carried out by staff previously trained and instructed.

All check, adjustment and maintenance steps of the electric system must be carried out by staff previously trained and instructed.

People on drugs, alcohol or drowsiness-causing medicines are not allowed to use or maintain the machine.

All check, adjustment, maintenance and lubrication steps must be carried out with the closing machine stopped and separated and split electric power supply.



The cap/cork feeding system has been designed and manufactured to be loaded by an automatic elevator with ground loading.

If the cork/cap loading is manual, it should be performed leaving the closing machine at a standstill, with separated and split electric power supply.



Before accessing the top of the machine, separate and split the energy sources and dissipate the residual energies.

Access at any height must occur using certified means according to the norm UNI EN ISO 14122.



5.4 SAFETY PROCEDURES FOR THE MAINTENANCE



BEFORE CARRYING OUT ANY MAINTENANCE OPERATIONS, the closing machine must be in safety conditions to avoid untimely screwing and/or electrocution.



It is therefore compulsory to insulate all energy sources dissipating residual energies.

The insulation for power sources must be charged to qualified technicians and be performed as follows:

- PNEUMATIC ENERGY - the slider valve must be in discharge position and then locked as it is.



Wrenches and tools used for the maintenance or the access to closing machine hazardous sections must not be left at the disposal of operators not suitably trained for the size change or the maintenance.

After the maintenance works and before starting the closing machine again it is always necessary to:

- check that the possibly replaced parts and/or used tools have been removed from the closing machine.
- check that all guards, protections and safety devices possibly removed during the intervention have been re-installed correctly and are operating.



Replace the electrical safety components within their life cycle, approximately 20 years.



The standard operating conditions will be restored only after the intervention and the checks.



5.4.1 GENERAL SAFETY WARNINGS

The safety for the closing machine and the operators also depends on a regular maintenance according to the manufacturer's indications.

The routine and extraordinary maintenance interventions must be exclusively charged to skilled staff, that is to technicians qualified according to the specifications in this manual.

If outsourced operators intervene, they must ensure the same performance of the inhouse qualified technicians and must work in safe operating conditions.

The access to high levels is allowed only after separating and disconnecting any residual energy and using means certified according to UNI EN ISO 14122 standard.

The qualified technicians must:

- respects the limits of their tasks (mechanical, electrical, etc.).
- Follow the procedures and warnings in the manual, within their tasks.
- Follow the times and intervals indicated in the manual for scheduled maintenance interventions.



The following general warnings must be noted:

- Electrocution hazard due to direct contact with the closing machine power supply terminal boards and in the electrical system connector blocks (the hazard is signalled by plates with the suitable triangle with yellow bottom).
- Be extremely careful during the interventions on the electrical system, if the system must be left live. Such interventions must be carried out by staff skilled in this type of maintenance only, trained and correctly equipped for the steps to be made. The intervention with live system must always be authorized in advance by the employer who must oversee its operations by assessing the specific risk situations.
- Carry out maintenance operations on pneumatic systems and components only after cutting off and separating the electric power and pneumatic sources and checking that the residual pressures have been relieved.
- It is forbidden to carry out maintenance and lubrication interventions on moving components.
- While working on heated parts, the operator must always wait until they cool to a temperature below 55°C to prevent serious burns.

5.5 RESIDUAL RISKS

The residual risks are shown in Annex VII B of the European Directive 2006/42/EC about "Technical documentation for partly completed machinery".



6 HOW TO ASK FOR INTERVENTIONS

AROL S.p.A. puts its own Technical Service at the disposal of its customers to solve any problem regarding the machine use and maintenance.

The interventions must be required after carefully evaluating the faults and their reasons, and stating:

- The details on the occurred faults.
- The performed checks.
- The performed adjustments and resulting effects.
- Any further information deemed useful.

7 HOW TO ORDER SPARE PARTS

All orders for spare parts must be sent to AROL S.p.A..

The installation of not original spare parts will invalidate the warranty.

To order spare parts, fill in the suitable form in the “*SPARE PART CATALOGUE*” for the machine.



The replacement of safety devices must be performed with the same components. The 2006/42/EC directive provides that the components supplied by the machine genuine manufacturer do not need the Declaration of Conformity. Vice versa for all components purchased on the market, the Declaration of Conformity is required, along with the control that the replacement component has the same features and performance of the replaced one. Unsuitable components might cause the decay of the safety performance expected by AROL S.p.A.. It is therefore recommended to contact AROL S.p.A..



8 INSTALLATION AND CONNECTIONS

8.1 POSITIONING OF THE BOTTLE PRESENCE SENSOR



For a correct distribution of the caps it is necessary to position suitably the bottle presence sensor for the cap release.

Position the sensor, operating as follows:

- carry out a closure test, with JOG, in such way to have a bottle under the capping head.
- Count the number of delivered caps starting from the closure head (where touches the bottle) going backwards until the last cap delivered on the transfer disc .
- Count the same number of bottles, starting from the closure point and going backwards into the load star-wheel.
- Position the presence sensor so that it detects the last counted bottle.

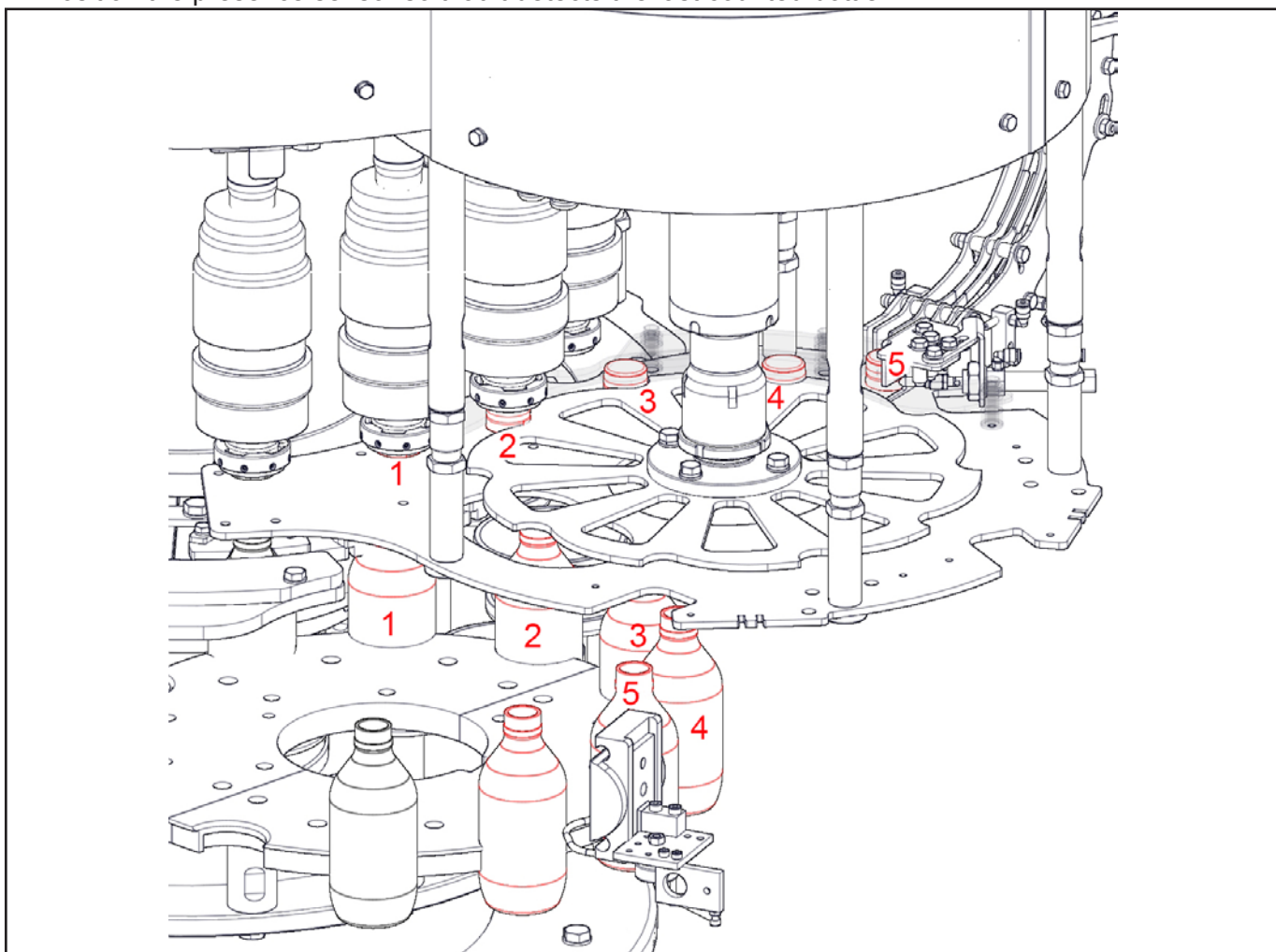


Fig. 14



Please note that the sensor must always keep the signal high so as not to result in a discontinuous operation of the cap stop cylinder in case of bottle continuous flow.



8.2 CAPS CHUTE INSTALLATION



To install the caps chute on the machine it is necessary:

- Loosen the threaded fastening placed on the caps sorter

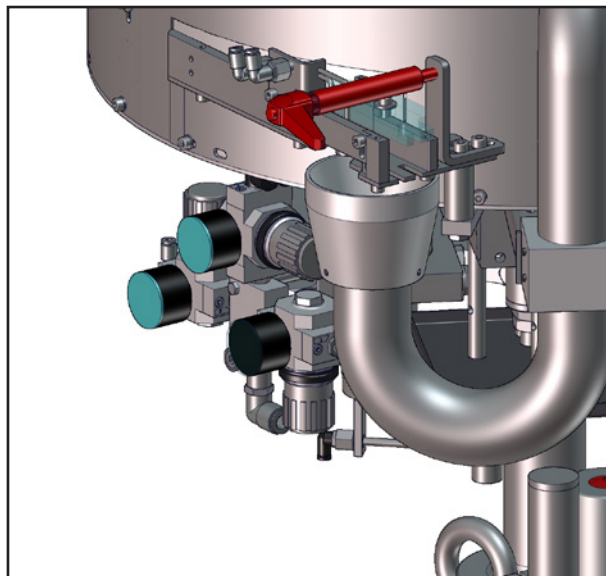


Fig. 15

- Loosen the threaded fasteners located on the support chute.

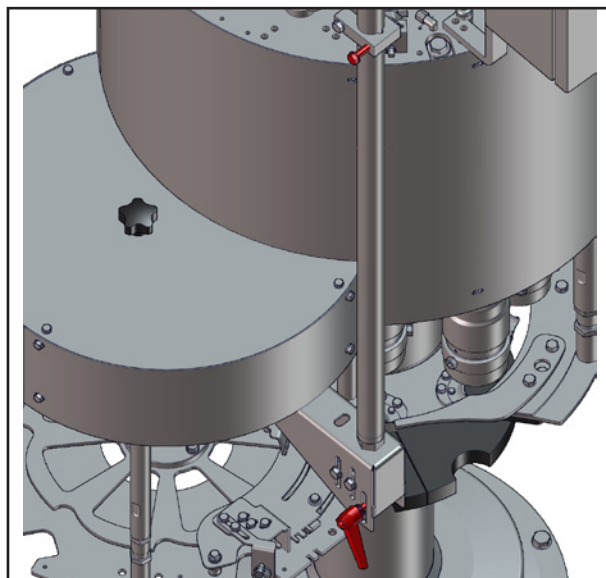


Fig. 16

- Place the caps chute so that the fixing brackets are positioned on threaded fasteners.

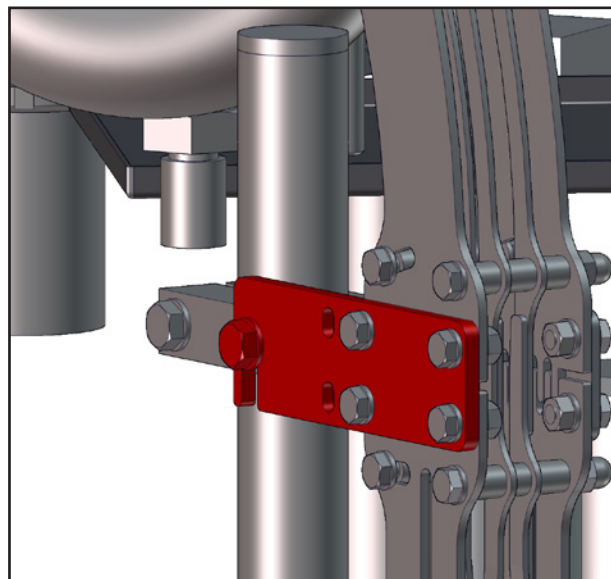


Fig. 17

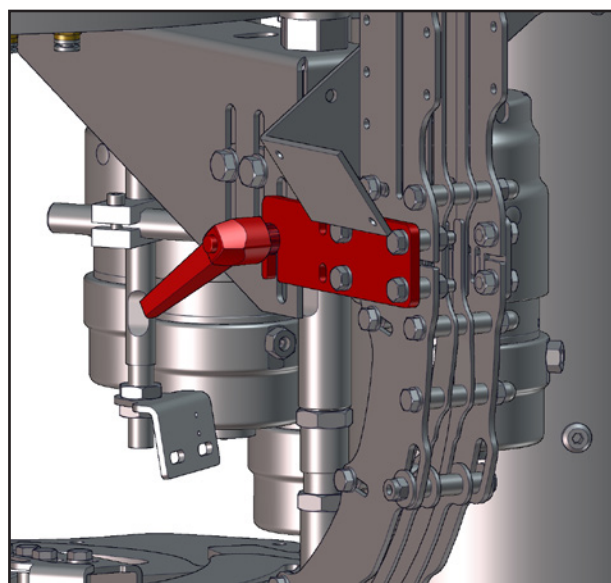


Fig. 18

- Check that the centering pin of the caps chute fits into the seat of a basic disk.

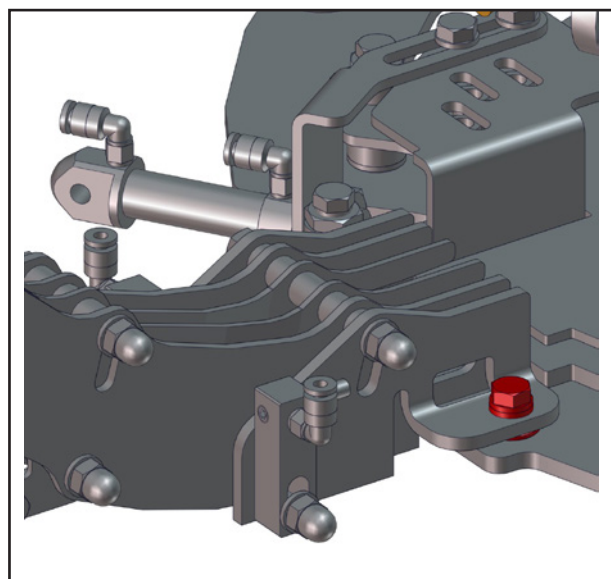


Fig. 19



- Fasten the caps chute to the base disk with the fastening bolt.

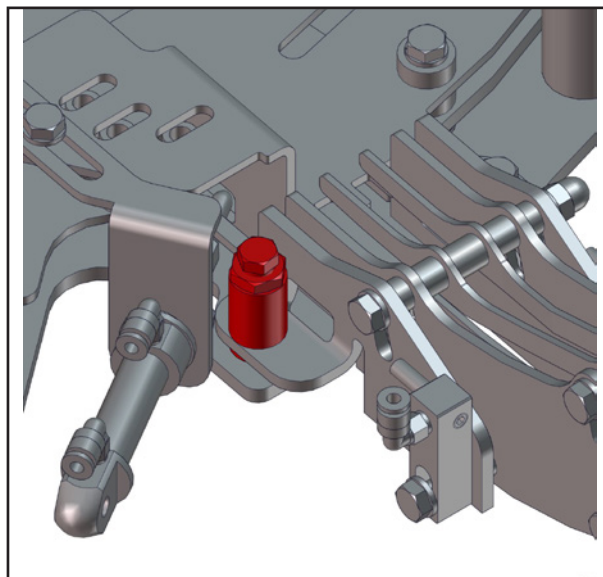


Fig. 20

- Lock threaded fasteners located on the support chute.

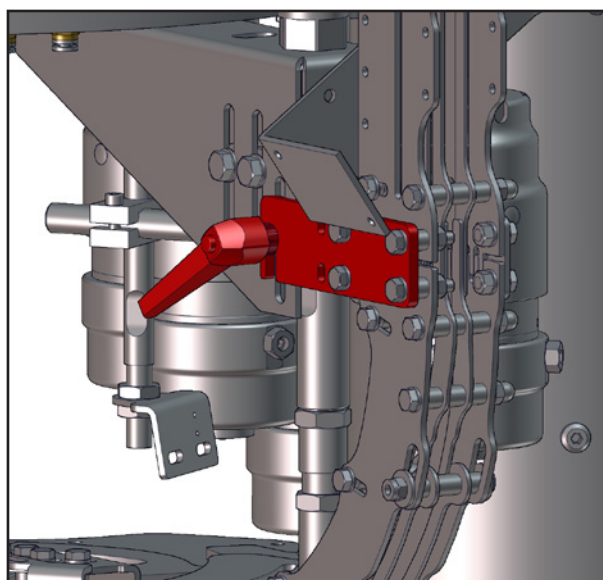


Fig. 21

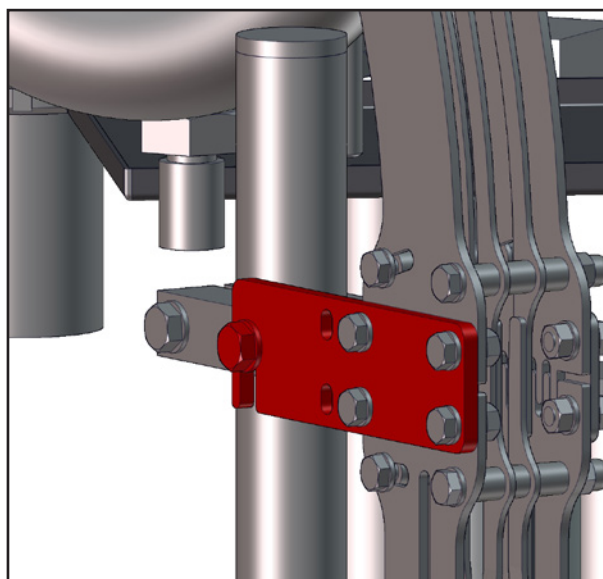


Fig. 22



- Connect the electrical connection of the caps presence photocell (one or more).

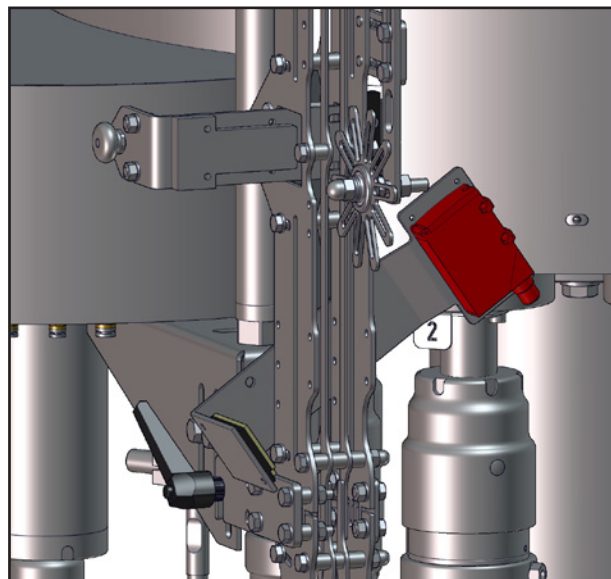


Fig. 23

- Connect the air compressing fitting to them connections.

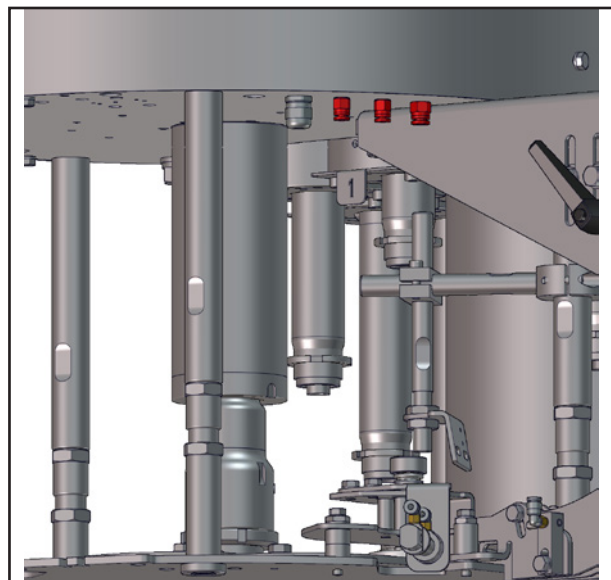


Fig. 24



8.3 "FLAT" INSTALLATION OF THE CENTRIFUGAL CAPS SORTER



For the installation of the centrifugal caps sorter see the listed below operations:

- Fasten belts or chains with hooks to the eyebolt outside the hopper.

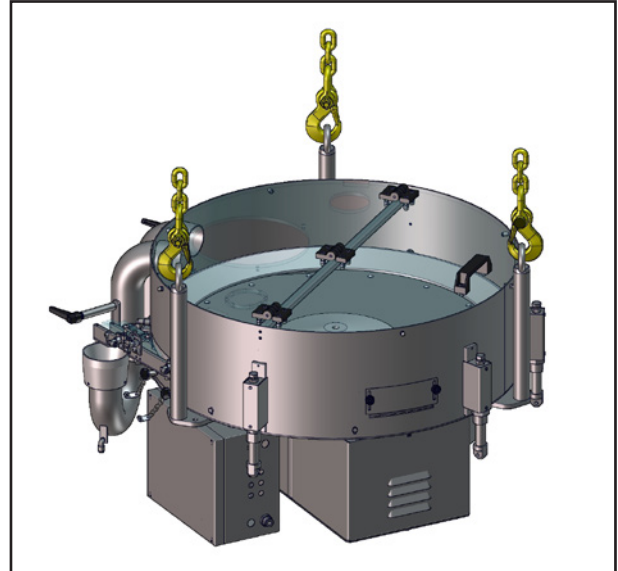


Fig. 25

- Unscrew the bolts that fasten the caps sorter to the transport platform.
- Lift the caps sorter with adequate lifting means.



The caps sorter mass is not symmetrically distributed so it overturns when lifted up. Do not stand or walk under the caps sorter when it is hoisted!

- Position the caps sorter complete with the support in such way that the references on the support and the references of the flange are aligned and that the exit mouth is aligned with the caps chute.
- Fix the caps sorter by bolts.

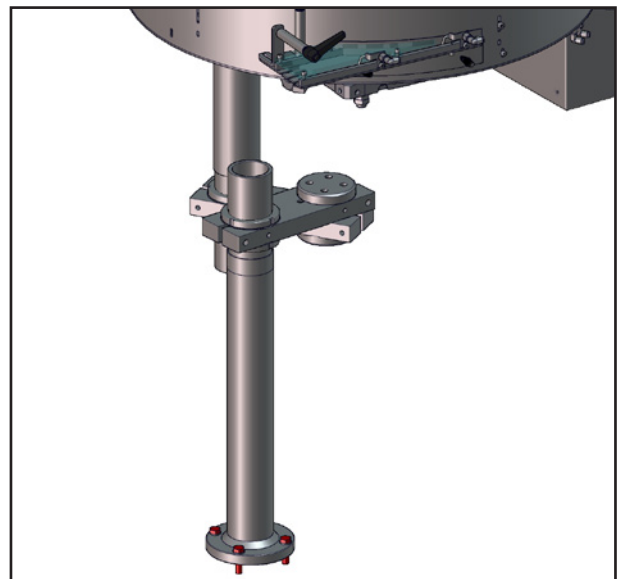


Fig. 26



- Unhook the belts or the chains with hooks from the eyebolts.

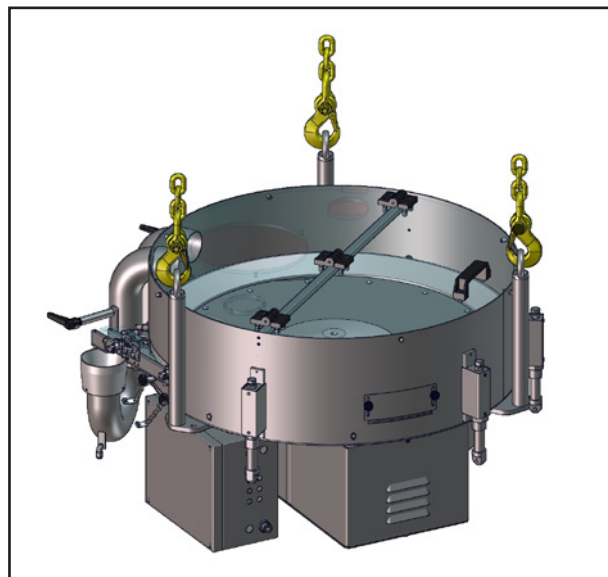


Fig. 27

- Lock threaded fasteners located on the support chute.

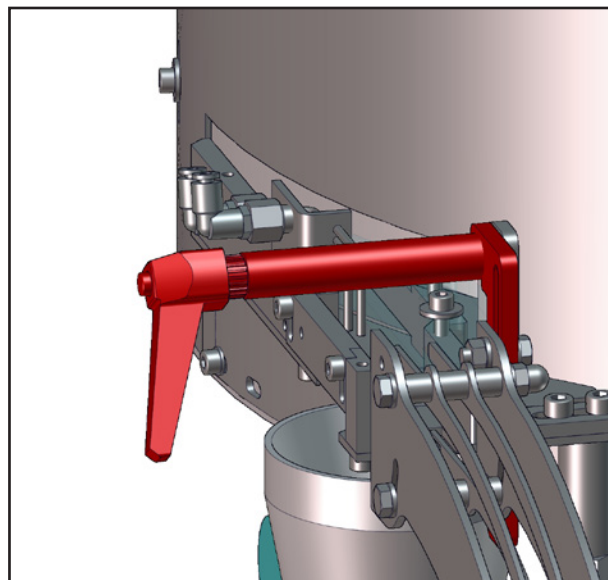


Fig. 28

- Place the caps level device on the caps sorter proceeding as described below:
 - Place the level blade in the hole of the cover and the rod fixing bush on the hopper.

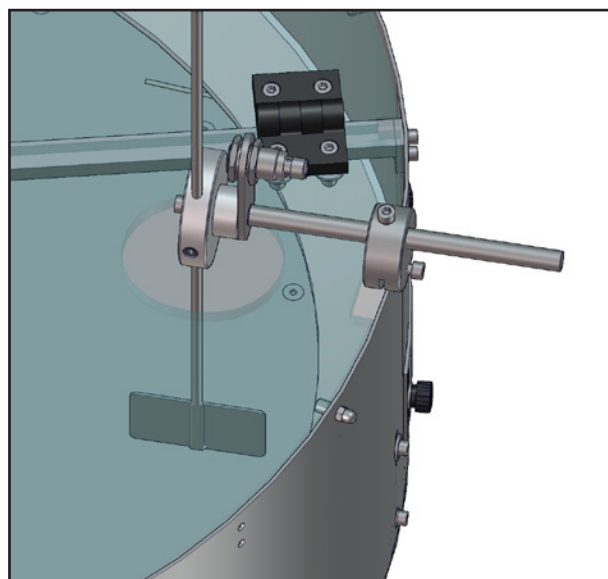


Fig. 29



- Fasten all things locking the screw against the hopper.

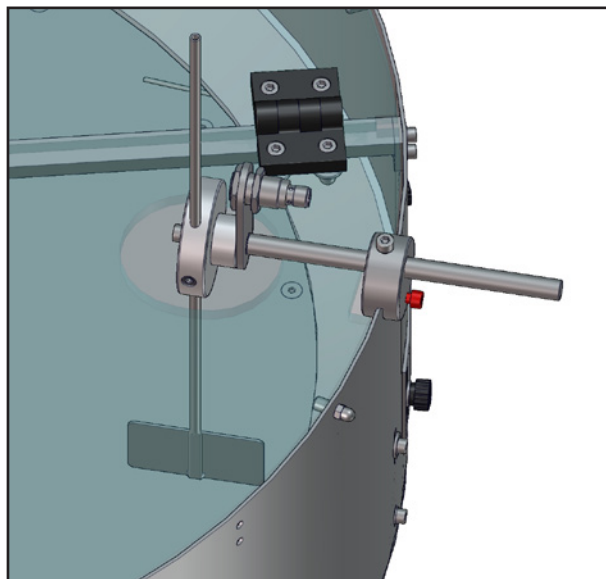


Fig. 30



8.4 PNEUMATIC CONNECTION



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



The air used in the pneumatic installation must be filtered, so without impurities, and dried.

For pneumatic data see the table page 14.

Connect with the mains as follows:

- connect the attack of the pneumatic assembly to the main distribution network.

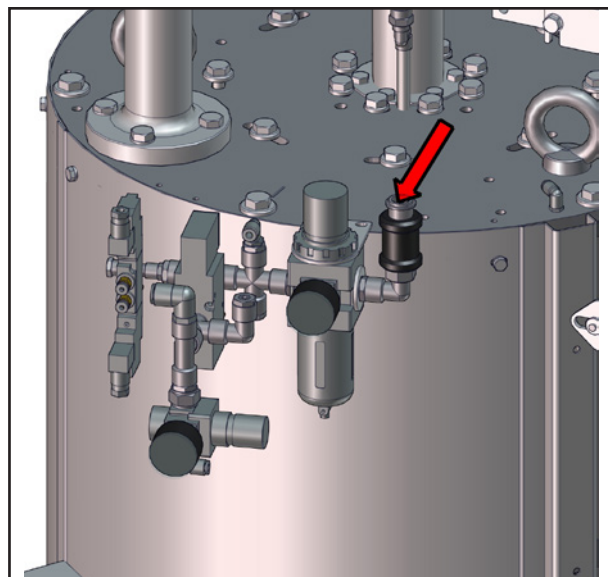


Fig. 31

- Connect the flow rate pipe coming from the main pneumatic unit to the caps orientation unit coupling

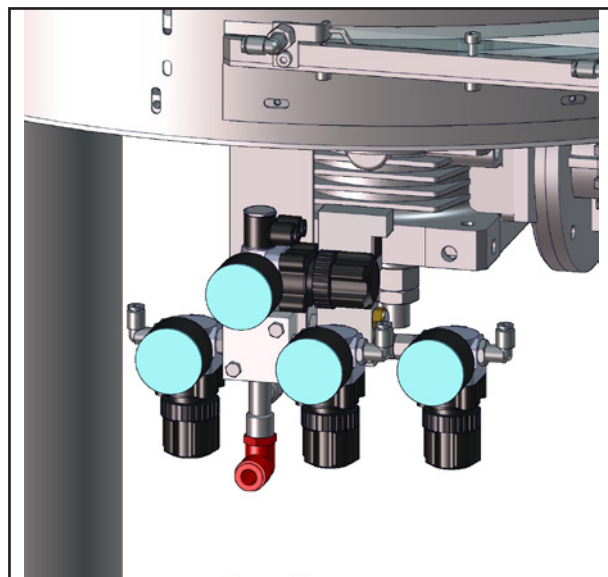


Fig. 32



9 CLOSING MACHINE USE

9.1 SETTING UP, CHECKS AND TEST FOR THE FIRST START UP



The setting up operations of the closing machine for the first start up must be carried out by qualified technicians.

The closing machine has been tested by AROL S.p.A. premises before the shipping; all the calibration and required adjustments have been carried out during the test.



Before carry out operations on the closing machine, it is necessary to install and activate the lubricant cartridge, proceeding as described in sub-chapter 11.6.6.2 CARTRIDGE REPLACEMENT page 130.

9.1.1 CHECKS BEFORE THE START UP



In order to prevent errors and/or accidents, before the start up of the closing machine it is necessary to carry out some checks:

- Check the sequence of the motor supply phases.
- Check that there are no leaks from the pneumatic circuit.



9.2 OPERATING OF THE "TWIN HOPPER" FEEDING SYSTEM

At the start up of the closing machine:

- The second caps chute is filled
- The main caps chute is filled; it is possible start the production
- The caps sorter operates with the usual system (start up at the same time of the closing machine rotation, stop delayed of about 1 minute)

During the normal operating cycle:

- the caps release takes place always by the first caps chute.
- The secondary caps chute is not used.

During the ordinary operating of the system it is necessary to empty the secondary caps chute periodically in order to avoid a too long permanence of the caps inside the system. As the caps sorter will be subject of continuous star/stop cycles, it is recommended the use of devices able to guarantee a soft start up.

In case of lack of caps into the caps chute or an error of the first transfer disk release check photocell

- The caps sorter is put in rotation inverting the motor rotation direction
- The release of the caps is carried out by the second caps chute; the error on the disk is restored by releasing the lacking cap/caps



The passage to the second caps chute must be temporary if the error that caused the change is the lack of caps into the caps sorter (at the restore of the operating condition it is possible to use again the first system). If the error has been noticed in the caps chute or on the transfer disk, the production will continue with the second caps chute, stopping the bottle flow at the beginning of the installation in order to stop and restore the priority caps chute.



9.3 PRELIMINARY CHECKS BEFORE STARTING



Before starting the closing machine up, check the following:

- Check that all other safety devices installed on the closing machine are operating properly. (See also sub-chapter 5.2.1 SAFETY DEVICES OF THE CLOSURE MACHINE page 24).
- Check all closing machine components are lubricated (See Chapter 11.6 LUBRICATION page 122).
- Make sure the specific installed equipment are suitable for the size to be processed.
- Check the correspondence between container and cork to be treated.



Each set of equipment is marked as indicated in the relevant tables enclosed in this manual. Each mark corresponds to a size to be processed.

- Check the height of the closure carousel according to the container height.
- Check the caps sorter is sufficiently full.
- Check that the working pressure is correct (the pressure is indicated on the pressure gauges).



9.4 PREPARATION FOR STARTING

9.4.1 CLOSING MACHINE HEIGHT ADJUSTMENT



The height must be adjusted exclusively by lowering the closure carousel. Before operating, take the closure carousel at a height higher than the one of the bottle to be processed.

For the adjustment and check, operate as described here below:

- Release the screw locking the central column.

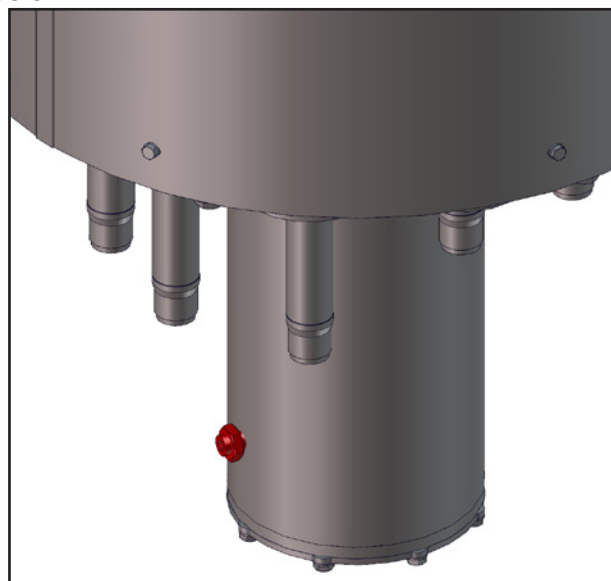


Fig. 33



- Lift the closure carousel using the control till passing the height of the bottle to be processed.
- Put a container to be closed under the cap transfer disk entering the closure machine.
- Fit the supplied spacer on the bottle.

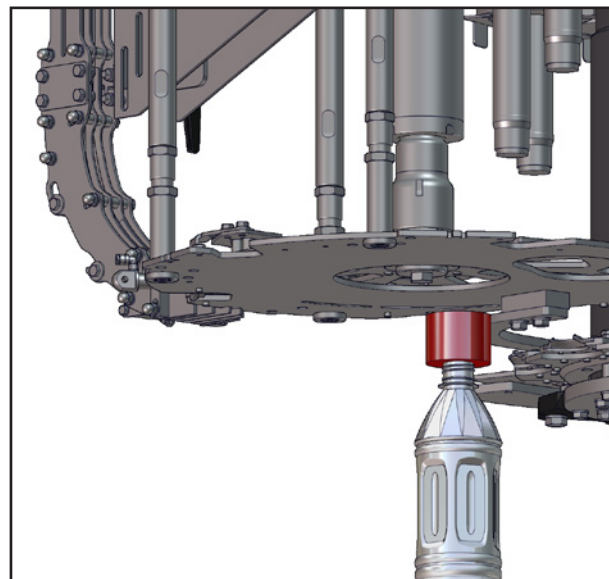


Fig. 34

- Lower the closure carousel as long as the base disk of the transfer disk just touches the spacer.

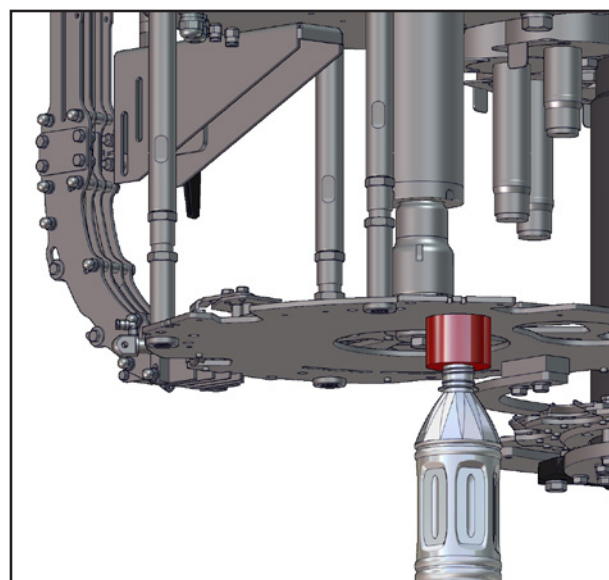


Fig. 35

- Lock the fixing screw of the central column.
- Remove the spacer and the bottle used for the adjustment.

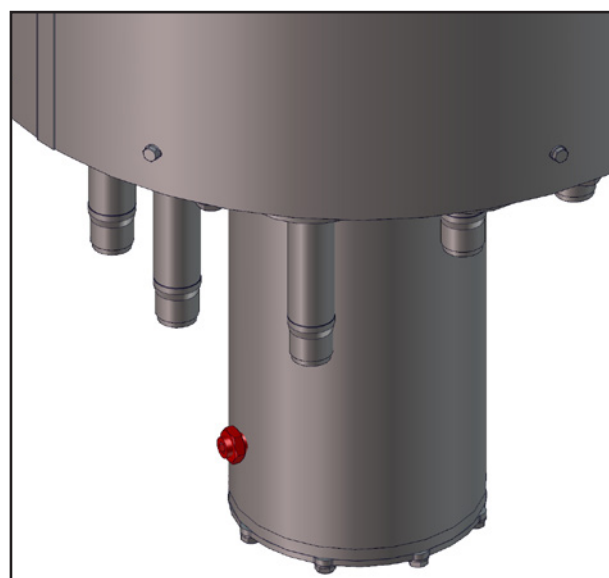


Fig. 36



9.4.2 CAPS SORTER LOADING



On the caps sorter is installed a caps level sensor having the task to signal to the automatic elevator when it is necessary to put a new quantity of caps.

For a correct operating, if the caps sorter is loaded with different system from the automatic elevator, it is necessary to supply a quantity of caps that arrives at least 5 mm under the lower plate of the device.

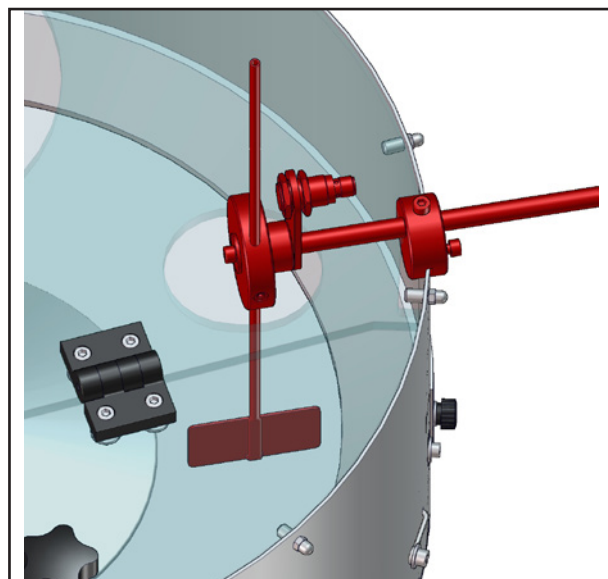


Fig. 37

9.5 EMPTYING OF THE CAPS DISTRIBUTION SYSTEM

9.5.1 CAPS SORTER EMPTYING



To discharge the caps sorter it is necessary to rotate it till totally empty.



9.5.2 CAPS CHUTE DISCHARGE



To discharge the caps chute, operate as follows:

- Insert a collection container under the caps chute, in proximity of the unloading door.
- Loosen the knob securing the unloading door.

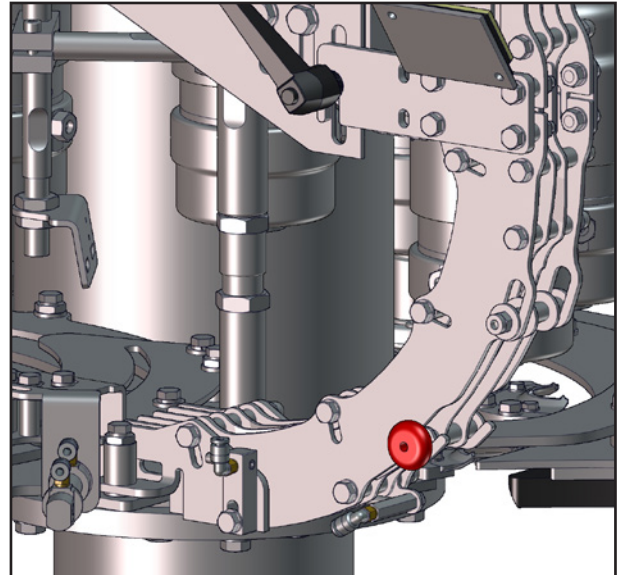


Fig. 38

- Push up the unloading door so as to release it from the fixing seat.
- Open the unloading door.
- If necessary, empty the cap sorter.
- Empty the cap chute.
- Restore the position of the unloading door and fasten by screwing the fixing knob.

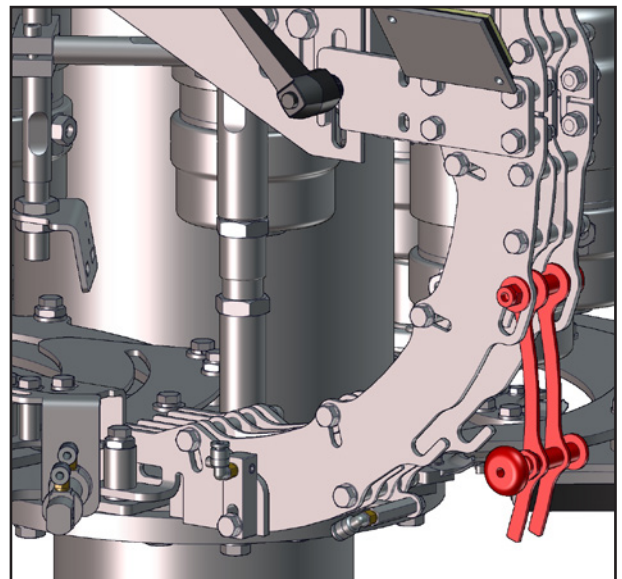


Fig. 39

9.5.3 OVERTURNED CAPS EJECTION



The caps chute has a overturned caps ejection automatic device **AED** (Automatic Ejection Device).

If caps are orientated in an incorrect way into the caps chute, a selection star-wheel stops the descent.

To discharge the overturned cap from the caps chute, it is necessary to open the caps ejection opening carrying out on the suitable control for the pneumatic cylinder operating.

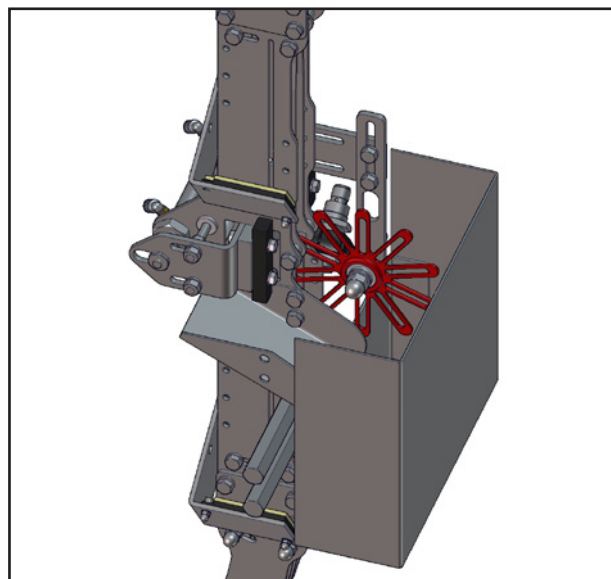


Fig. 40

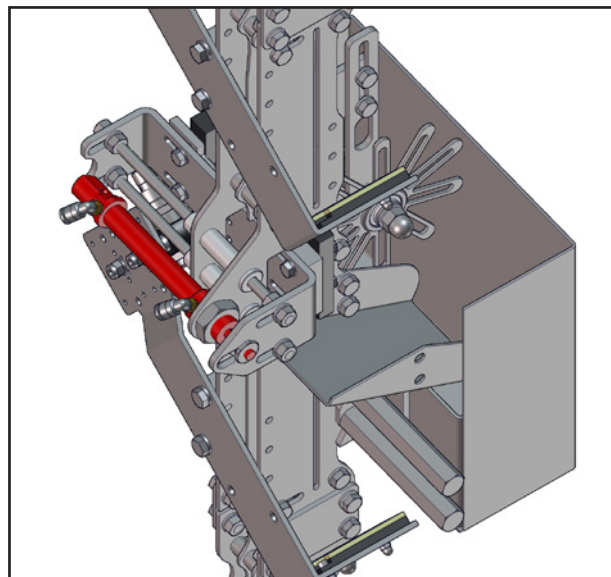


Fig. 41

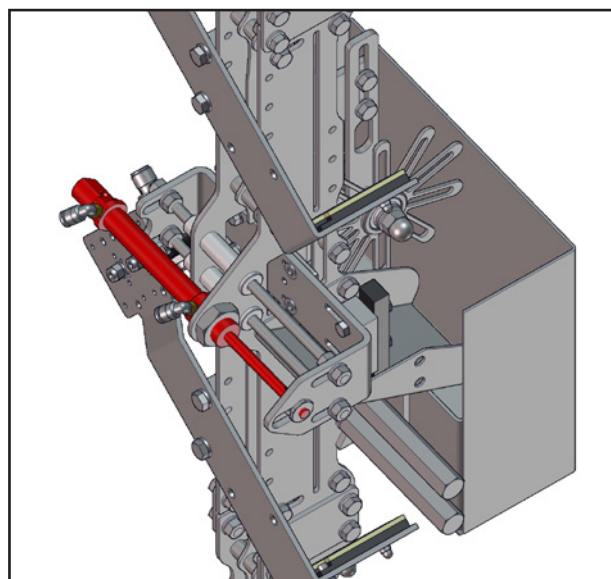


Fig. 42



9.6 CLOSING MACHINE CLEANING



The determination of the frequency of cleaning operations during the closing machine continuous use is to be checked by the user, depending on the environment is being operated.

The following subchapter describes the cleaning intervals of the closing machine and its components, when considering their use in a clean industrial production area.

9.6.1 CLOSING MACHINE CLEANING INTERVALS

The following table shows the cleaning intervals of the closing machine, expressed in hours.

Cleaning	Work shift (hours)		
	1 shift	2 shifts	3 shifts
Closing machine external washing. See sub-chapter 9.6.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 47.	8	16	40
External washing of the closure heads. See sub-chapter 9.6.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 47.	8	16	40
External washing of the closure chucks. See sub-chapter 9.6.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 47.	8	16	40
Cork/cap sorter cleaning. See sub-chapter 9.6.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 47.	120	120	120



9.6.2 EXTERNAL WASHING OF THE CLOSING MACHINE



Cleaning and washing are allowed with cold closing machine only.

The description below concerns: cleaning areas, cleaning systems and specifications, products allowed for cleaning the closing machine and its components.

Legend:

- 1 Closure area
 - 2 Bottle-guide and conveying area
-  The washing jet should be directed from top to bottom. Do not direct air jet towards area "1".
- 4 Upper area of the closure carousel
 - 5 Closure system direction area

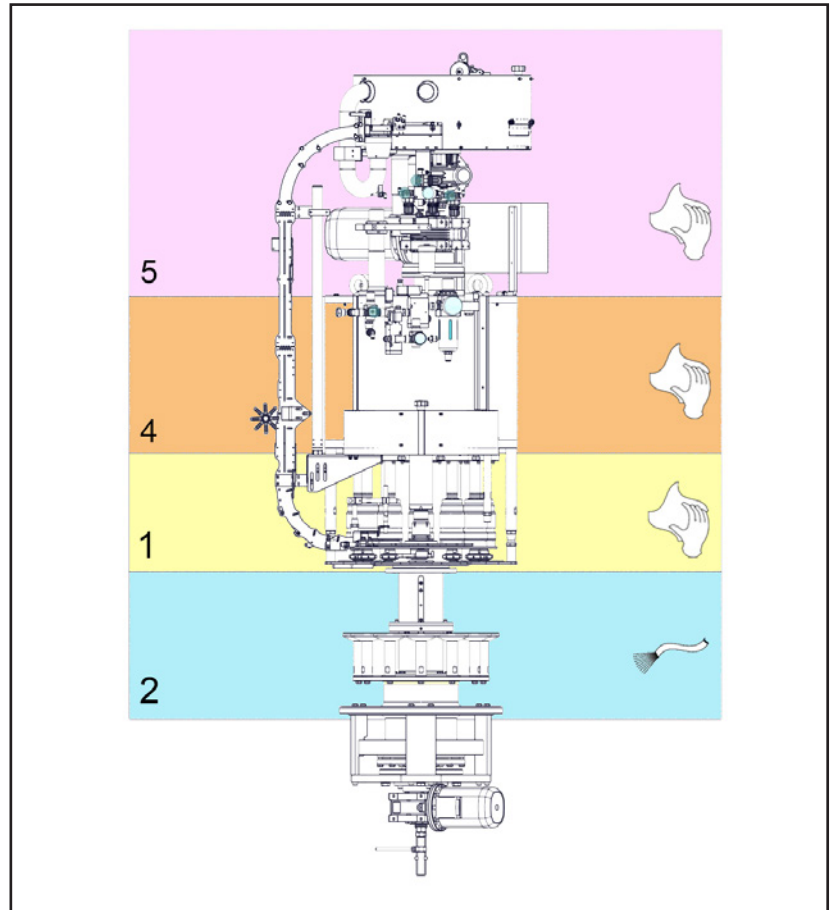


Fig. 43

CLEANING SYMBOL		MAX PRESSURE OF THE CLEANING [bar]	DISTANCE FROM THE SURFACE [mm]	WASHING SYSTEM TILTING [°]
	washing hose	≤ 2,5	-	
	cloth.	-	-	-



To clean all components, use the products listed in the table below. For cleaning systems, see the table on page 47.

PRODUCT	MAX. TEMP. (°C)	pH
Water	35	
Neutral detergents	35	5 ÷ 9

It is absolutely forbidden to clean the closing machine with:

- **water jets under pressure.**
- **aggressive detergents.**
- **steam.**



Before carrying out cleaning, you should refer to the product data sheets relating to detergents.



Before starting production, the closing machine should be suitably rinsed, to remove all the detergents used, then dried up.



IMPORTANT : AROL S.p.A. supplies instructions for washing in order to clean the parts are remove coarse grease stains, powder and scale. On the other hand, it does not guarantee the hygiene of the surfaces, which must be checked by the user depending on the type of treated products, pollutants and disinfection, sanitisation or sterilisation procedures.



Detergents residuals or residuals of liquids used for the cleaning of the closing machine must be disposed according the laws in force of the country in which it is used.



10 SIZE CHANGE



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



The closing machine has been designed to handle containers and caps / corks as indicated in the relevant tables enclosed in this manual.



Every container and/or cap must be processed exclusively with the equipment there indicated.

Each series of equipment is marked according to the size to be processed:

- The cap guide equipment, in case of multiple sizes, are marked with an identification initials.

In order to run properly and without obstructions the format change operations, it is necessary to raise the closing turntable at its maximum height.



10.1 CONTAINERS NECK GUIDES EQUIPMENTS SIZE CHANGE



Having to change the size of the container should be treated:

- lift the closure carousel so as to allow the removal of bottles neck guide units.
- Remove the external neck guide. (See sub-chapter 10.1.1 EXTERNAL BOTTLE NECK GUIDE page 51).

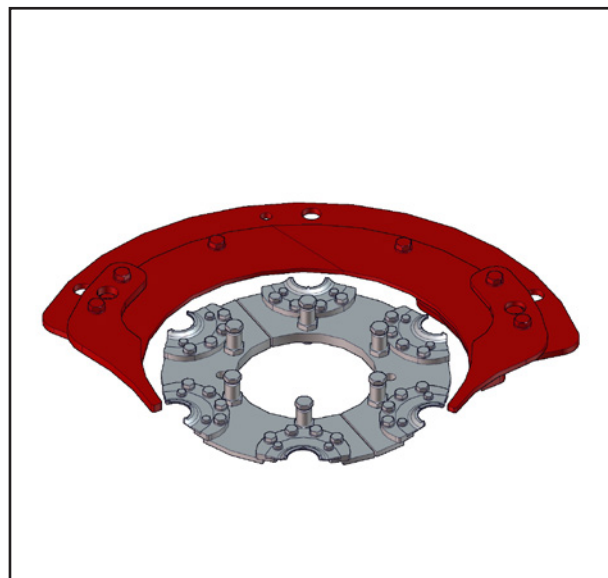


Fig. 44

- Remove the flange for neck guide centering devices. (See sub-chapter 10.1.2B O T T L E NECK CENTERING DEVICE SUPPORT page 52).

Replace the bottles guide equipments with the suitable ones to the size to proceed, proceeding in the reverse way to what described above



To identify the correspondence between container guide unit and the same container, see the TOOLING COLOUR TABLE attached to this manual.

Adjust the closure machine height for the size of the container to be treated.

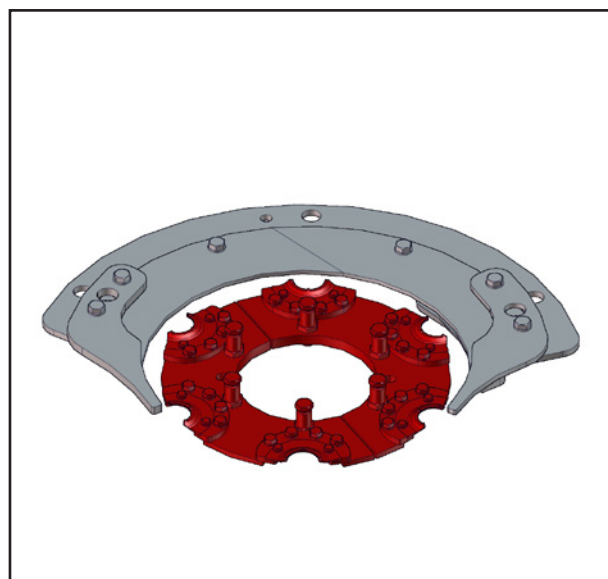


Fig. 45



10.1.1 EXTERNAL BOTTLE NECK GUIDE



10.1.1.1 REMOVAL

To remove proceed in the following way:

- Loosen and remove the fixing bolts.

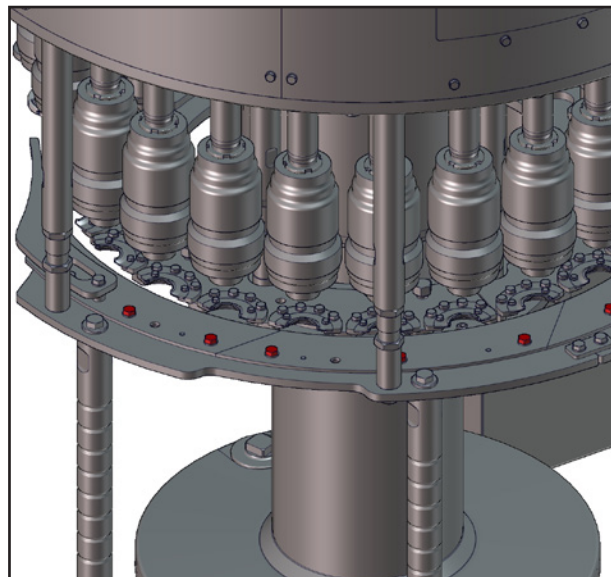


Fig. 46

- Remove the external neck guide sectors.

10.1.1.2 INSTALLATION

For installation proceed as follows:

- Position the external neck guide sectors on the supporting guide.
- Fasten the external neck guide sectors by the bolts.

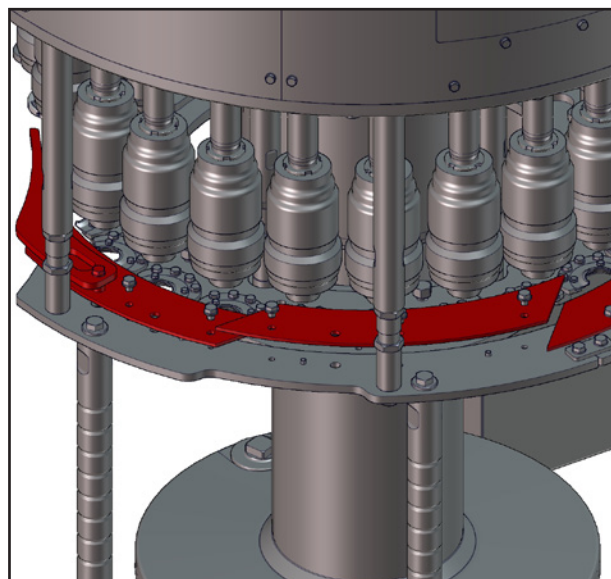


Fig. 47



10.1.2 BOTTLE NECK CENTERING DEVICE SUPPORT



10.1.2.1 REMOVAL

To remove proceed in the following way:

- Make the closing machine rotate in manual jog until a centering device holder sector is located between the caps transfer disk and the input bottle-neck guide.
- Unscrew fastening bolts of the centering device holder sector with a 19 mm hexagonal wrench, and remove bolts.

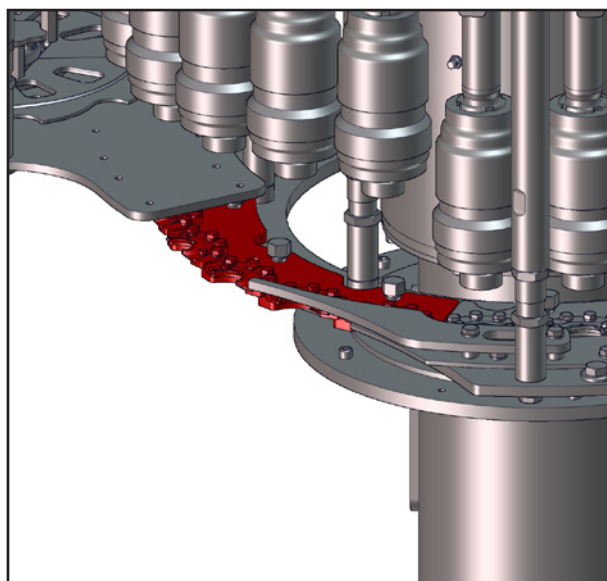


Fig. 48

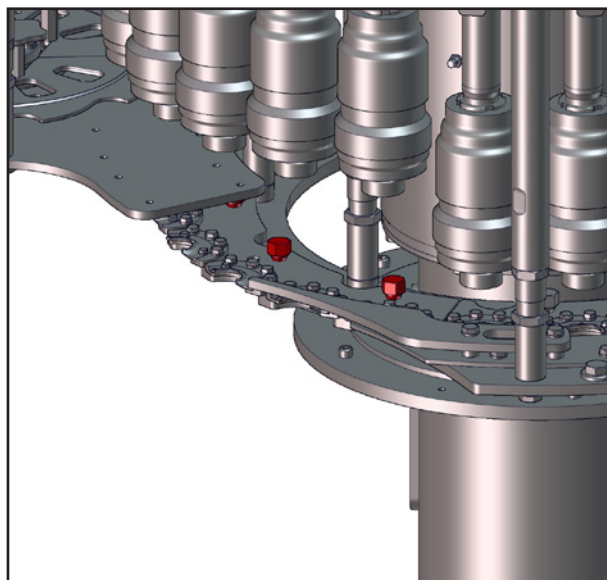


Fig. 49

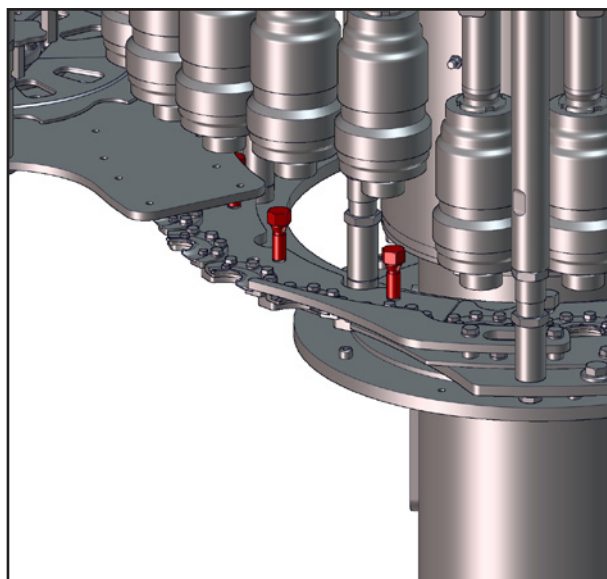


Fig. 50



- Remove the centering device holder sector.
- Make the closing machine rotate in manual jog until having the next centering devices holder sector next in the removal position.
- Proceed with removal of the centering devices holder sector as previously indicated.

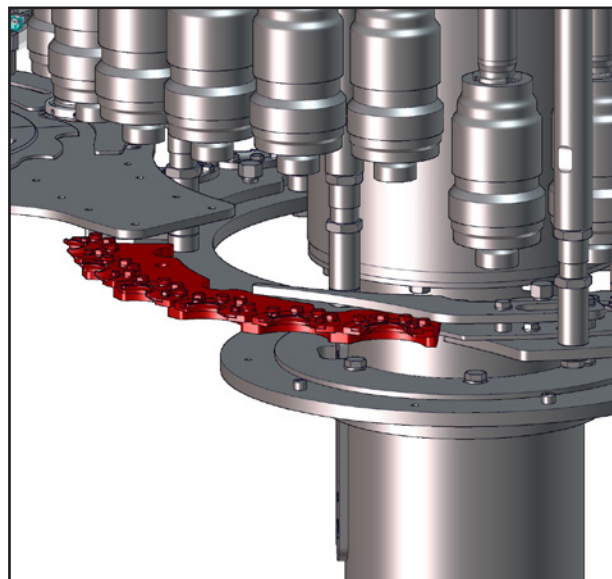


Fig. 51

10.1.2.2 INSTALLATION

For installation proceed as follows:

- Place the centering devices holder sector on the support.

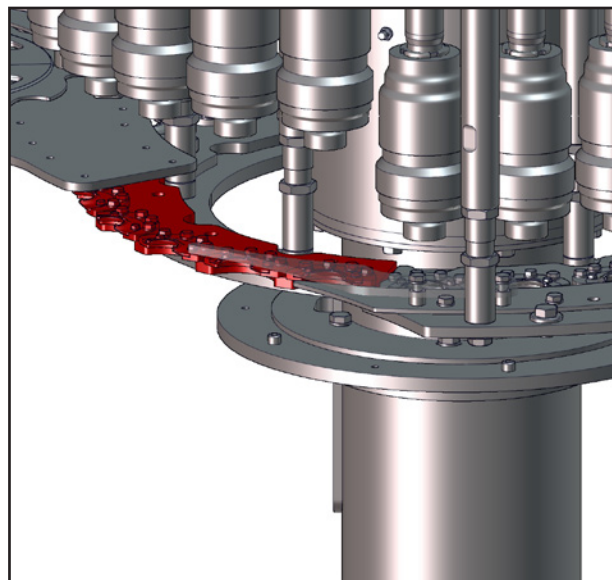


Fig. 52

- Fasten the centering devices holder sector by fixing bolts. Screw bolts with an hexagonal wrench.
- Make the closing machine rotate in manual jog to install the next centering devices holder sector.
- Proceed with installation of the centering devices holder sector as previously indicated.

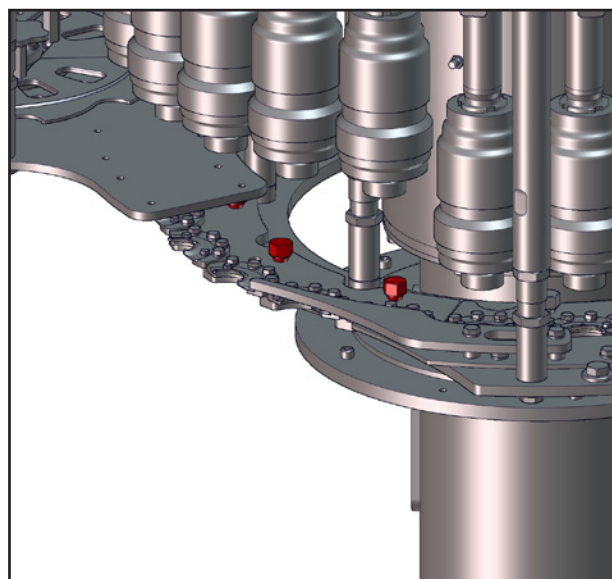


Fig. 53



10.1.3 BOTTLE NECK GUIDE SECTOR



10.1.3.1 REMOVAL

To remove proceed in the following way:

- Loosen and remove the fixing bolts.

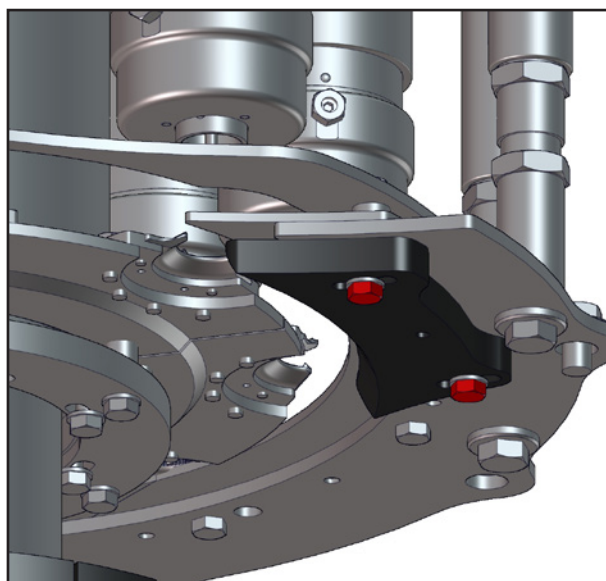


Fig. 54

- Extract the bottle-neck guide sector.

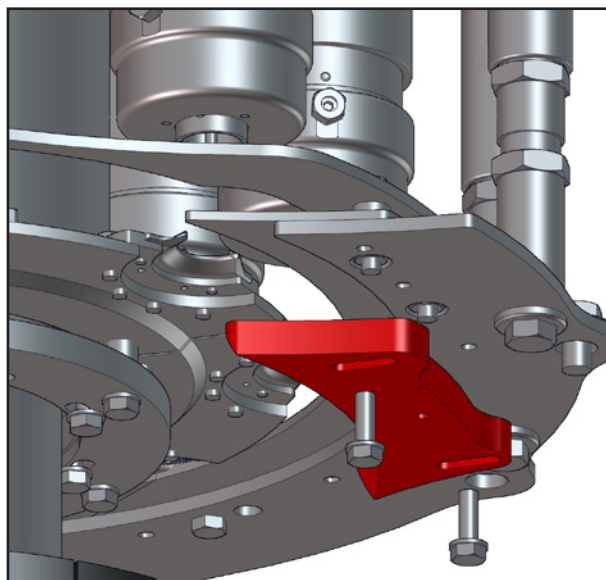


Fig. 55

10.1.3.2 INSTALLATION

For installation proceed as follows:

- Position the bottle-neck guide sector under the bottle-neck guide support.
- Fasten the bottle-neck guide sector using the bolts.



After the installation of the bottle-neck guide sector, it is necessary to adjust its position in such way it is in contact with the neck of the bottle. On the guide there are adjustment slots.



10.2 CAP SIZE CHANGE



Having to change the format of the caps to be treated, must be replaced:

- replace the closure cones. (See sub-chapter 10.2.4 REMOVAL AND INSTALLATION OF THE CLOSURE CHUCK page 57).
- Replace the closure heads, if provided. (See sub-chapter 10.2.5 REMOVAL AND INSTALLATION OF THE CLOSURE HEADS page 60).
- Replace the caps delivery chute, if provided. (See sub-chapter 10.2.6 PICK AND PLACE CAPS CHUTE page 64).
- Replace the caps transfer group on the transfer disk, if provided. (See sub-chapter 10.2.7 CAPS TRANSFER GROUP page 70).

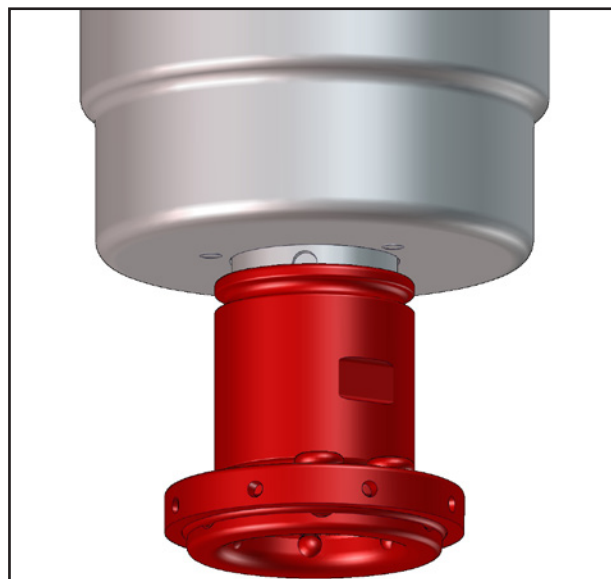


Fig. 56

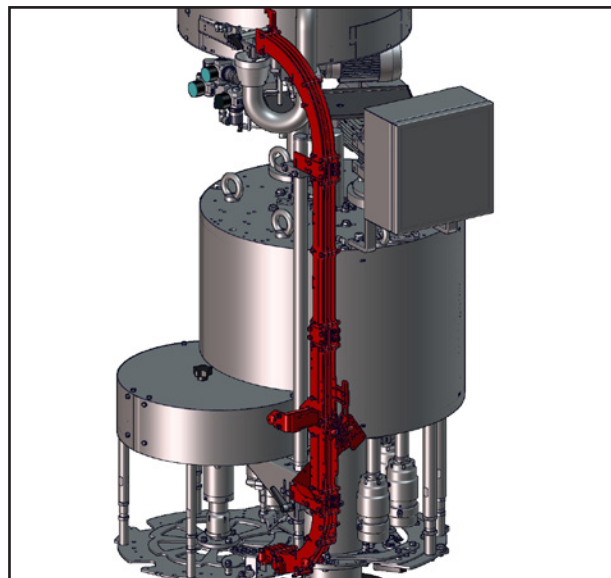


Fig. 57

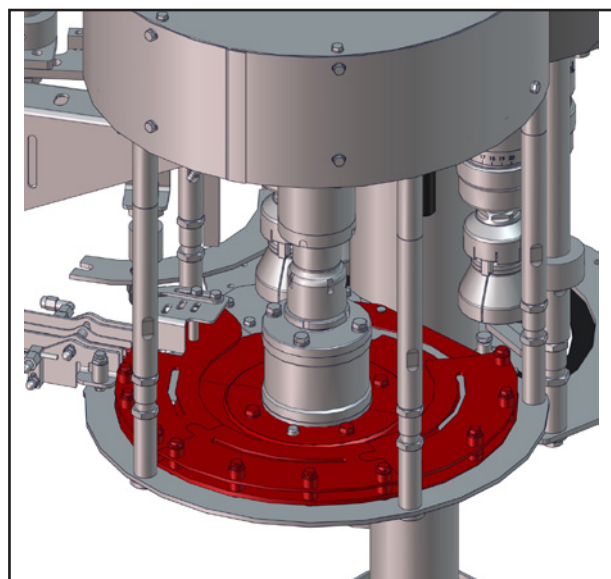


Fig. 58



- Replace the caps selection units, if provided. (See sub-chapter 10.2.8 CAPS SELECTION EQUIPMENT FOR CENTRIFUGAL CAPS SORTER page 77).

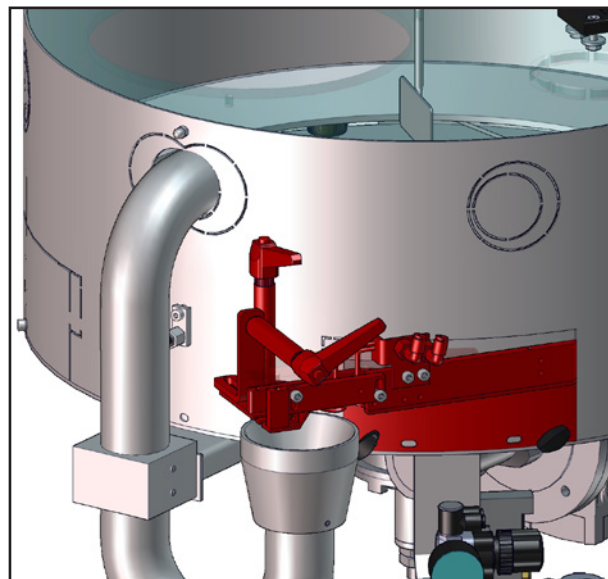


Fig. 59



10.2.4 REMOVAL AND INSTALLATION OF THE CLOSURE CHUCK

10.2.4.1 REMOVAL OF THE CLOSURE CHUCK "PK 314"

The table below shows the dedicated tools for disassembly and assembly of the closure chuck.

TOOLS	
	Hook wrench slim
	Hexagonal wrench 34 mm

To remove the closure chuck, operate as follows:

- fit the slim wrench for ring nut in the hole of the ejector guide bush.

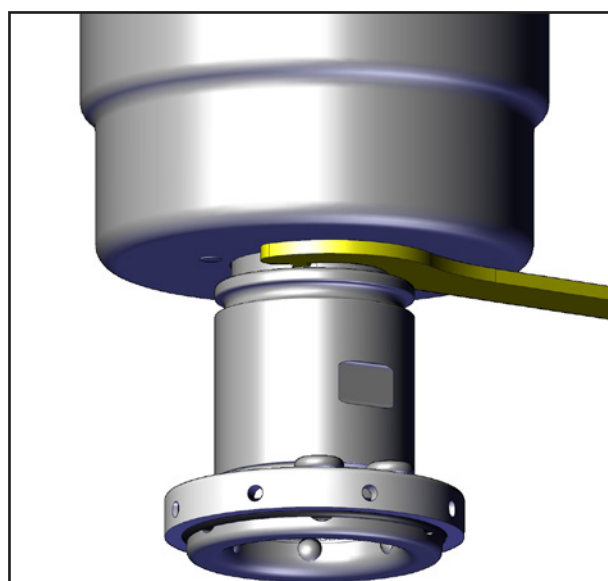


Fig. 60

- Fit the 34 mm hexagonal wrench in the seats of the closure chuck.
- By means of the two wrenches, unscrew the closure chuck; if the closure chuck is locked, it is sufficient to give a light stroke on the handle of the hexagonal wrench.

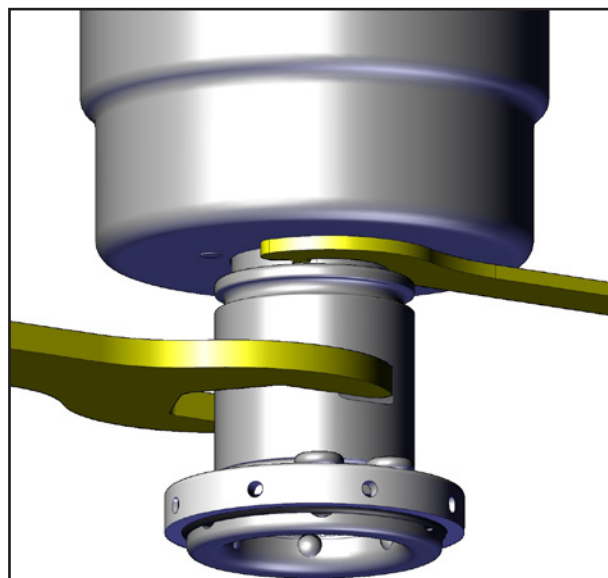


Fig. 61



- Remove the closure chuck unscrewing by hand.

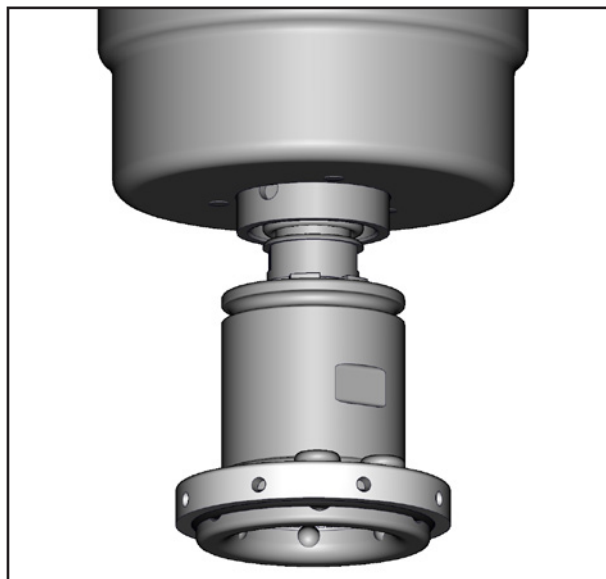


Fig. 62



10.2.4.2 INSTALLATION OF THE CLOSURE CHUCK "PK 314"

To install the closure chuck, proceed as follows:

- Screw by hand the chuck on ejector guiding bush.

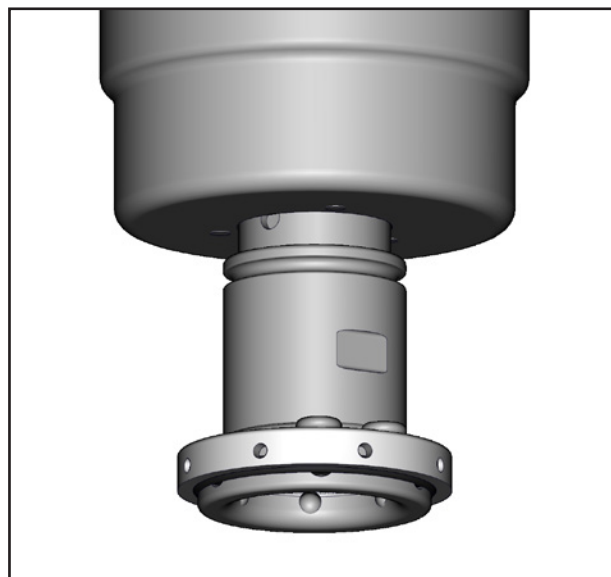


Fig. 63

- fit the slim wrench for ring nut in the hole of the ejector guide bush.

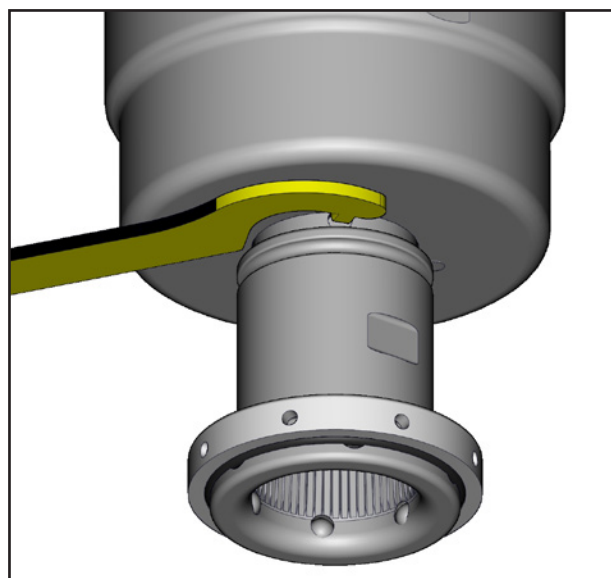


Fig. 64

- Fit the 34 mm hexagonal wrench in the seats of the closure chuck.
- Using the two wrenches, lock the closure chuck.

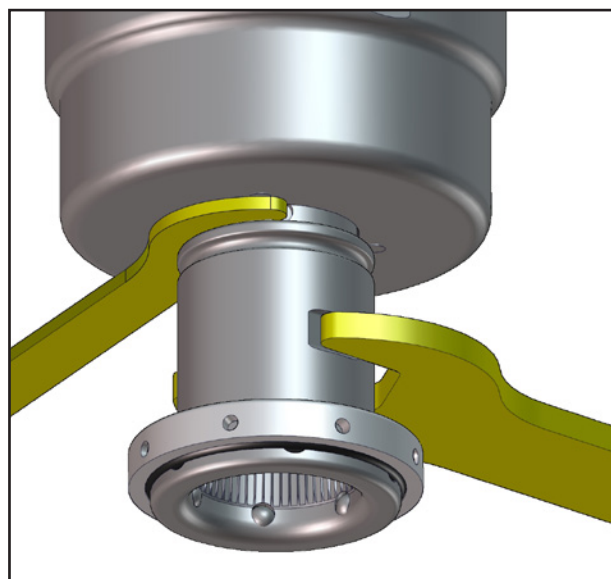


Fig. 65



10.2.5 REMOVAL AND INSTALLATION OF THE CLOSURE HEADS



Protect the centering device upper surface with rubber or similar material.

10.2.5.3 CLOSURE HEADS REMOVAL "PK 314"

The table below shows the dedicated tools for disassembly and assembly of the closure head.

TOOLS	
	Hook wrench 58-62
	Hook wrench 45-50

To remove the closure head from the spindle, proceed as follows:

- Insert 45-50 hook wrench in a seat of the head fixing ring-nut.
- Unscrew the head fastening ring nut with the 45-50 hook wrench till releasing the closure head.

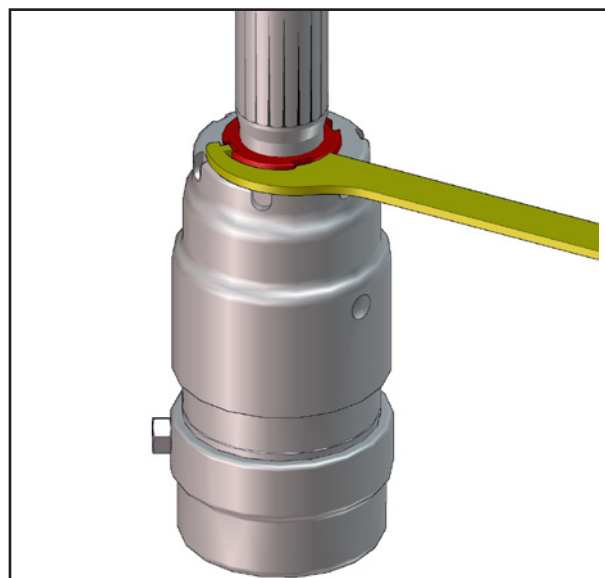


Fig. 66



- Insert the 58-62 hook wrench in the seats of the closure head.
- Unscrew the closure head with the 58-62 hook wrench and remove it.

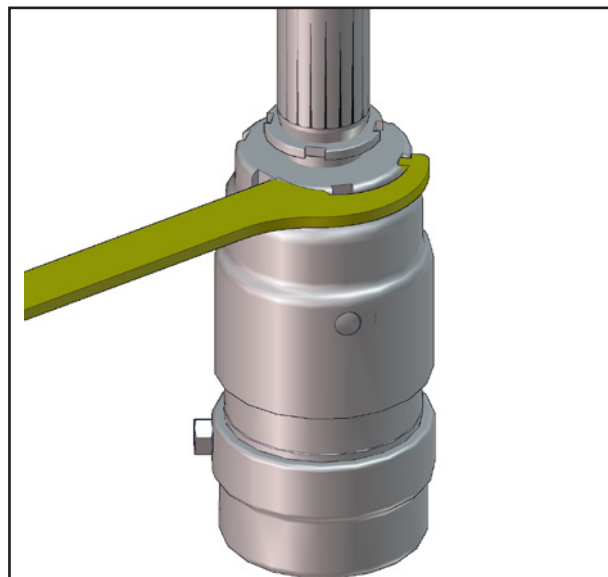


Fig. 67



If the upper ejector unit has not come out from the spindle together with the head, remove it manually.

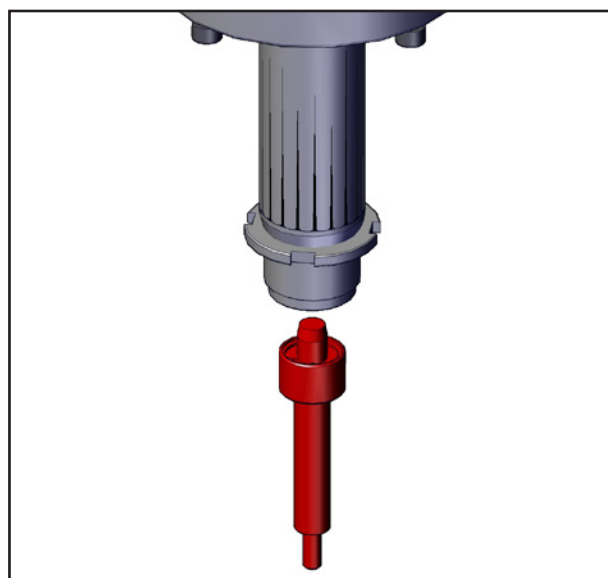


Fig. 68



10.2.5.4 INSTALLATION OF THE CLOSURE HEADS "PK 314"

To install the closure head on the spindle, proceed as follows:

- Fit the closure upper together with the ejector unit inside the spindle.

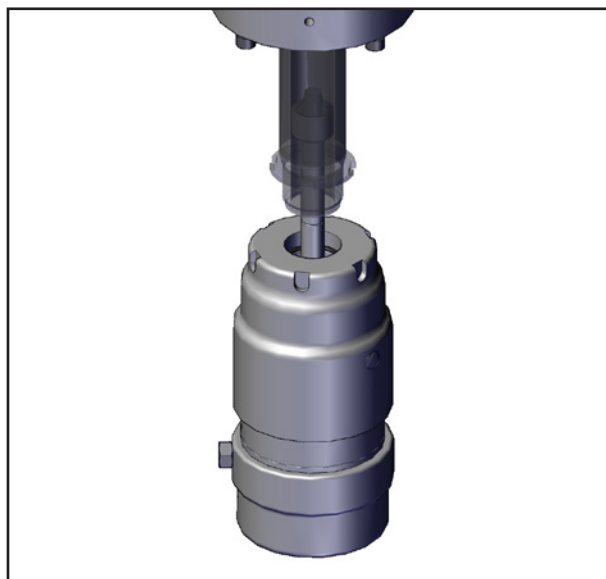


Fig. 69

- Screw the head on the spindle, manually as far as possible.

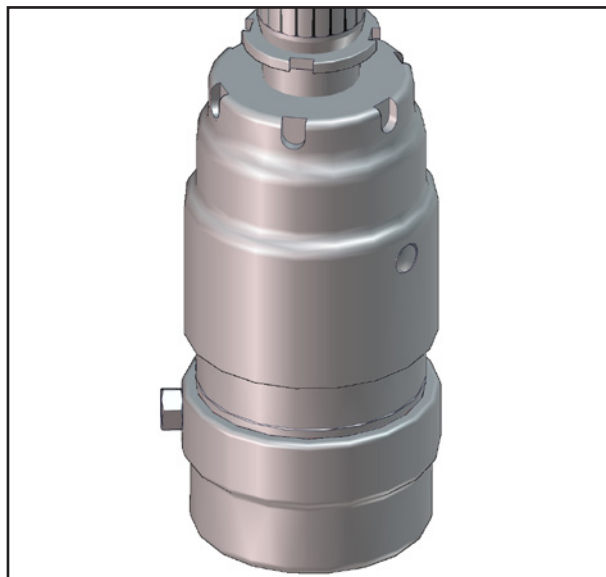


Fig. 70

- Insert the 58-62 hook wrench in the seats of the closure head.
- Complete the screw of the closure head to the bottom of the thread with the 58-62 hook wrench.

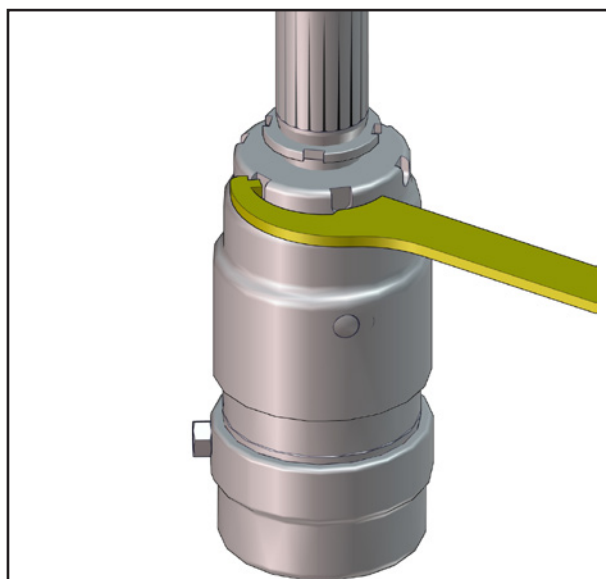


Fig. 71



- Tighten the head fixing ring-nut on the spindle by hand until possible to.

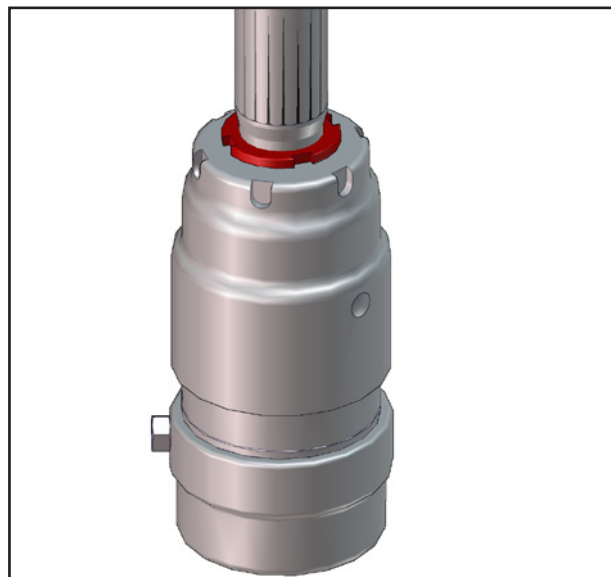


Fig. 72

- Insert 45-50 hook wrench in a seat of the head fixing ring-nut.
- Complete the screw of the head fixing ring-nut with the 45-50 hook wrench.

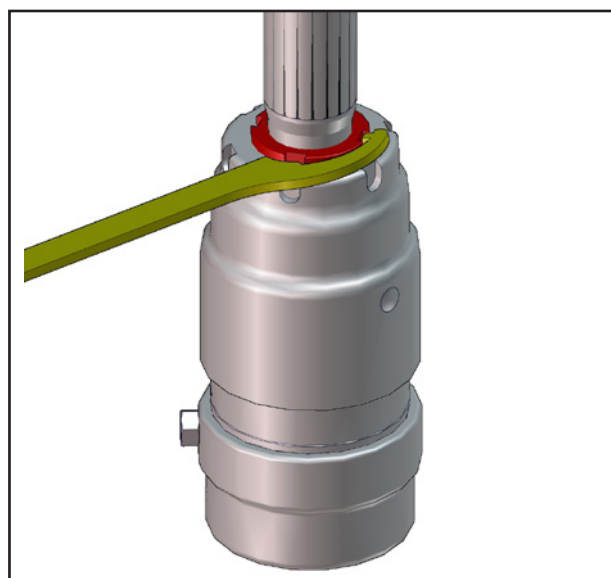


Fig. 73



10.2.6 PICK AND PLACE CAPS CHUTE



10.2.6.1 CAPS CHUTE REMOVAL

Remove the caps chute in the following way:

- Disconnect the compressed air fitting.
- Disconnect the electrical connection from the caps presence photocell (one or more).



Fig. 74

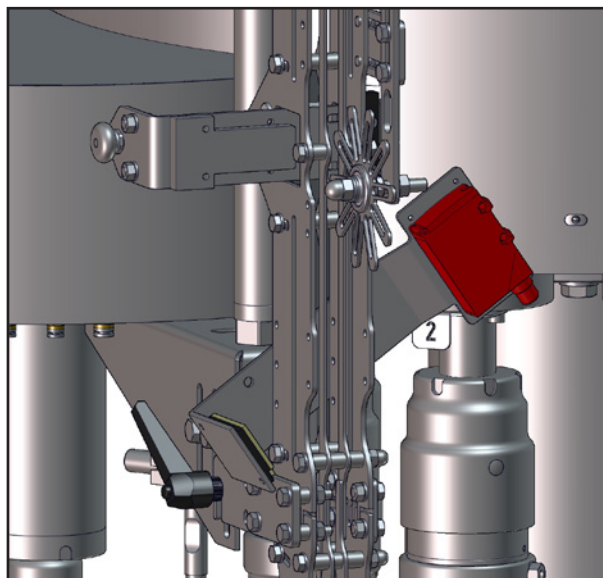


Fig. 75



- Unscrew the bolts that fasten the delivery chute to the base disk.

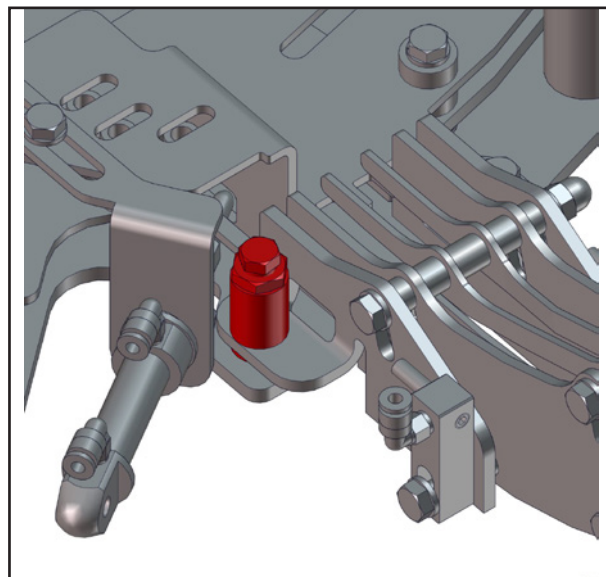


Fig. 76

- Loosen the threaded fasteners located on the support chute.

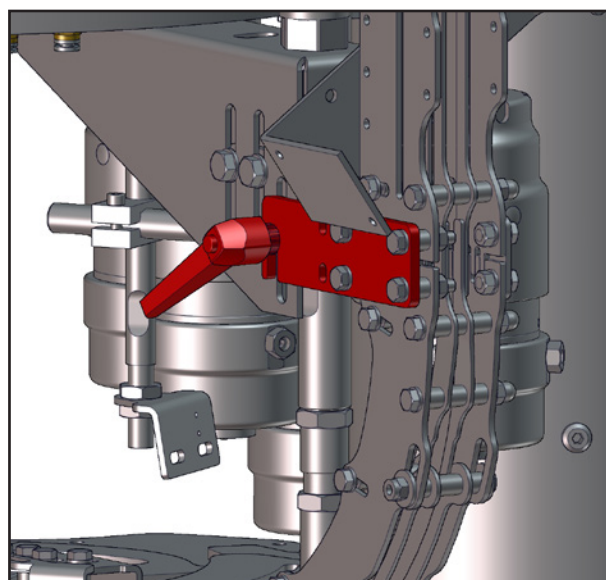


Fig. 77

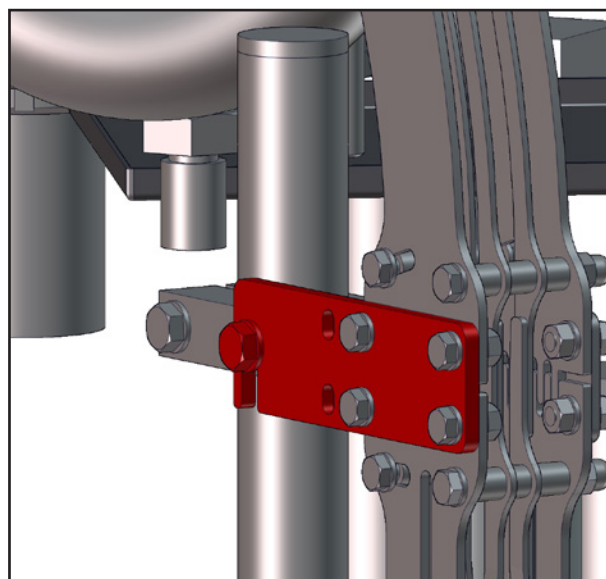


Fig. 78



- Loosen the threaded fastening placed on the caps sorter

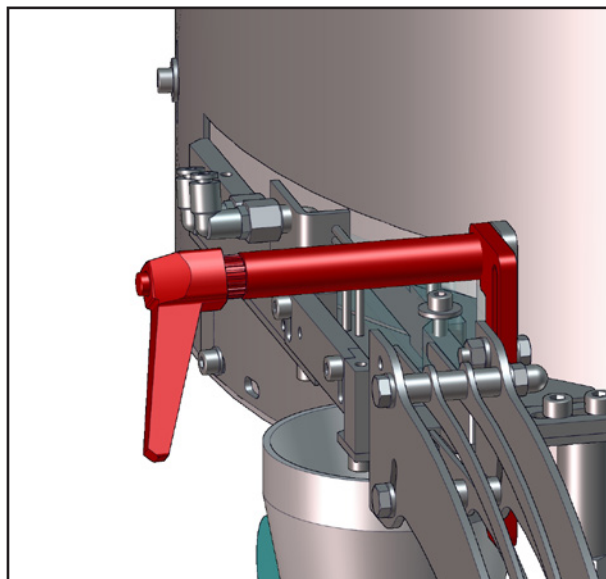


Fig. 79

- Lift the caps chute complete, so as to extract it from fasteners and remove it.

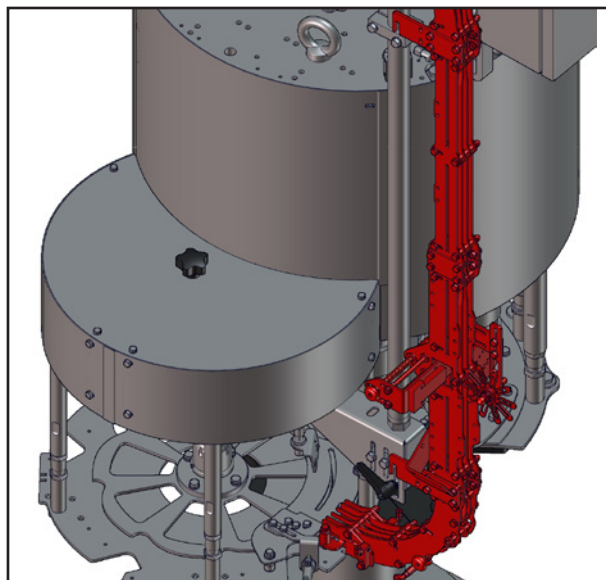


Fig. 80



10.2.6.2 CAPS CHUTE INSTALLATION

To install the caps chute on the machine it is necessary:

- Place the caps chute so that the fixing brackets are positioned on threaded fasteners.

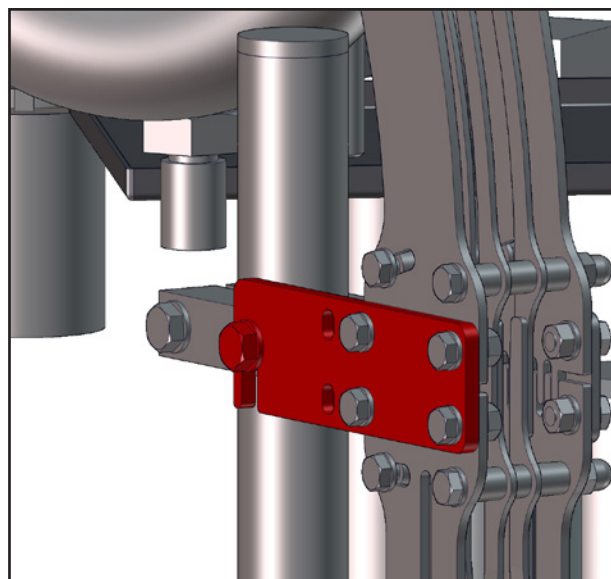


Fig. 81

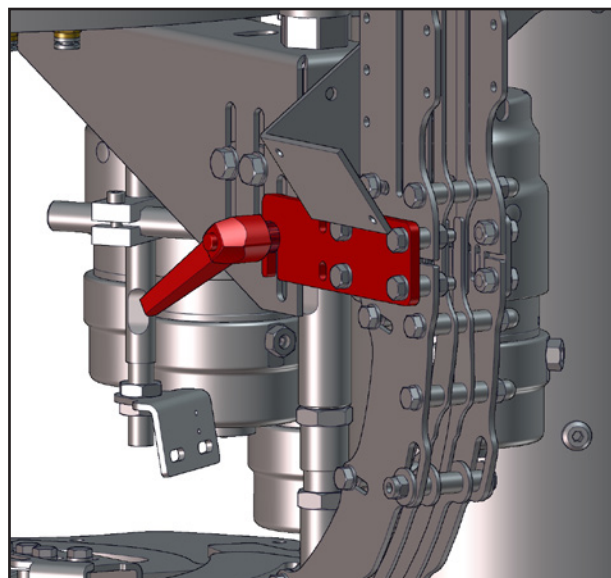


Fig. 82

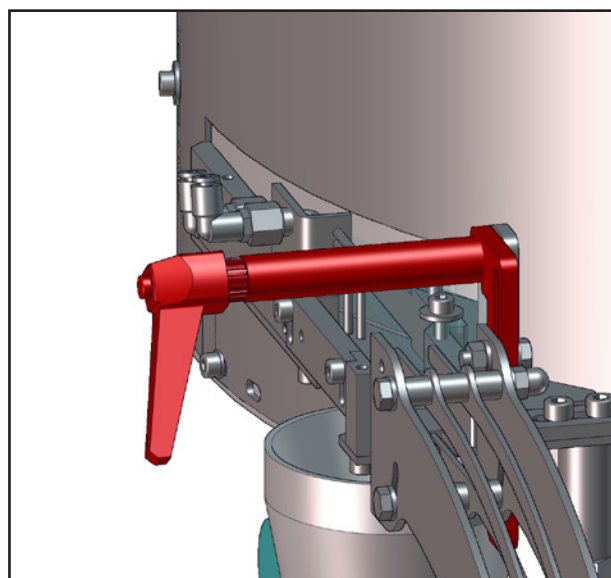


Fig. 83



- Check that the centering pin of the caps chute fits into the seat of a basic disk.

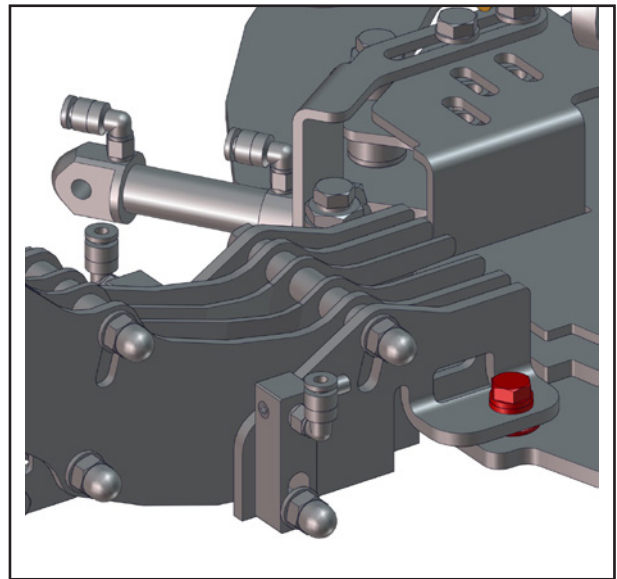


Fig. 84

- Fasten the caps chute to the base disk with the fastening bolt.

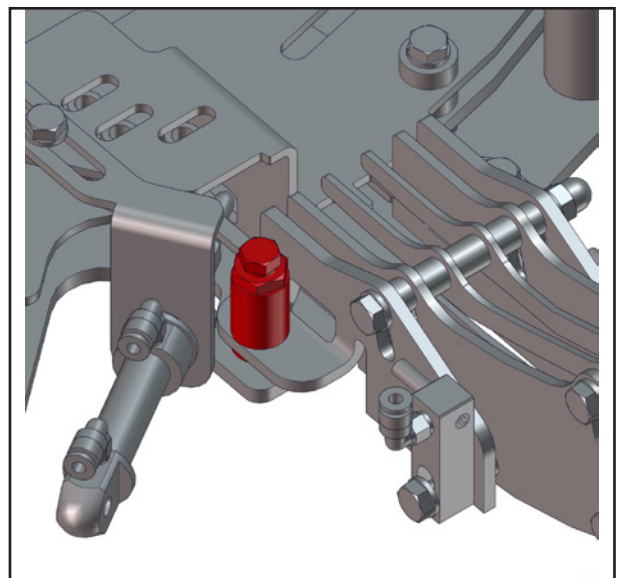


Fig. 85

- Lock threaded fasteners located on the support chute.
- Lock the threaded fastener located on the caps sorter
- Connect the electrical connection of the caps presence photocell (one or more).

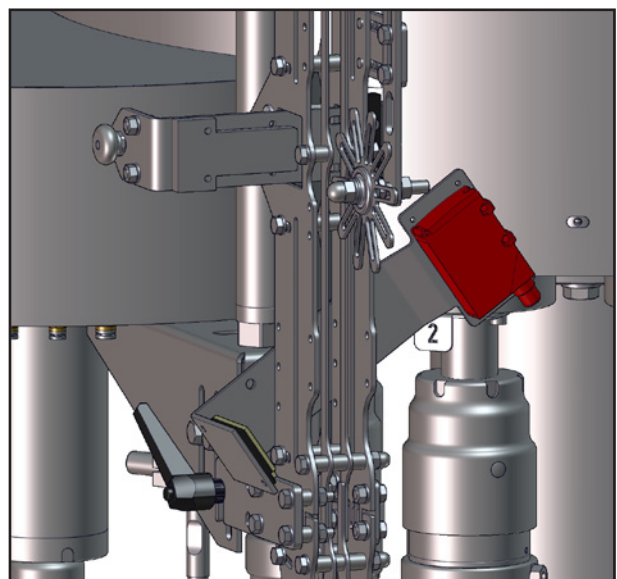


Fig. 86



- Connect the air compressing fitting to them connections.



Fig. 87



10.2.7 CAPS TRANSFER GROUP



10.2.7.1 CAPS GUIDE SECTORS

10.2.7.1.1 REMOVAL

Proceed as follows:

- Disconnect the caps stopping cylinder actioning air fittings.
- Unscrew bolts that fastening the caps guide sectors of the base disc

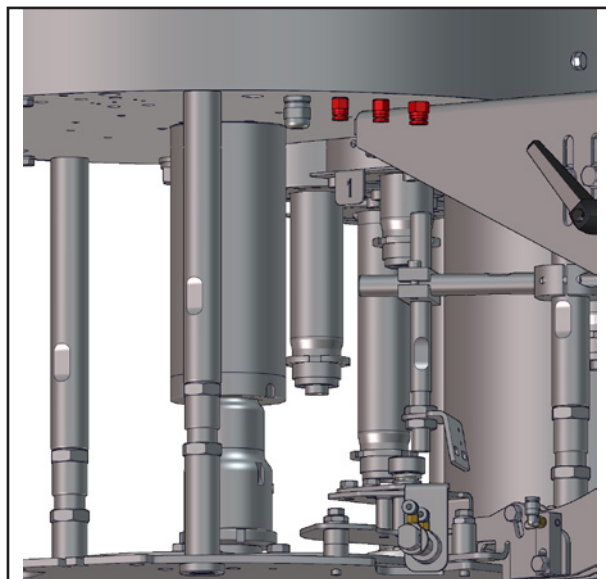


Fig. 88

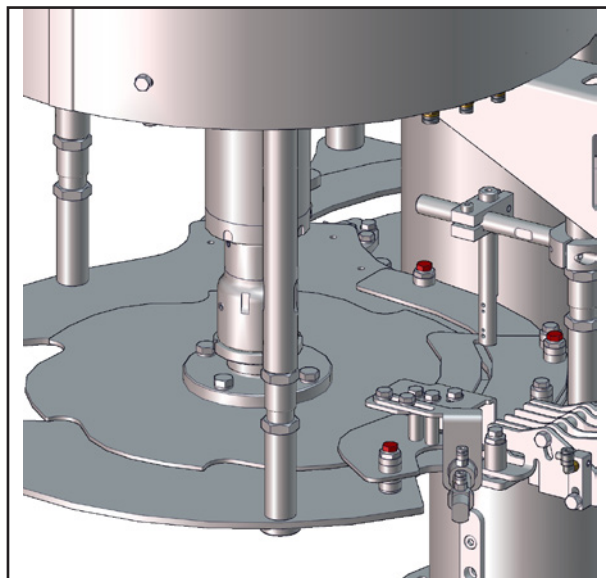


Fig. 89



- Lift and remove the caps guide sectors.

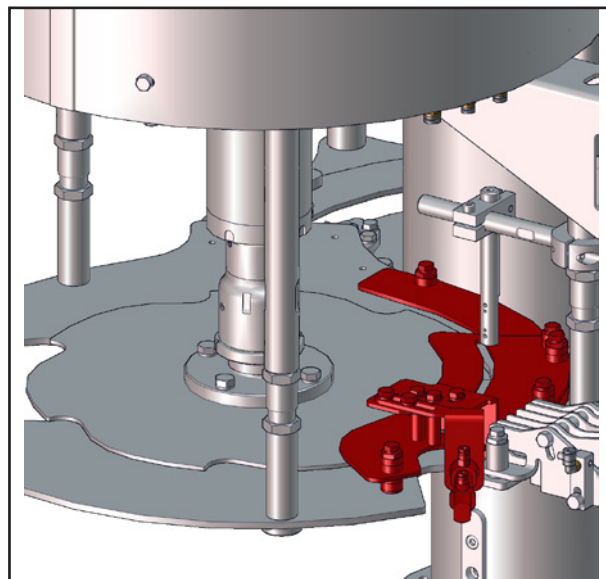


Fig. 90

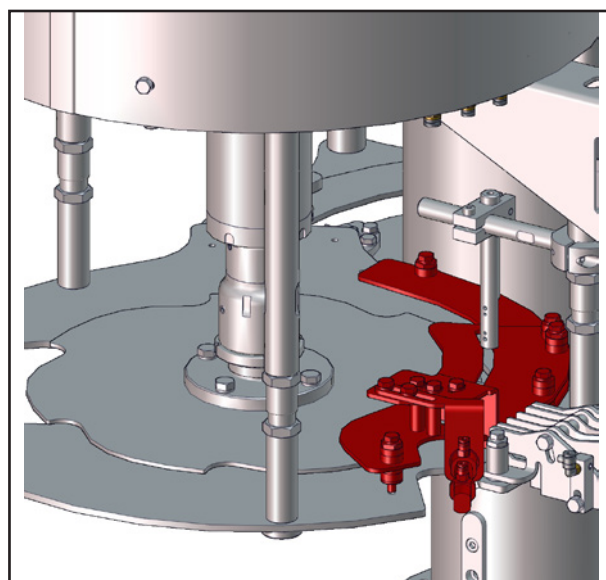


Fig. 91



10.2.7.1.2 INSTALLATION

Proceed as follows:

- Place the caps guide sectors on the base disk.

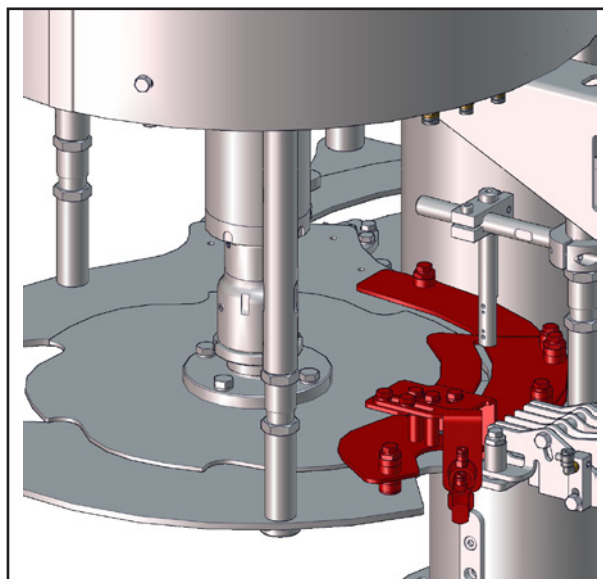


Fig. 92

- Screw the bolts to the base disc.

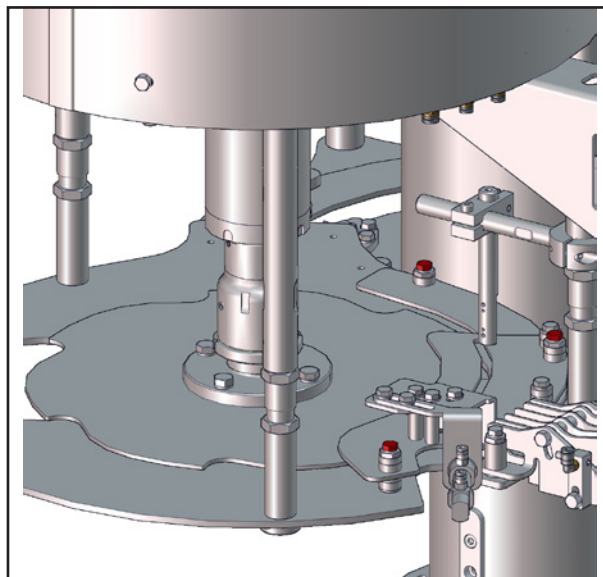


Fig. 93

- Connect the compressed air fittings.



Fig. 94



10.2.7.2 CAPS TRANSFER STAR-WHEEL

10.2.7.2.1 REMOVAL

Proceed as follows:

- Remove the caps guide sectors as described in subsection 10.2.7.1 CAPS GUIDE SECTORS page 70.

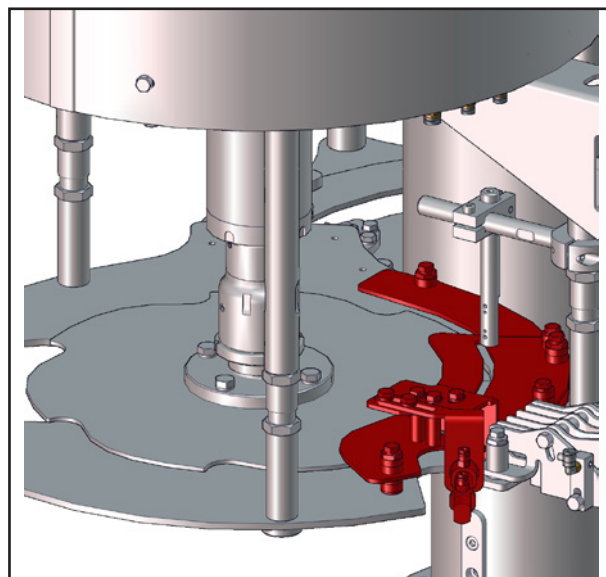


Fig. 95

- Remove the caps delivery chute (See sub-chapter 10.2.6 PICK AND PLACE CAPS CHUTE page 64).

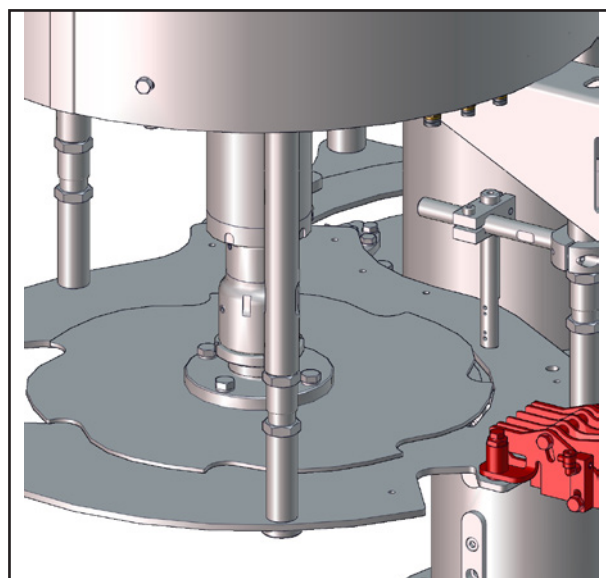


Fig. 96



- Unscrew and remove the screws that fastening the base disk to the supporting columns.

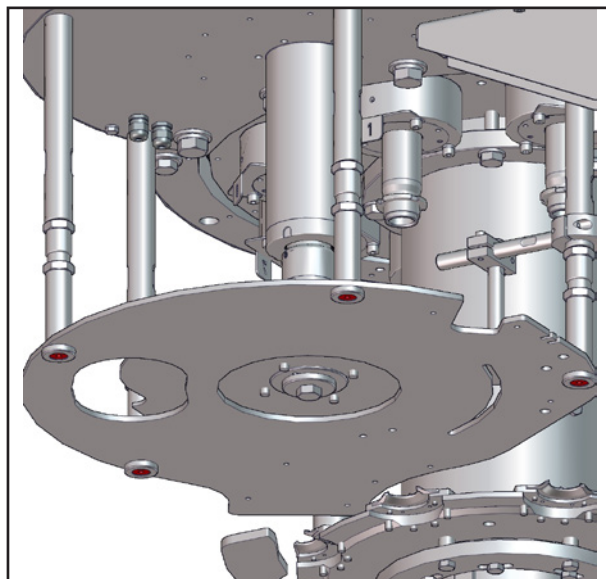


Fig. 97



The fixed base disk is completely supported by these screws; loosening them support the fixed base disk.

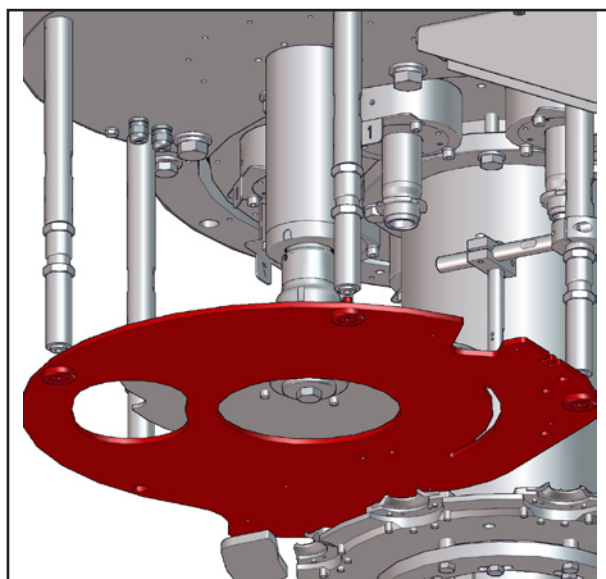


Fig. 98



- Unscrew and remove the fixing bolt.

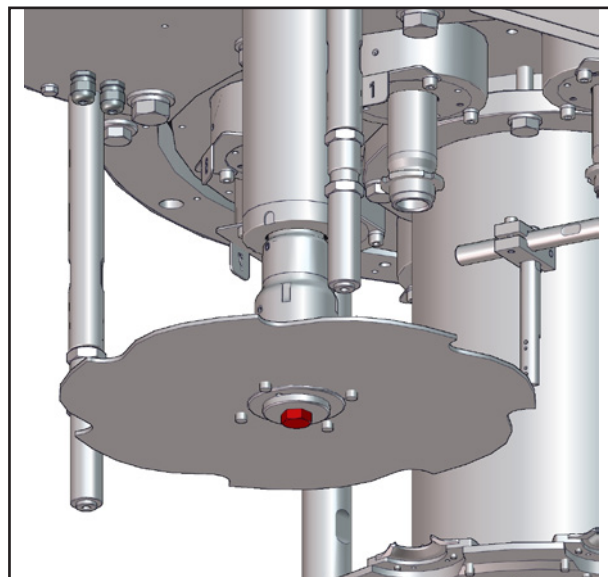


Fig. 99



The transfer star-wheel unit is completely supported by this bolt; loosening it support the star-wheel unit.

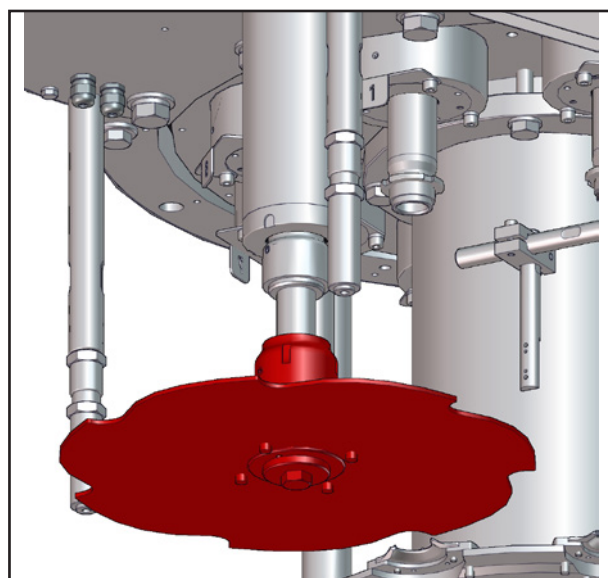


Fig. 100



10.2.7.2.2 INSTALLATION

Proceed as follows:

- Position the transfer star-wheel unit on the hub of the transfer disk and tighten it strongly by the bolt.
- Place the base disk under the supporting columns and fasten it by the suitable washers and screws.

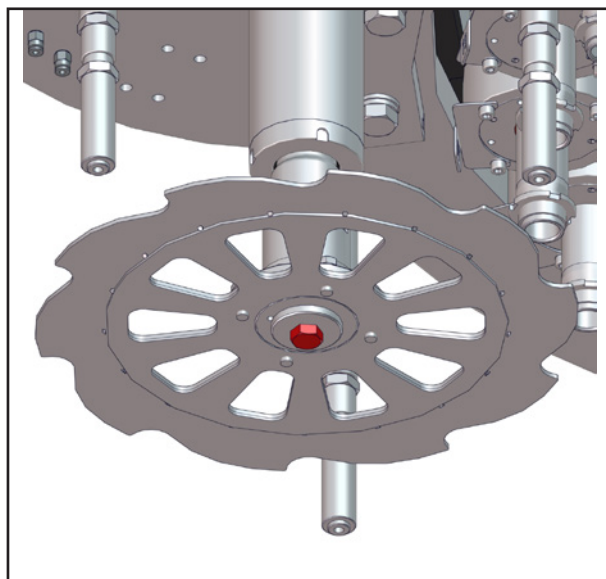


Fig. 101

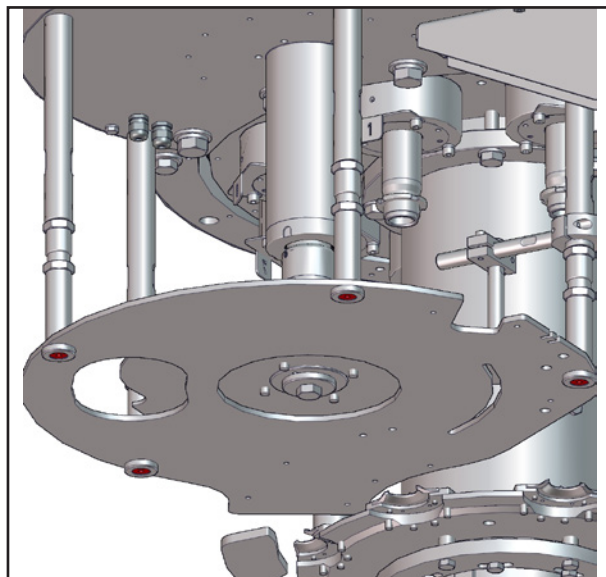


Fig. 102



10.2.8 CAPS SELECTION EQUIPMENT FOR CENTRIFUGAL CAPS SORTER



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

10.2.8.1 CAPS EXIT MOUTH



10.2.8.1.1 REMOVAL

To remove proceed as follows:

- disconnect the pneumatic connections from the caps exit-mouth.

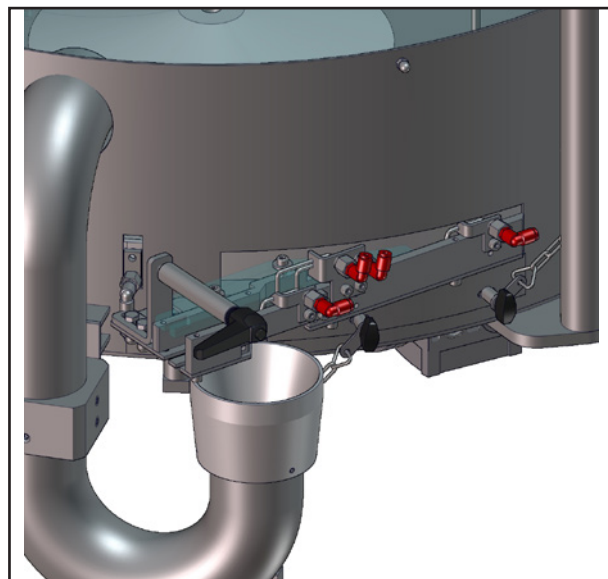


Fig. 103

- Unscrew and remove the fasten knobs of the caps exit-mouth.

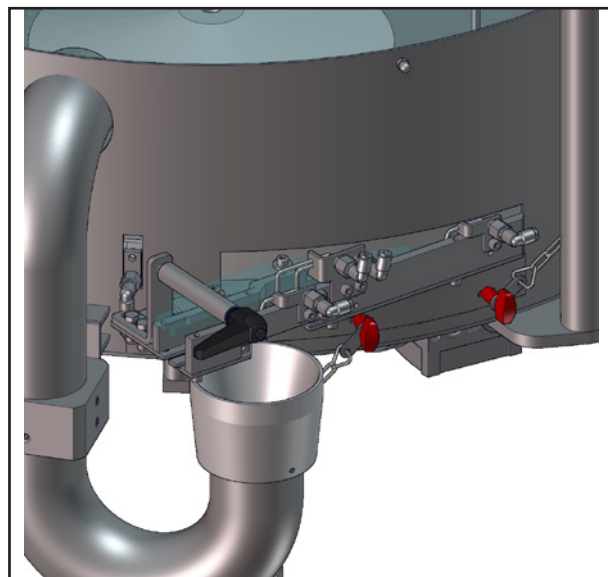


Fig. 104



- Unscrew and remove the return handle that fixes the caps exit-mouth to the caps sorter.

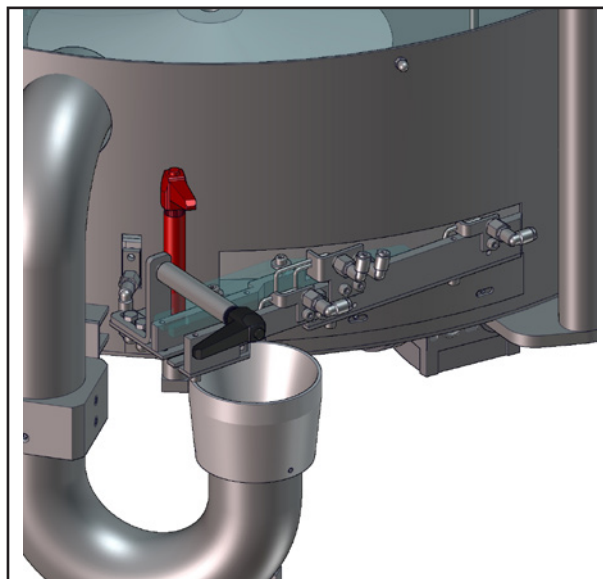


Fig. 105

- Remove the whole caps exit-mouth.

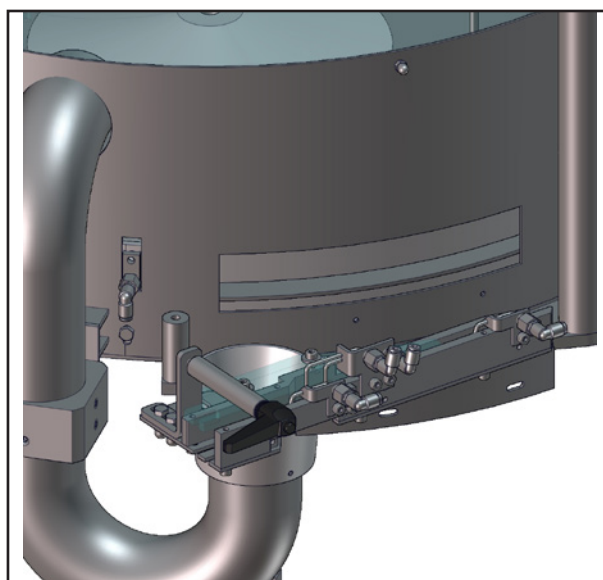


Fig. 106



10.2.8.1.2 INSTALLATION

To install proceed as follows:

- Position the exit mouth in its seat and fix with the fasten knobs.

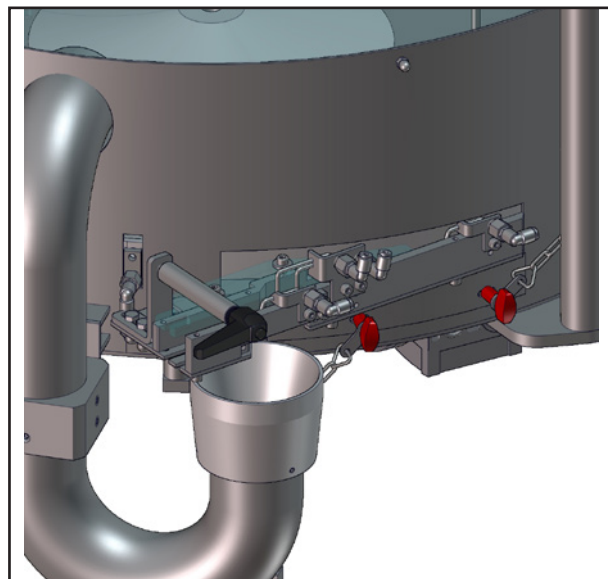


Fig. 107

- Tighten the return handle that fixes the caps exit-mouth to the caps sorter.

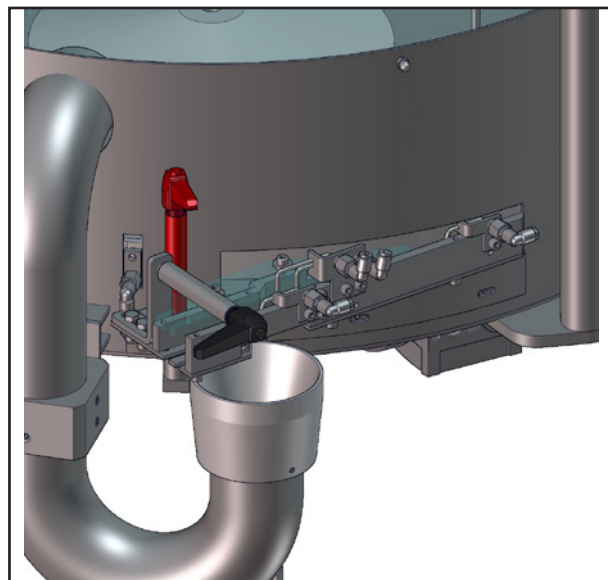


Fig. 108

- Connect the pneumatic connections to the caps exit-mouth.

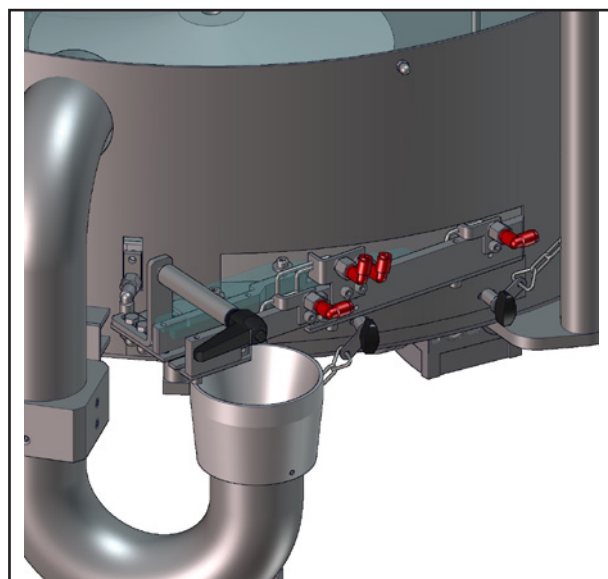


Fig. 109



10.2.8.2 ADJUSTMENT OF THE CAPS SORTER FOR SIZE CHANGE



- If varies the height of the cap to be treated must be adjust the height of the inner hopper ring in the following way:
 - loosen the screws located on the outer perimeter of the hopper.

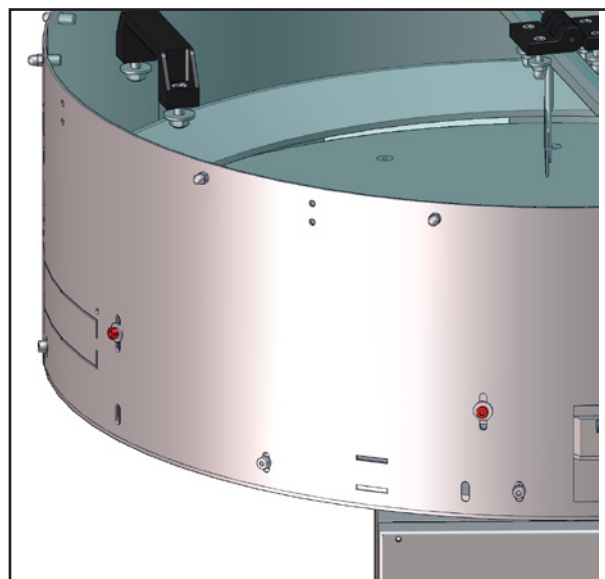


Fig. 110

- Adjust the height of the hopper ring in such a way between the new cap to be treated and the ring there is a clearance of about 3 mm.
- Tighten the screws previously loosened.

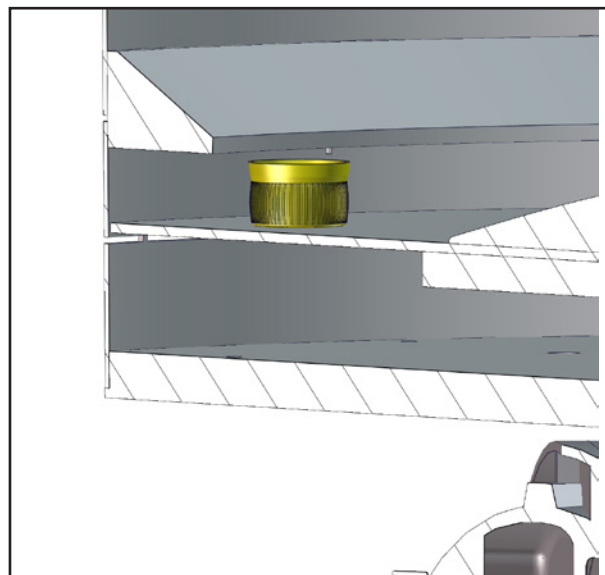


Fig. 111

- On the caps sorter there is a door that allows, thanks to the centrifugal force, the ejection of caps without seaming and seaming on them.
- If varies the height of the cap to be treated must adjust the height of the adjustable door in the following way:
 - loosen the fixing knobs.

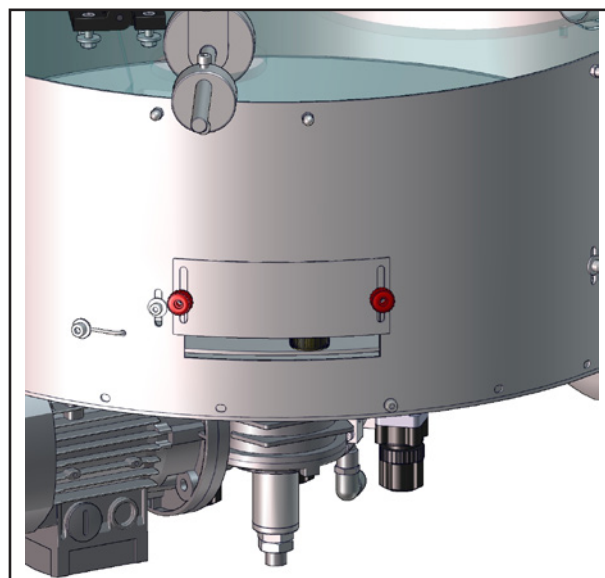


Fig. 112



- Adjust the height of the adjustable door so that they can be expelled only the caps without seal.
- After adjustment, tighten the fixing knobs.

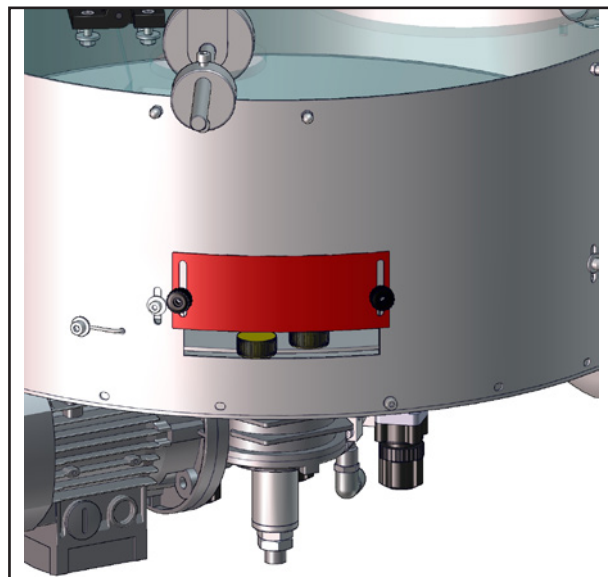


Fig. 113

If the cap is not correctly shifted by the nozzles it is sufficient to increase or decrease the air pressure by the fitting pressure reducers.



10.3 ADJUSTMENTS OF THE CLOSURE HEAD

10.3.1 CLOSURE HEAD "PK 314"

10.3.1.1 TIGHTENING TORQUE ADJUSTMENT

This step can be carried out with the head installed on the closure machine. The adjustment is made by a graduated scale engraved on the slotted ring-nut (in Pound x Inch) and a notch engraved on the magnet adjustment ring nut:



Fig. 114

- Insert a 4 mm allen screw wrench in the seats of the fastening grub screw.
- Insert a 10 mm hexagonal wrench in the seats of the grub screw protection nut.
- Hold the fastening grub screw with the 4 mm allen screw wrench and unscrew the fastening grub screw with the 10 mm hexagonal wrench.



Fig. 115



- Unscrew the fixing dowel with a 4 mm Allen wrench.



Fig. 116

- Rotate the magnet adjustment ring-nut till the values of the wished tightening torque and the notch are aligned.



Fig. 117



The adjustment must be made on all closure heads with the same tightening torque.



10.3.1.2 VARIATION IN THE VERTICAL LOAD

To change the vertical load, remove the head from the closure machine and replace the compensating spring:

- To remove the closure head, operate as indicated in sub-chapter 10.2.5.3 CLOSURE HEADS REMOVAL "PK 314" page 60.
- Push the head upper bush downwards.



Fig. 118



Fig. 119

- Turn the head upper bush counterclockwise, while maintaining the lower part, and pull upward.



Fig. 120



- Remove the compensating spring.
- Replace the spring with a fit to the vertical load to be achieved.
- Remount proceeding reversely than the removal, taking care to grease the spring.

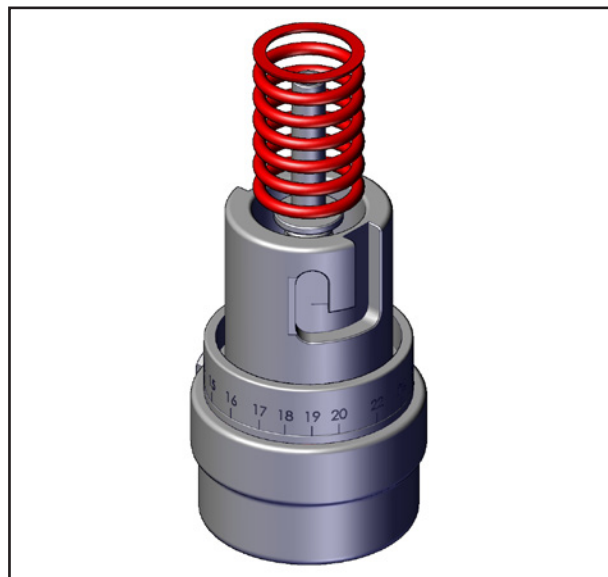


Fig. 121



Are available as standard, the springs shown in the table indicated below ($1 \text{ kg}_f = 9.81 \text{ N}$).

Load Kg_f	Colour	Code
4	None	P07D00621300
8	YELLOW	P07D00621700
12	BROWN	P56D02555900
16	WHITE	P07D00621900
20	RED	P03D00363500
25	BLUE	P52D02240100
32	GREEN	P52D02240300
61	VIOLET	P52D02240400
100	BLACK	P52D02240700



10.3.2 CLOSURE APPLICATION DATA TABLE

To facilitate the correct collection of the information on the closure, at the end of this manual and for example purposes only, is attached a table useful for the purpose.



The user shall anyway refer to his own quality department and to the specifications given by the cap manufacturers, following them.

For the compilation of attached table, proceed as follows:

- indicate in the table heading:
 - the model of the closure machine
 - the serial number
 - the used container type
 - the used cap type
 - the set static torque
 - the number of bottles per minute
 - the type of compensating spring
 - the ejector spring type
 - the type of closure terminal
- indicate in the “data collection” table:
 - the number of bottles relevant to the closure head taken at the exit of the closure machine
 - the value indicated by the torquemeter checking each single bottle and, in the last column, the average value.



Photocopy the enclosure table to get the number of modules required for collection of data over time.



11 MAINTENANCE OF THE CLOSING MACHINE

11.1 MAINTENANCE GENERAL INFORMATION



11.1.1 MAINTENANCE SCHEDULE

From the constructive point of view, interventions concern mechanical, electrical and pneumatic parts.

The envisaged operations are grouped according to work time intervals. Each intervention or group of interventions can concern about mechanical, electrical and pneumatic aspects.

The ordinary maintenance is divided in two categories:

- scheduled ordinary maintenance (or preventive).
- Maintenance according to the conditions (caused by wear and tear).

The scheduled ordinary maintenance (so called periodical or preventive) includes inspections, checks and interventions that, to prevent stops and faults, keep under control:

- the lubrication state of the closing machine.
- The state of parts that can be affected by wear and tear.

The maintenance according to special conditions recommends a regular inspection of components, to check their conditions and perform the maintenance only if it is required.

11.1.2 MAINTENANCE INTERVALS



The following tables show the maintenance deadlines and the operations needed for a correct operation and long life of the closing machine.

The interventions described below have been arranged at intervals of time expressed in hours.

Interval (hours)	Working in 1 shift	Working in 2 shifts	Working in 3 shifts
8	-	-	-
40	1 week	-	-
120	3 weeks	-	1 week
250	6 weeks	3 weeks	2 weeks
500	3 months	6 weeks	1 month
1000	6 months	3 months	2 months
3000	12 months	9 months	6 months
6000*	*	*	12 months

(*) every 6000 hours or anyway once a year.



11.1.3 GENERAL INDICATIONS ON THE SCHEDULED MAINTENANCE PLAN

A logbook of the scheduled maintenance and lubrication interventions must be arranged according to the maintenance indications in the sub-chapters below and to the closing machine working conditions.

Example:

MAINTENANCE SCHEDULED AT 1000 HOURS					
INTERVENTION	DATE	MAINTENANCE MAN	SIGNATURE	INTERVENTION RESULT	NOTES
Check of the height sliding of the closure carousel.	XX/XX/XX	Mario Rossi		positive	
Control gear lubrication	XX/XX/XX	Mario Rossi		lubricated	

11.2 MAINTENANCE AND LUBRICATION PLANNING



Interval	Intervention type
40 hours	Ordinary maintenance
120 hours	lubrification
	Ordinary maintenance
250 hours	lubrification
500 hours	lubrification
	Ordinary maintenance
1000 hours	lubrification
	Ordinary maintenance
3000 hours	Ordinary maintenance
5000 hours	Ordinary maintenance
6000 hours	lubrification
	Ordinary maintenance
12000 hours	Ordinary maintenance
18000 hours	Ordinary maintenance
20000 hours	Ordinary maintenance



11.3 CLOSURE HEAD WEARING PARTS PK 314



It is recommended to stock a spare part kit.

SPARE PART KIT	Working hours	
	3000	6000
Compensating spring	x	
Ejector rod supporting spring	x	
Central oblique bearing	x	
Thrust bearing on the ejector rod	x	
Bush for dowel		x
Magnetic rings		x
O-ring	x	



Carry out the head overhaul, following the instructions in the chapter referring to the capping head.

11.4 PK 314 CLOSURE CHUCK WEARING PARTS



It is recommended to stock a spare part kit.

SPARE PART KIT	Working hours	
	1000	
Spheres	x	
O-ring	x	



Carry out the head overhaul, following the instructions in the chapter referring to the capping head.



11.5 SCHEDULED MAINTENANCE



11.5.1 ROUTINE MAINTENANCE

11.5.1.1 EVERY 40 WORKING HOURS

OPERATION	AIM	NOTES
Pneumatic system check.  	Check the air supply pressure to the various utilization points. Check by the gauges, that the pressure reducers are adjusted at the work pressure (the pressure is indicated on the pressure gauges) Drain the possible condensate collected under the discharge filter. Checking the filter clogging. If necessary, replace it.	



11.5.1.2 EVERY 500 WORKING HOURS

OPERATION	AIM	NOTES
Lubrication of the closure head movable parts. 	Lubricate the springs. Lubricate the upper bush and the seat of the head central body. Lubricate the plug for bush.	See sub-chapter 11.6.7 CLOSURE HEAD LUBRICATION page 131.
Lubrication of the closure chuck. 	Lubricate the chuck.	See sub-chapter 11.6.8 CLOSURE CHUCKS LUBRICATION page 135.
Verify the conditions of the bottle centering devices. 	Check if the teeth of the centering devices are worn and, if necessary, replace them.	See sub-chapter 11.5.5 REPLACEMENT OF TOOTHED CENTERING DEVICE page 119.
Pneumatic system check. 	Check the correct operation and possible leaks (from fittings, lines, etc.): ▪ verify the air pressure at the closing machine entry. ▪ Supplying the pneumatic system with compressed air, check there is no leak, broken or disconnected pipe. ▪ Check that the closing machine pressure gauges indicate the applied pressure on the suitable data plate.	



11.5.1.3 EVERY 1000 WORKING HOURS

OPERATION	AIM	NOTES
Check the conditions of the closure chuck. 	Verify the state of wear of the chuck.	See sub-chapter 11.5.4 MAINTENANCE OF THE CLOSURE CHUCK page 116.
Check of the turret vertical sliding.  	Check the correct vertical sliding, making the turret slide from the min. to the max. value, so as to ensure a correct component sliding in time.	
Electric connection check.   	Carefully check all the connectors and connection cables and if necessary replace all the damaged parts.	
Check of the head control spindles.  	Check the clearance of the spindle pads.	See sub-chapter 11.5.2.1 SPINDLE PADS CLEARANCE ADJUSTMENT page 96.
Sight and sound check of possible noises, clearances and unbalancing.    	Check, starting and stopping the closing machine several times without bottles, for possible noises, faults, clearances and vibrations in the motor-drive components (motors, reducers, gears, bearings). Replace possible worn components. Lubricate the new installed components. Clean and lubricate the disassembled components. Report any anomalies you need to contact <i>Technical Support</i> .	



11.5.1.4 EVERY 3000 WORKING HOURS

OPERATION	AIM	NOTES
Overhaul of the closure heads. 	Overhaul the closure head, replacing the wear components.  The component being replaced are indicated in sub-chapter 11.3 CLOSURE HEAD WEARING PARTS PK 314 page 89. Check the head correct tightening torque.	See sub-chapter 11.5.3 MAINTENANCE OF THE CLOSURE HEAD page 98.
 To avoid production line long stops at the complete head overhaul, AROL S.p.A. offers its customers the “HEADS EXCHANGE PROGRAM” service. The service includes: ▪ Temporary supply of substitute heads for the time strictly necessary for the overhaul. ▪ Complete head overhaul carried out at AROL S.p.A. by its technicians. ▪ Additional 1-year guarantee on the heads. Contact the <i>TECHNICAL SERVICE</i> for info.		
Lubrication of the closure head movable parts. 	Lubricate the springs. Lubricate the upper bush and the seat of the head central body. Lubricate the bush for dowel.	See sub-chapter 11.6.7 CLOSURE HEAD LUBRICATION page 131.



11.5.1.5 EVERY 6000 WORKING HOURS

OPERATION	AIM	NOTES
Check of the head control spindles. 	Check the clearance of the spindle pads.	See sub-chapter 11.5.2.1 SPINDLE PADS CLEARANCE ADJUSTMENT page 96.
Overhaul of the closure heads. 	Overhaul the closure head, replacing the wear components.  The component being replaced are indicated in sub-chapter 11.3 CLOSURE HEAD WEARING PARTS PK 314 page 89. Check the head correct tightening torque.	See sub-chapter 11.5.3 MAINTENANCE OF THE CLOSURE HEAD page 98.
Lubrication of the closure head movable parts. 	Lubricate the springs. Lubricate the upper bush and the seat of the head central body. Lubricate the bush for dowel.	See sub-chapter 11.6.7 CLOSURE HEAD LUBRICATION page 131.
Check the wear of the cap/cork driving punch unit cam race. 	Visually check the wear of the roller sliding race.	See sub-chapter 11.7.1 SPINDLES CONTROL CAM WEAR CHECK page 138.
Closing machine general check. 	Check the correct operation of the machine; if faults are found, contact the <i>Technical Service</i> .	



11.5.1.6 EVERY 18000 WORKING HOURS

OPERATION	AIM	NOTES
Overhaul of the closure heads. 	Overhaul the closure head, replacing the wear components.  The component being replaced are indicated in sub-chapter 11.3 CLOSURE HEAD WEARING PARTS PK 314 page 89. Check the head correct tightening torque.	See sub-chapter 11.5.3 MAINTENANCE OF THE CLOSURE HEAD page 98.
Lubrication of the closure head movable parts. 	Lubricate the springs. Lubricate the upper bush and the seat of the head central body. Lubricate the bush for dowel.	See sub-chapter 11.6.7 CLOSURE HEAD LUBRICATION page 131.
Overhaul of the transfer disk shaft unit. 	Overhaul the transfer disk by replacing the components below: ▪ Bearings. ▪ Snap rings. ▪ Seals. ▪ Cork stopping cylinder.	For the cork transfer disk revision, you need to contact <i>Technical Support</i> .
Overhaul of the heads rotation motor group. 	Overhaul by replacing the following components: ▪ Bearings. ▪ Seals. ▪ Movement drive gears.	Contact the <i>Technical Service</i> .

11.5.1.7 EVERY 24000 WORKING HOURS

OPERATION	AIM	NOTES
Overhaul of the cork inserting pistons. 	Replacement of the inner sleeve group.	For the overhaul of the piston, contact the <i>Technical Service</i>



11.5.2 HEADS CONTROL SPINDLES MAINTENANCE



11.5.2.1 SPINDLE PADS CLEARANCE ADJUSTMENT

To adjust the pad clearance, operate as follows:

- make the closing machine rotate in jog until the spindle to be adjusted is in the flat (rear) section of the cam.
- Lift the ends of the pads fixing safety plates so as to release the bolts.

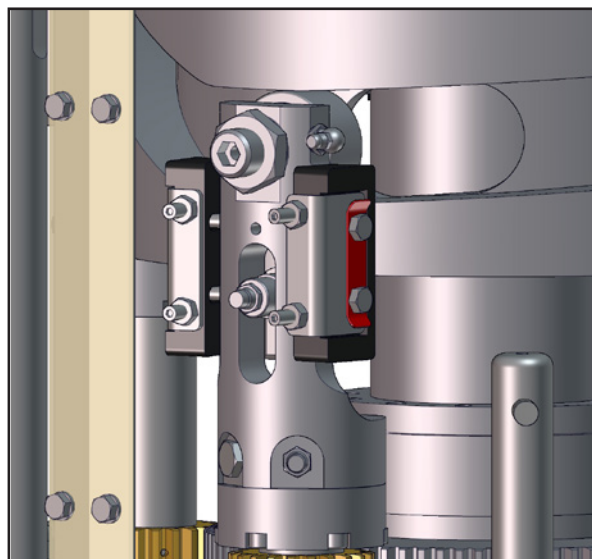


Fig. 122

- Loosen the fixing bolts of both pads.

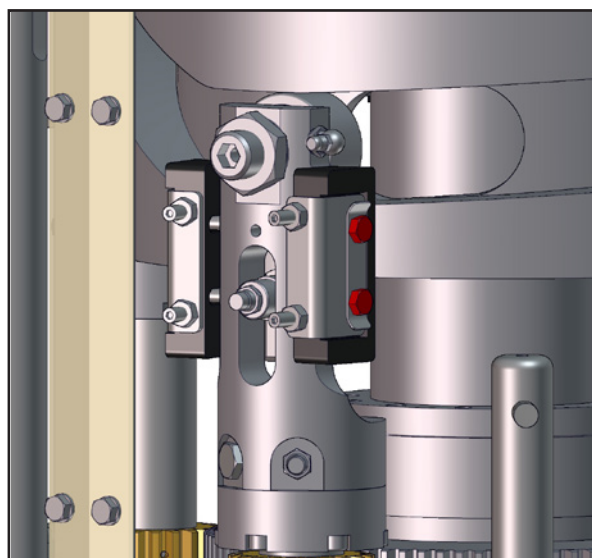


Fig. 123

- Loosen nuts and adjustment grub screws for both pads.

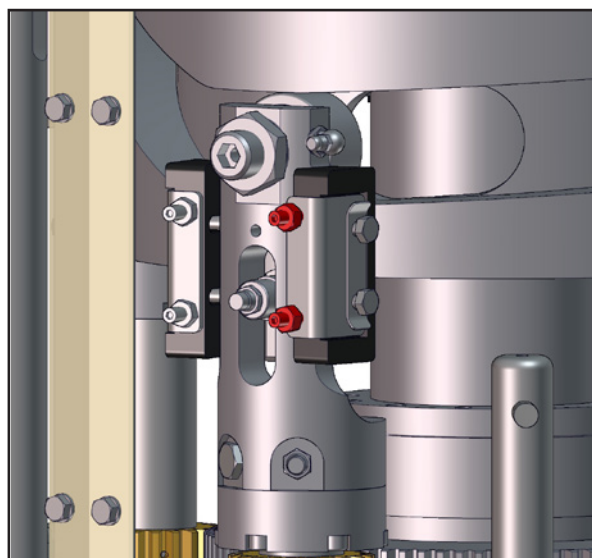


Fig. 124



- Fit a 0.1 mm shim between the two pads and the spindles driving cam.

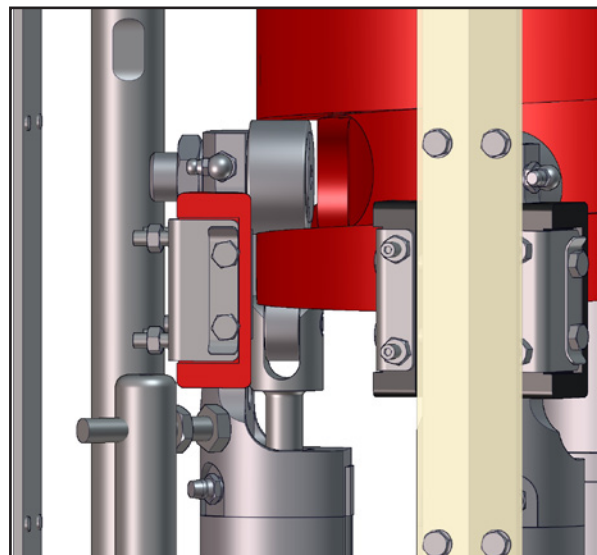


Fig. 125

- Push the pads against the shim.
- Register the position of the head driving piston so that the distance between the pad and the cam is the same for both pads.
- Verify the distances and lock the previously loosened bolts.
- Screw the adjusting screws until they are in contact with the pads.
- Screw all the grub screws locking nuts.

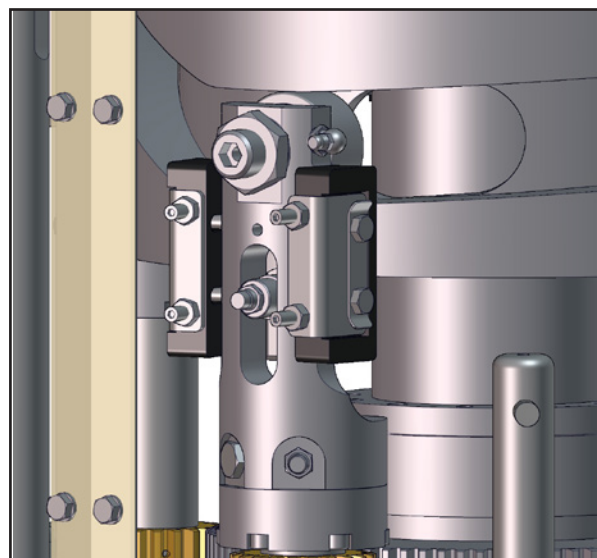


Fig. 126



A clearance between 0.1 and 0.2 mm must be kept all along the race cam.

This condition must be checked on the cam circumference in at least four diametrically opposed points (e.g. rear and front parts, left and right side).

- After adjustment, again bend the ends of the pads fixing safety plates so as to prevent that bolts may loosen.

Repeat the operation for the other possible spindles to be adjusted.



11.5.3 MAINTENANCE OF THE CLOSURE HEAD

11.5.3.1 CLOSURE HEAD "PK 314"

11.5.3.1.1 CLOSURE HEAD OPENING

To open the closure head it is necessary to:

- remove the closure head from the closure machine as described in sub-chapter 10.2.5.3 CLOSURE HEADS REMOVAL "PK 314" page 60.
- Push the head upper bush downwards.
- Turn the head upper bush counterclockwise, while maintaining the lower part, and pull upward.



Fig. 127



Fig. 128



11.5.3.1.2 REPLACING THE SPRINGS

11.5.3.1.2.1 COMPENSATING SPRING REPLACEMENT

When it is necessary to replace the compensating spring, proceed as described below:

- Open the closure head as described in sub-chapter 11.5.3.1.1 CLOSURE HEAD OPENING page 98.



Fig. 129

- Remove the compensating spring.

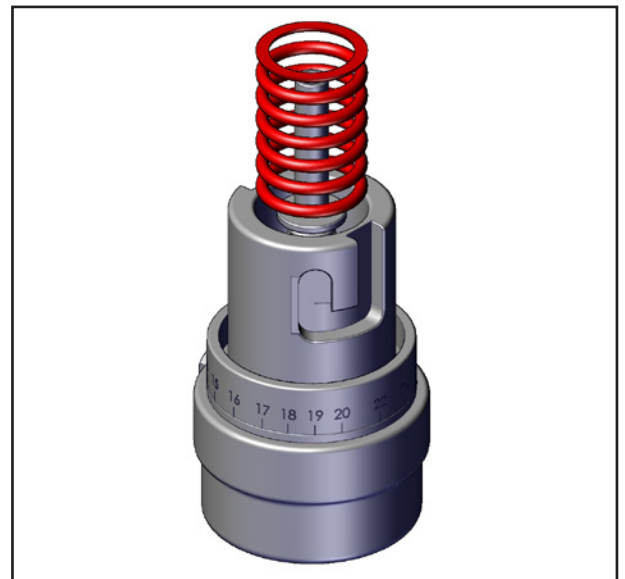


Fig. 130



11.5.3.1.2.2 REPLACEMENT EJECTOR ROD SUPPORTING SPRING

When it is necessary to replace the ejector rod holder spring, proceed as described below:

- Open the closure head as described in sub-chapter 11.5.3.1.1 CLOSURE HEAD OPENING page 98.



Fig. 131

- Remove the compensating spring.

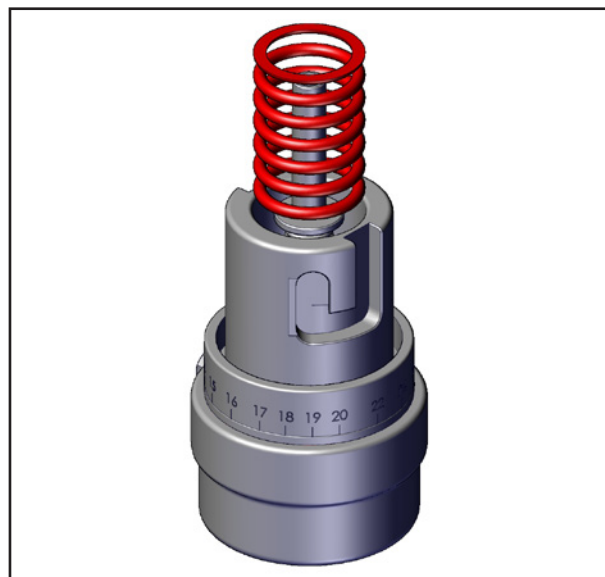


Fig. 132

- Pull out the ejector driving rod.

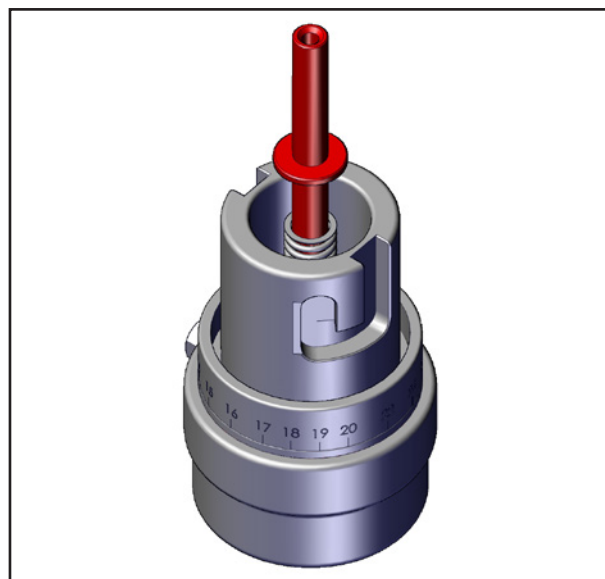


Fig. 133



- Pull out the ejector rod holder spring.

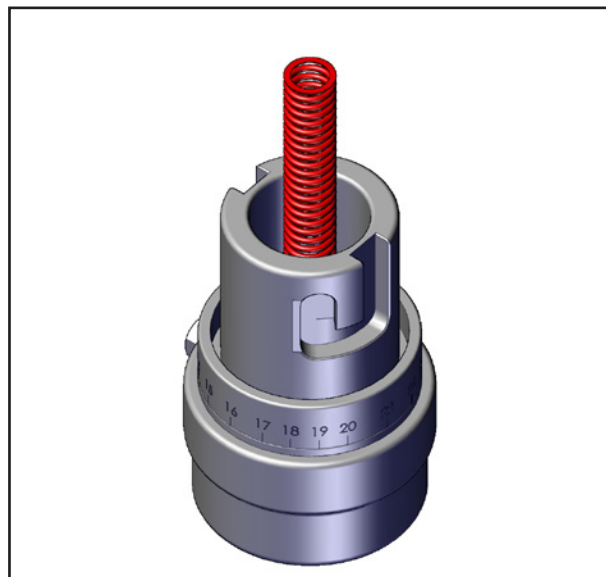


Fig. 134



11.5.3.1.3 MAGNETIC RING OVERHAUL

11.5.3.1.3.1 MAGNETIC RING CHECKING

Magnetic rings are components that tend to lose their original features in time; thus, they must be replaced one or more times during the head life.

Check using a torquemeter that the torque corresponds to the static torque value set on the slotted ring nut.



Tolerance $\pm 5\%$ of the stroke end value engraved on the slotted ring nut. (Example: if the graduated scale has values between 10 and 36 Lbs x inch and want to get a screwing torque equal to 30, the admitted tolerance value is between 28,2 - 31,8).

If the obtained value is higher than the tolerated value, proceed to replace of the magnetic rings.

11.5.3.1.3.2 LOWER MAGNETIC RING OVERHAUL



Note down the value of the screwing torque set on the graduated scale etched on the slotted ring nut.



Fig. 135

When it is necessary to overhaul the lower magnetic ring, proceeding as described below:

- Insert a 4 mm allen screw wrench in the seats of the fastening grub screw.
- Insert a 10 mm hexagonal wrench in the seats of the grub screw protection nut.
- Hold the fastening grub screw with the 4 mm allen screw wrench and unscrew the fastening grub screw with the 10 mm hexagonal wrench.



Fig. 136



- Unscrew the fixing dowel with a 4 mm Allen wrench.



Fig. 137

- Loosen and remove the magnet adjustment ring-nut from the head central body.



Fig. 138

- Loosen the Allen screws fastening the lower magnetic ring on the magnet adjustment ring nut with a 3 mm Allen wrench.

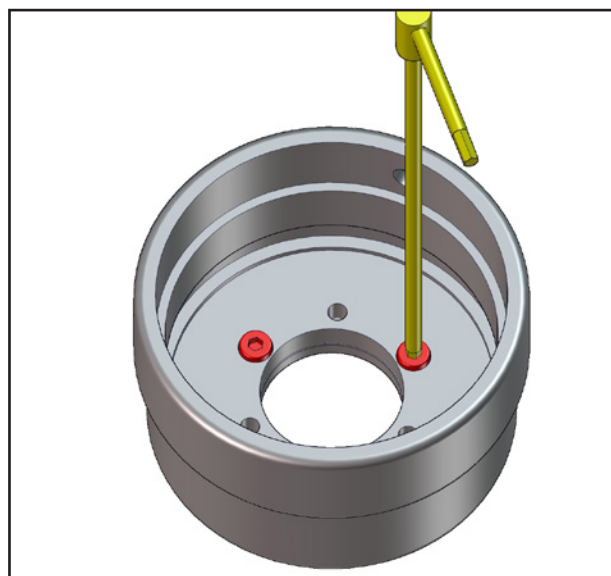


Fig. 139



- Remove the lower magnetic ring.

Remove the shim, if installed.

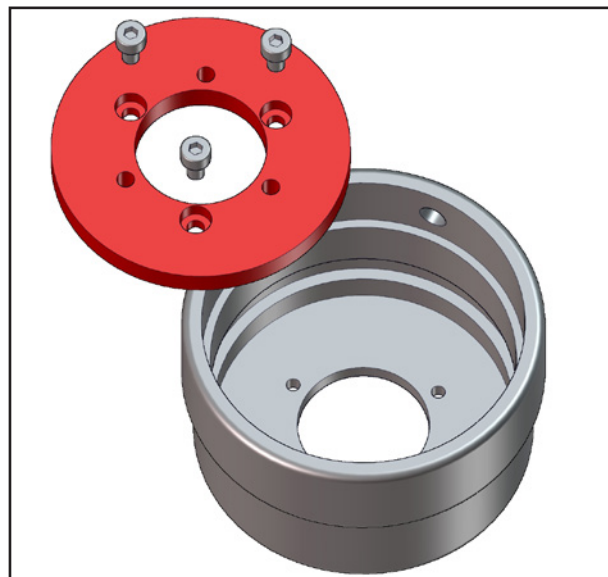


Fig. 140

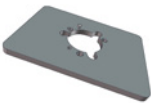


Use a medium resistance thread locking agent when tightening the Allen screws during the installation of the magnetic rings.



11.5.3.1.3.3 UPPER MAGNETIC RING OVERHAUL

The table below shows the necessary tools for adjustment of the closure head.

TOOLS	
	Supporting tool

When it is necessary to remove the upper magnetic ring, proceed as follows:

- Open the closure head as described in sub-chapter 11.5.3.1.1 CLOSURE HEAD OPENING page 98.



Fig. 141

- Remove the compensating spring.

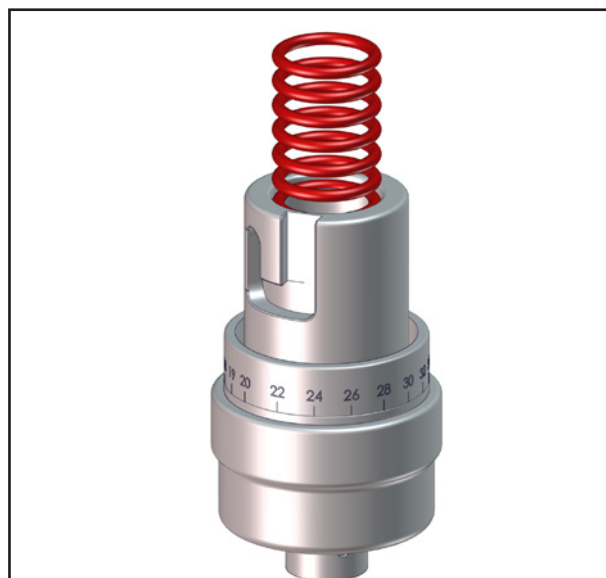


Fig. 142



- Loosen and remove the magnet adjustment ring-nut from the head central body as described in sub-chapter 11.5.3.1.3.2 LOWER MAGNETIC RING OVERHAUL page 102.



Fig. 143

- Fit the supporting tool fitting the pin in one of the holes of the upper magnetic ring.



Fig. 144

- Position the supporting tool so that the Allen screws of the upper magnetic ring can be seen from the suitable openings.

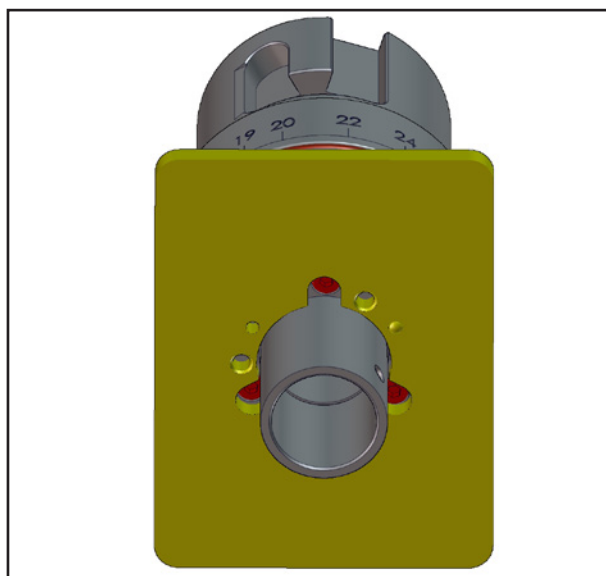


Fig. 145



- Holding in position the supporting tool and loosen the Allen screws that secure the magnetic ring with a 3 mm Allen wrench.

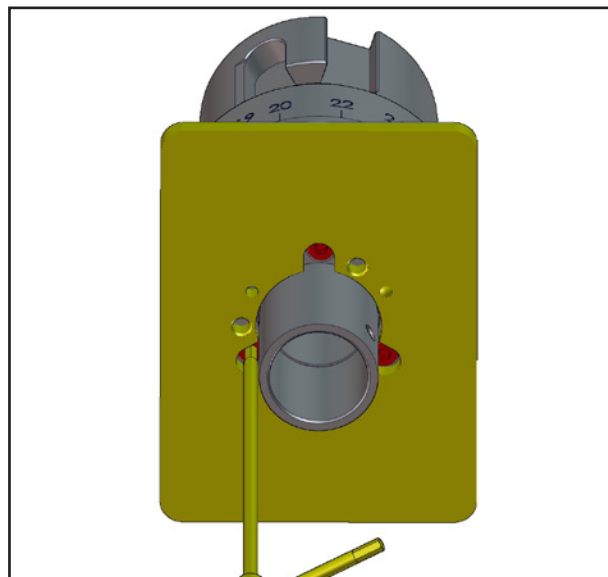


Fig. 146

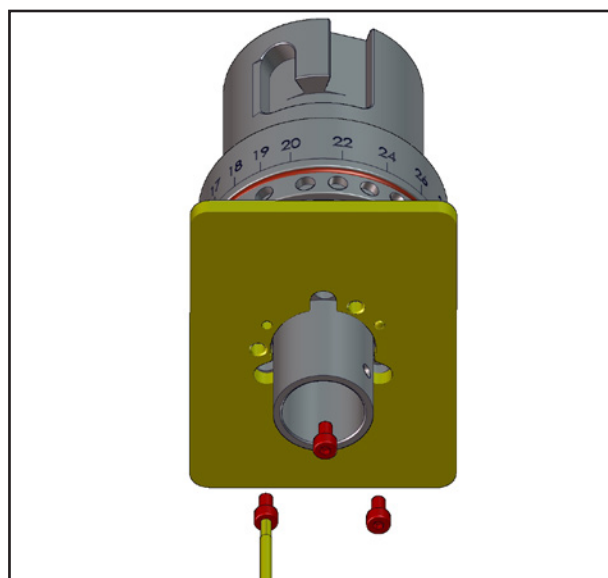


Fig. 147

- Remove the magnetic ring from the head central body.



Use a medium resistance thread locking agent when tightening the Allen screws during the installation of the magnetic rings.



Fig. 148



11.5.3.1.4 BEARINGS OVERHAUL

The table below shows the necessary tools for maintenance of the closure head.

TOOLS	
	Hook wrench 45-50 with ball
	Supporting tool

11.5.3.1.4.1 REMOVING OF CENTRAL BEARING

When it is necessary to remove the central bearing, proceed as follows:

- Open the closure head as described in sub-chapter 11.5.3.1.1 CLOSURE HEAD OPENING page 98.
- Pull out the upper bush from the lower part of the head.



Fig. 149

- Remove the compensating spring.

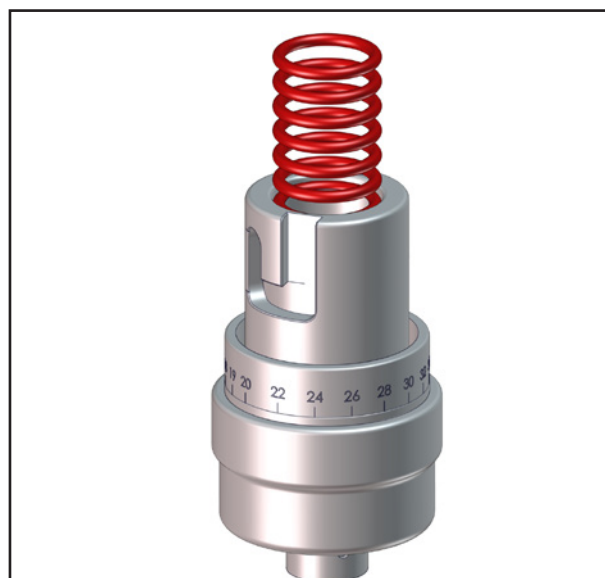


Fig. 150



- Remove the upper magnetic ring as described in sub-chapter 11.5.3.1.3.3 UPPER MAGNETIC RING OVERHAUL page 105.



Fig. 151

- Remove the snap ring with a snap ring pliers.

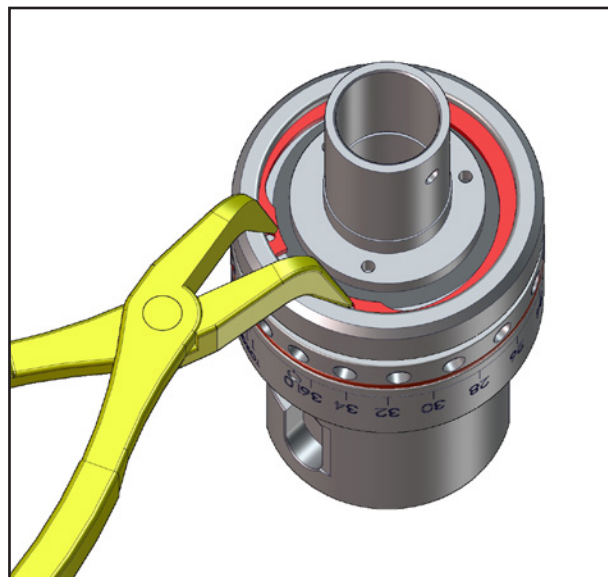


Fig. 152

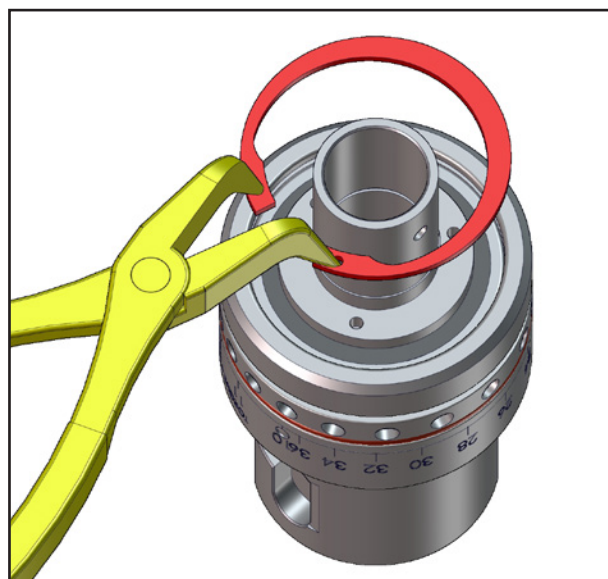


Fig. 153



- Remove the ejector guide bush complete of bearing from the head central body.



Fig. 154

- Fit the supporting tool on the ejector guide bush and tighten three Allen screws in the hole with a the 3 mm Allen wrench.

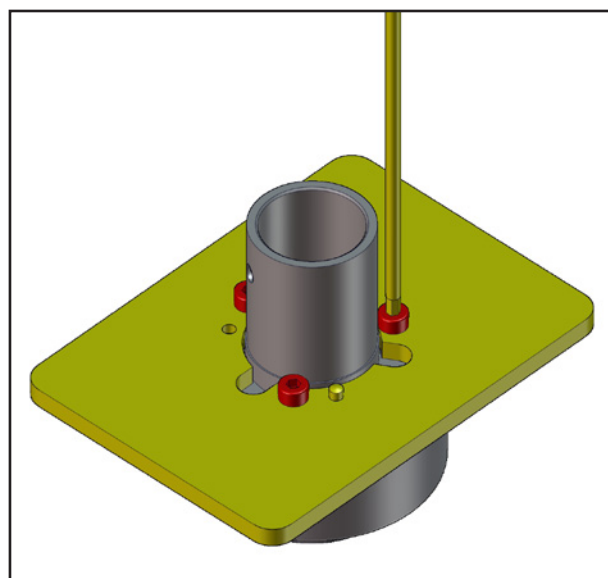


Fig. 155

- Fit the supporting tool locked in this way on the bearing by a vice.

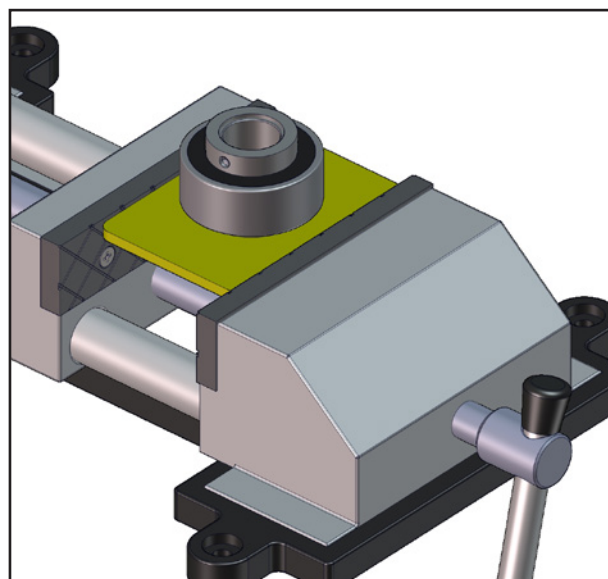


Fig. 156



- Unscrew the bearing locking ring-nut with the 45-50 with ball hook wrench, and remove the bearing locking ring-nut.

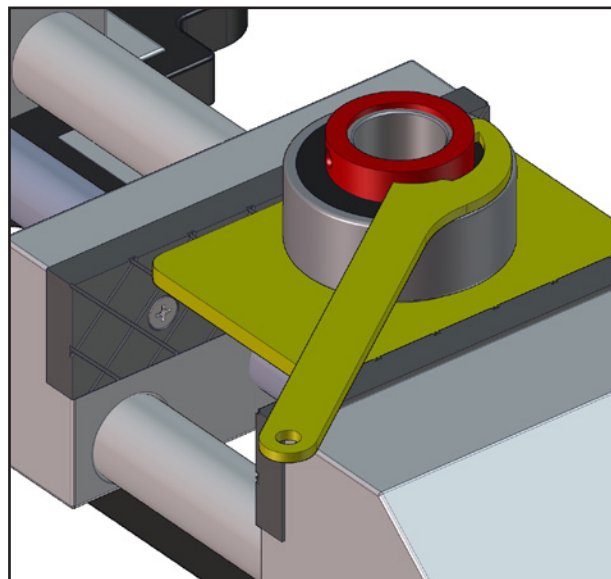


Fig. 157

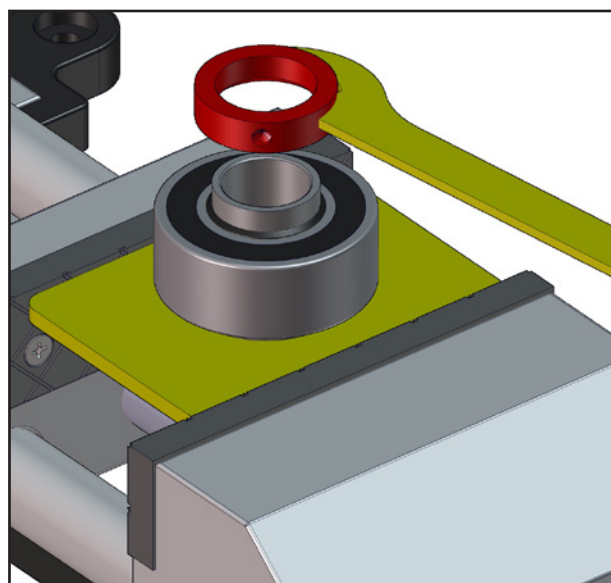


Fig. 158

- Remove the O-ring.

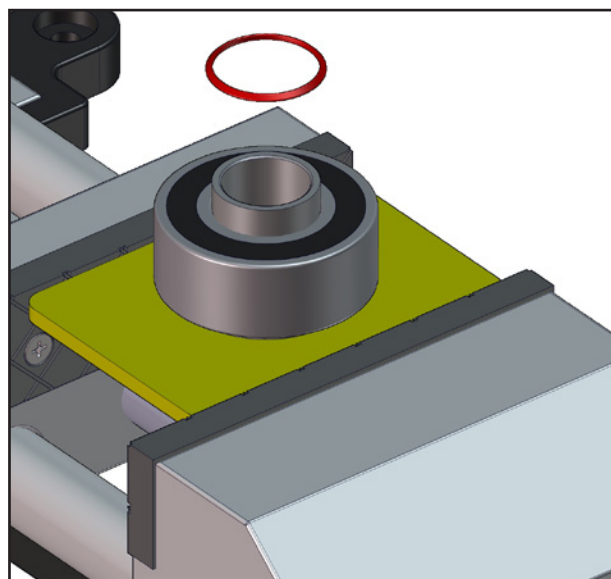


Fig. 159



- Extract the central bearing from the ejector guide bush.

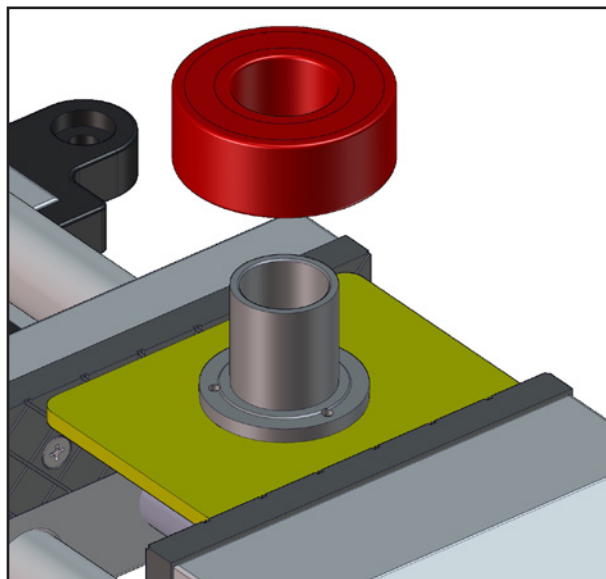


Fig. 160



11.5.3.1.4.2 THRUST BEARING ON THE EJECTOR ROD OVERHAUL

When you need to overhaul the thrust bearing, proceed as follows:

- Remove the snap ring with a snap ring pliers.

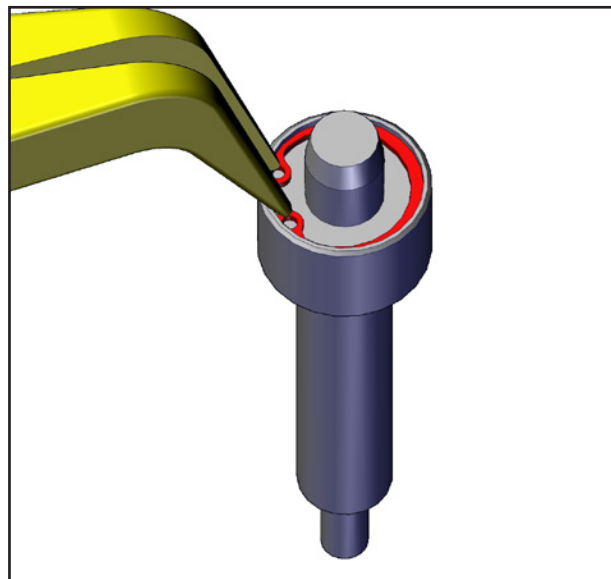


Fig. 161

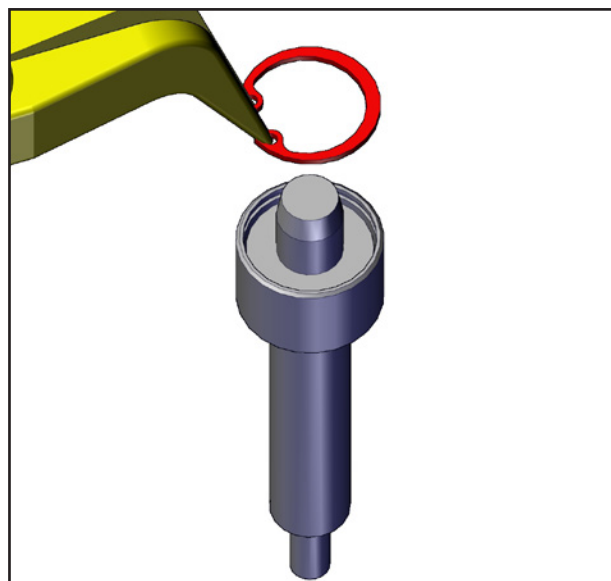


Fig. 162



- Remove the bearing fixing washer.

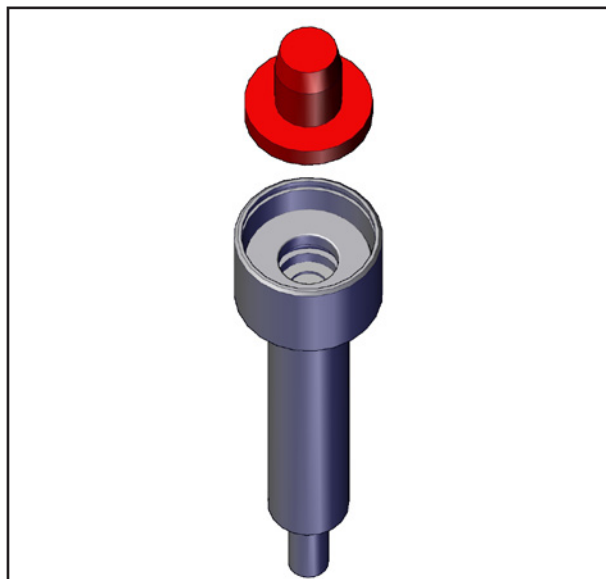


Fig. 163

- Remove the 3 parts of the thrust bearing.

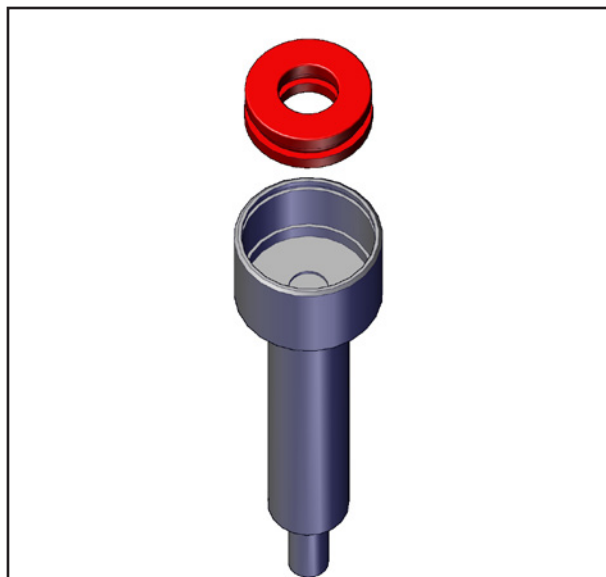


Fig. 164



11.5.3.1.5 REMOVING OF BUSH FOR DOWEL

When it is necessary remove the bushes, proceed as follows:

- Open the closure head as described in sub-chapter 11.5.3.1.1 CLOSURE HEAD OPENING page 98.
- Pull out the upper bush from the lower part of the head.



Fig. 165

- Remove the dowel bush pulling it out from the dowel.

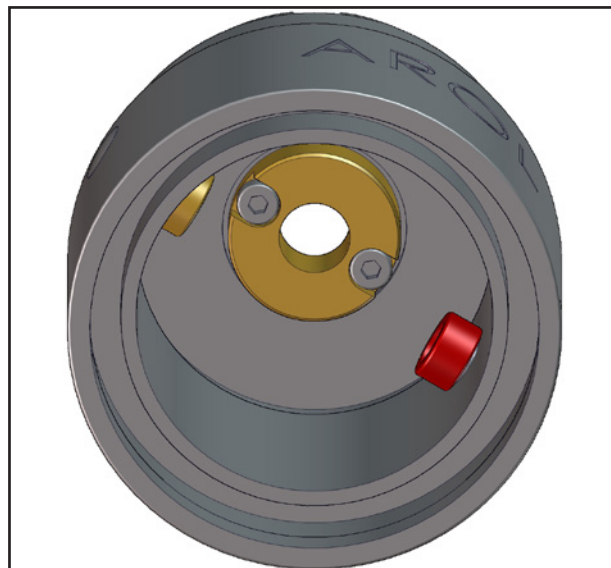


Fig. 166

- Re-install the bush caring to position it correctly, with the most curved side facing the head inner side.



The dowel bushes must be both replaced in the same intervention period.

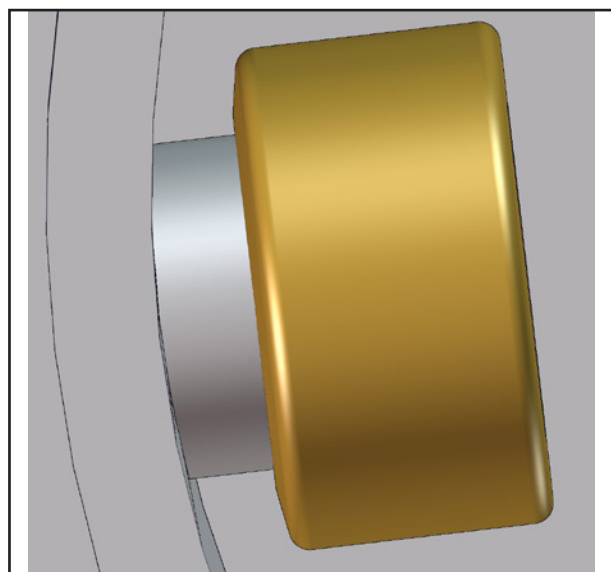


Fig. 167



11.5.4 MAINTENANCE OF THE CLOSURE CHUCK

11.5.4.1 "PK 314" CLOSURE CHUCK

11.5.4.1.1 REMOVING OF THE O-RINGS AND SPHERES

To remove the O-rings and the spheres , proceed as follows:

- Remove the cone lid blocking ring with a pliers for elastic ring.

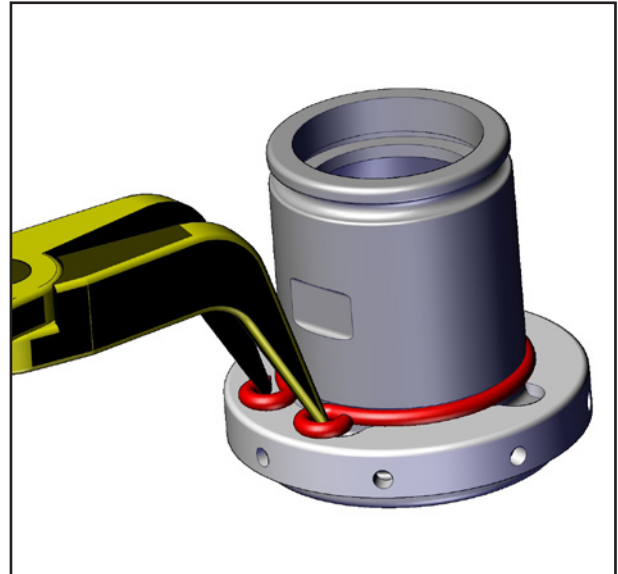


Fig. 168

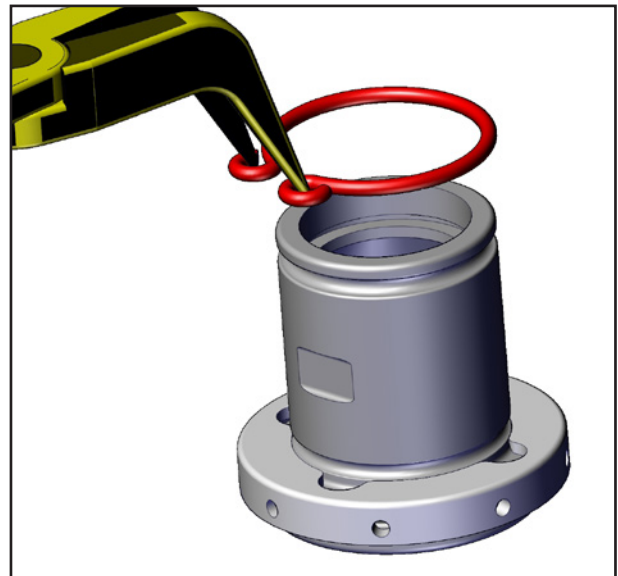


Fig. 169



- Slide out the cone protecting lid.

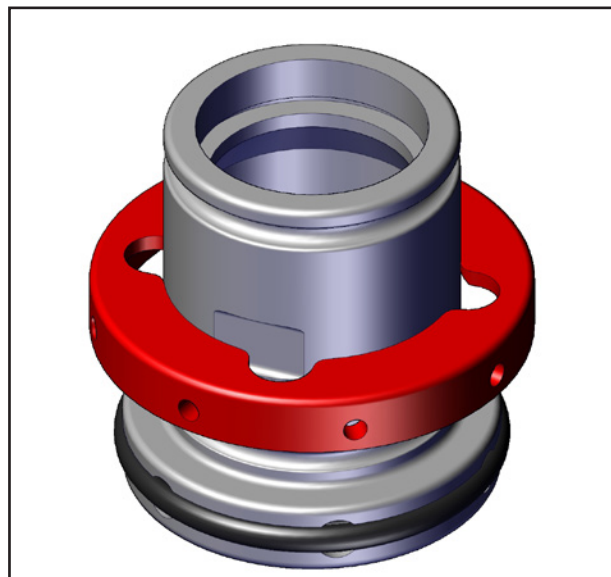


Fig. 170

- Remove the O-Rings (one or more depending on the type cap treated) from their seat with a screwdriver or a suitable tool.



If are damaged, replace the O-Rings.



If the O-rings are not damaged, replacing them has to happen from time to time as described in sub-section 11.4 PK 314 CLOSURE CHUCK WEARING PARTS page 89.

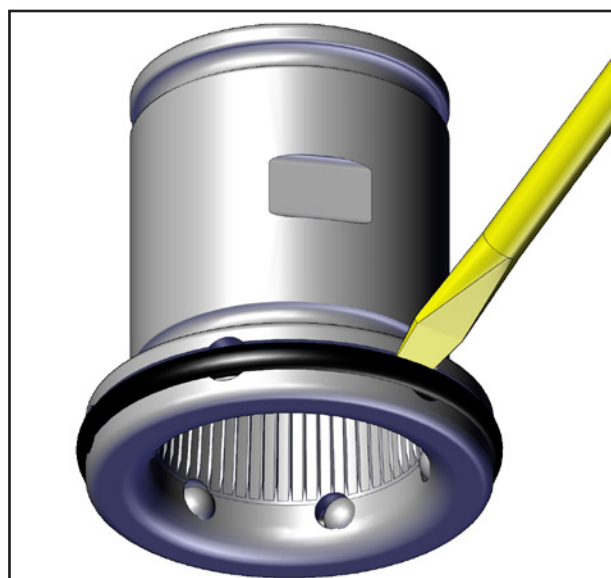


Fig. 171

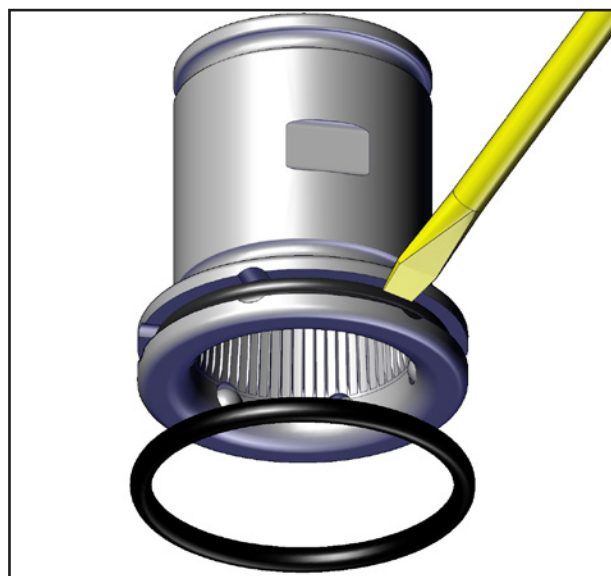


Fig. 172



Removing the O-rings, the chuck spheres are free and can scatter.

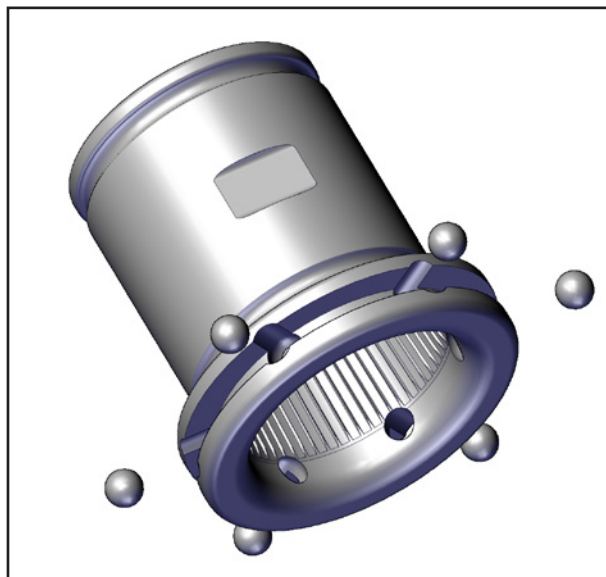


Fig. 173



11.5.5 REPLACEMENT OF TOOTHED CENTERING DEVICE



The teeth of the centering devices to carry out their bottle anti rotation task have a triangular shape. If the teeth are no longer sharp, it is necessary to replace them.

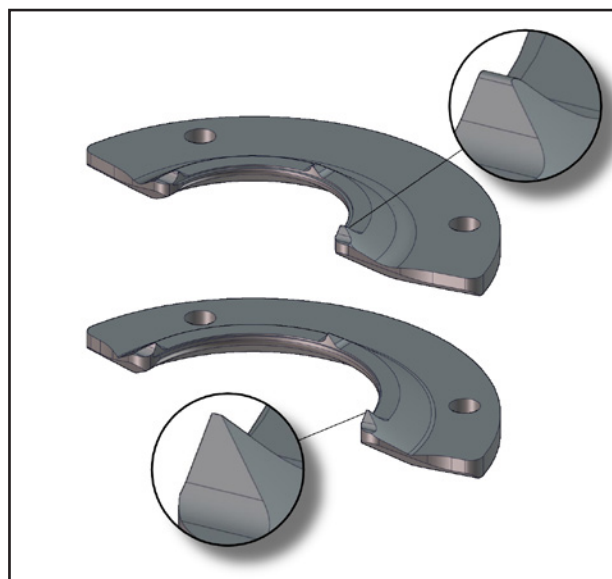


Fig. 174

To replace the worn centering devices, operate as follows:

- unscrew the fastening bolts.

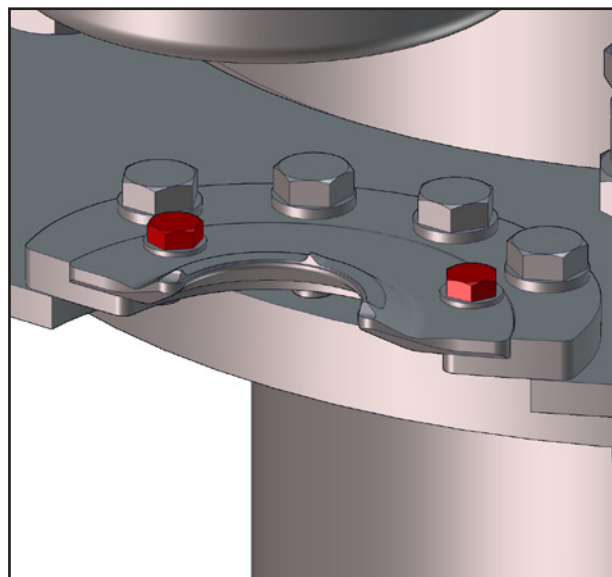


Fig. 175

- Remove the bottle centering devices

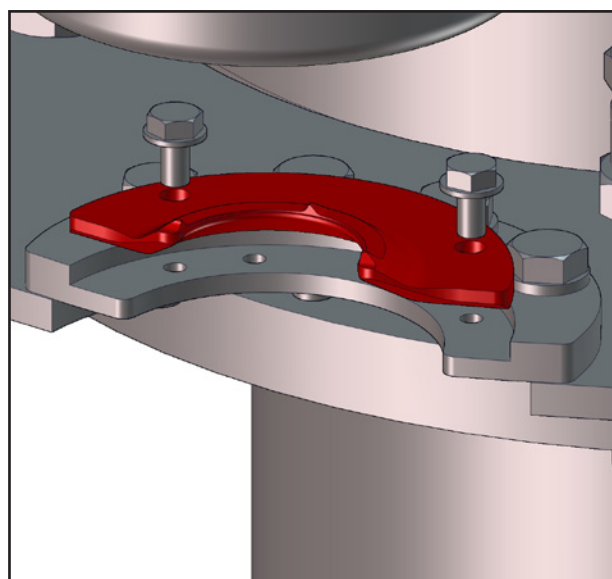


Fig. 176



11.5.6 MAINTENANCE OF THE "FLAT" CENTRIFUGAL CAPS SORTER



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

If there is an anomalous distribution, it is necessary to carry out the following operations:

- check that the nozzles located on the caps exit-mouth are not obstructed and are in the correct position to push the caps into the exit-mouth.
- Check that the nozzles located on the hopper circumference are not obstructed.

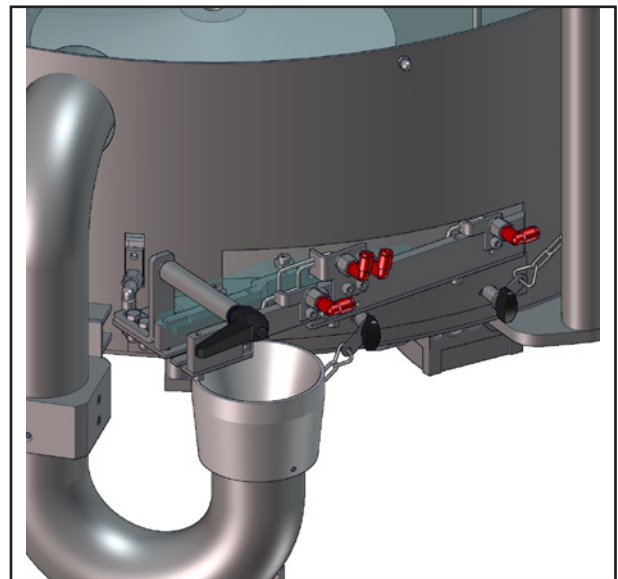


Fig. 177

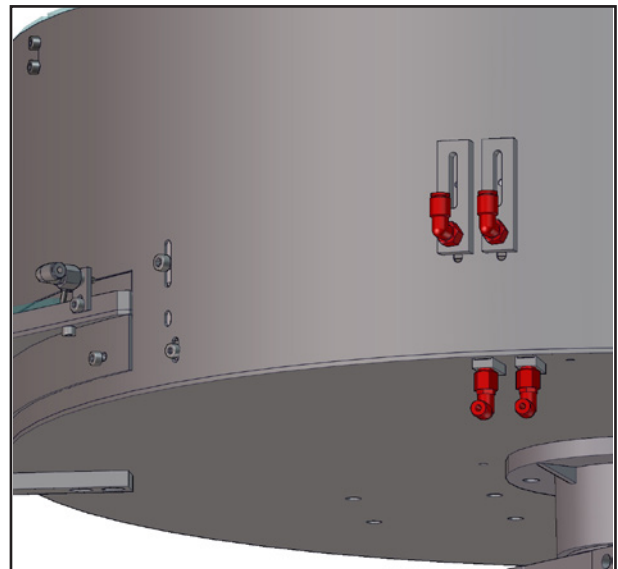


Fig. 178



- Check that the nozzle placed on the discharge tube is not obstructed.

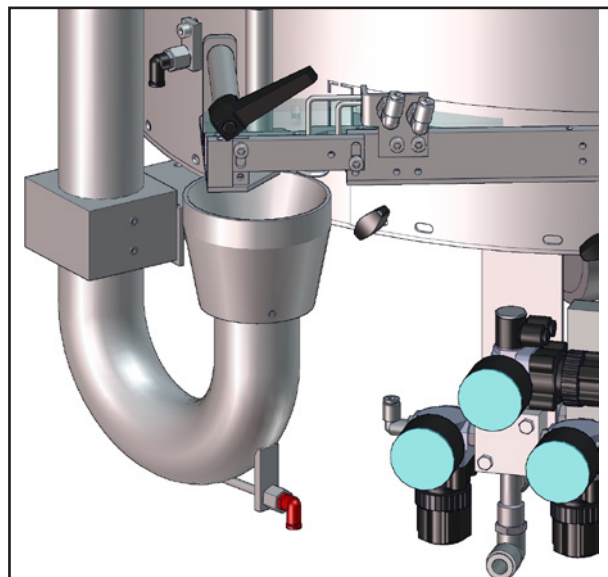


Fig. 179

- Check that in the tube there are no caps; if there are caps it is necessary to remove them.
- For the removal of the caps from the discharge tube it is necessary:
 - unscrew the fixing grub screw placed on the holding clamp;

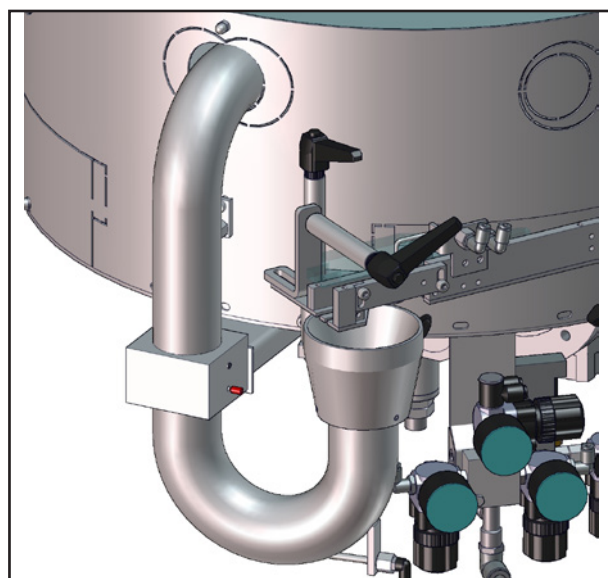


Fig. 180

- Pull down the lower discharge tube.

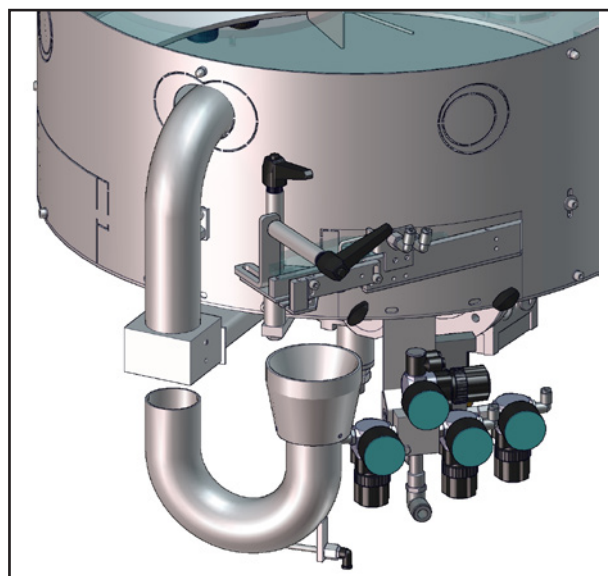


Fig. 181



11.6 LUBRICATION



11.6.1 GENERAL INFORMATION



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



Before starting the closing machine, check the lubrication of all components described in the following subchapters.

If the closing machine is not used for a long time, lubricate it again.

Mineral oils, when not used for over six months, lose their properties and must be changed.

11.6.2 LUBRICANT TYPE-APPROVAL ACCORDING TO NSF

The type-approval according to NSF includes two categories: NSF - H1 and NSF - H2.

- **H1** indicates *Food grade lubricants*, that can thus be used in all friction points of machines and systems for the food and pharmaceutical industries, where occasional, and technically unavoidable, contacts can occur between lubricant and food.
- **H2** indicates lubricants that are recommended for the general use in the food and pharmaceutical industries, supposing that a contact with the food is totally to be excluded.



11.6.3 TYPES OF LUBRICANTS

AROL S.p.A., after careful studies and functional tests, recommends to use only the lubricants indicated in the table below to lubricate its machines:



The lubricants indicated in the table are classified **NSF-H1**.

SIGNS		PRODUCT	NLGI	Base oil	Thickener [Viscosity Index]	Base oil viscosity [cSt]	Worked penetration [0.1mm]	T. Min [°C]	T. Max [°C]
A		Klubersynth UH1 64-62	2	Synth-PAO	Gel silica	60	265-295	-40	150
		G4501	1	Synth-PAO	AIK	100	310-340	-40	150
B		Kluberlub NH1-11-222	2	MO	AIK	220	265-295	-15	110
		Klubersynth UH1-14-222	2	Synth-PAO	AIK	220	265-295	-2	120
		Kluberfood NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
C		Paraliq P150		Synth+MO		150		-15	100
D		Klubersummit FG46		Synth-PAO	[130]	46		-45	135
		L0346FG		MO	[100]	46		-21	
		Purity AW 46		HT/MO	[108]	46		-15	
E		Purity FG Synth	2	Synth-PAO	Complex sulfonate calcium	46	294	-45	200
F		Kluberpaste UH1 84-201	1	Synth-PAO	PTFE	200			
		P1900	1	MO	AIK	85	290-340	-30	300
G		Barrierta L55 / 1	1	Synth-PFPE	PTFE	400		-40	260
		Krytox GPL 226	2	Synth-PFPE	PTFE [155]	240		-30	288
H		Klubersynth UH1 64-62	2	Synth-PAO	Gel silica	60	265-295		
		G4501	1	Synth-PAO	AIK	100	310-340	-40	150
I		Purity MF Spray		HT/MO	[150]	151		-9	150



SIGNS		PRODUCT	NLGI	Base oil	Thickener [Viscosity Index]	Base oil viscosity [cSt]	Worked penetration [0.1mm]	T. Min [°C]	T. Max [°C]
J		Klubersynth UH1 64-62	2	Synth - PAO	Gel silica	60	265-295	-40	150
		G4501	1	Synth - PAO	AIK	100	310-340	-40	150
K		Paraliq GB 363	3	Synth+MO	Bentonite	2400	215-245	-30	140
L		Paraliq GB 363	3	Synth+MO	Bentonite	2400	215-245	-30	140
M		Kluberlub NH1-11-222	2	MO	AIK	220	265-295	-15	110
		Klubersynth UH1-14-222	2	Synth - PAO	AIK	220	265-295	-25	120
		Kluberfood NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
N		Kluberoil 4 UH1-220N		Synth - PAO	[150]	220		-25	120
		L1122FG		Synth - PAO / MO	[142]	197		-33	
		Purity FG Gear Oil 220		HT/MO	[92]	206		-18	
O		G4501	1	Synth - PAO	AIK	100	310-340	-40	150
P		Klubersynth UH1 64 1302	2	Synth	Silicate	1300	265-295	-10	150
Q		Purity FG EP 150		MO	[95]	144		-18	247
R		Kluberfood NH1 4-220N Spray		Synth - PAO	[150]	220		-30	120
S		Klubersynth UH1 64 1302	2	Synth	Silicate	1300	265-295	-10	150
		Kluberfood NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
T		Cassida Grease GTS 2	2	Synth - PAO	Complex sulfonate calcium	46	265-295	-45	170



11.6.3.1 LUBRICANT RE-ORDERING

The lubricants used for the machine can be directly ordered to *AROL S.p.A.* or to the local distributors. See the table here below for the international addresses.

	http://www.klueber.com
	http://www.molykote.com
	http://www.petro-canada.ca
	http://www.krytox.com
	http://www.fuchs-oil.com

11.6.4 LEGEND FOR LUBRICATION SCHEMES

SYMBOL	DESCRIPTION
	Lubricate with manual pump for grease
	Lubricate with brush for grease
	Lubrication glass with disposable cartridge
	Lubricate with spray
	Lubricate with manual pump for oil
	Lubrication glass



11.6.5 POINTS TO BE LUBRICATED



To protect the closing machine from wear, seizure or other severe damage to the mechanisms, it is necessary to lubricate and grease periodically all indicated points.



AROL S.p.A. guarantees the maintenance intervals indicated hereby and the correct operation of the mechanisms exclusively with the recommended lubricants.

11.6.5.1 INSTRUCTIONS FOR READING THE "LUBRICATION POINT LIST" TABLE

The following pages show:

- the description table (*Lubrication point table*) relevant to time intervals, methods and used lubricants for the maintenance interventions on the closing machine herein.
- the position of the lubricating points (*Lubrication point scheme*) described herein.

The table "*Lubrication point list*" is designed to find immediately the operations to be carried out for the closing machine lubrication.

The "*Lubrication point scheme*" indicates the position of the points described in the table on the closing machine.

The points marked by a double frame indicate that the lubrication is carried out in a centralized way. Thus, the quantity of lubricant indicated in the table must be filled; then, the system will distribute the correct quantities to the different components.

POINT CODE	The column indicates the reference of the point to be lubricated, indicated on the closing machine next to the greaser and in the " <i>Lubrication point scheme</i> "
POINT DESCRIPTION	This column briefly describes the greasing point.
LUBRICANT TYPE	The reference letter for the lubricant to be used is indicated in this column. See the table for the lubricant type to be used.
METHOD	This column shows the tool by which the relevant greasing point must be greased.
INLET NUMBER [NO.]	It indicates the number of greasing points on the closing machine.
QUANTITY/POINT [gr.]	It indicates the lubricant quantity to be fed in each greasing point.
TOTAL QUANTITY [gr.]	It indicates the total lubricant quantity to be used.
INTERVAL [h]	It indicates every how many hours the different lubrication points must be greased.



After the lubrication of each points by grease, check that there are not leaks from seal gaskets or from the ditribution pipes.



11.6.5.2 LUBRICATION POINT LIST AND SCHEMES

LUBRICATION POINTS LIST

Order Ack.	S/N	Version	Reference	Typ	Model	P.D.	H.N.
	15610	0		T	EURO PK TWIN CHUTE D	720	20
Point	Description	Lubr. Type	Lubr. Method	Number of points	Qty per point [g]	Total qty [g]	Lubrication interval
6	Central bearings or central slewing bearing	M		3	15	45	120
9A	Lower bearing or heads rotation footstep bearing	M		2	20	40	120
2	Transfer disc bush	M		1	20	20	500
3	Heads rotation motor reducer	M		1	20	20	500
8A	Piston upper roller	A		20	6	120	500
8B	Piston lower roller	A		20	3	60	500
8C	Piston fork bearing	A		20	6	120	500
8D	Piston liner	A		20	15	300	500
4	Pistons controlling cam	M		2	15	30	1000
11	Heads rotation gearing	M					1000
17	Central column inner bearing	N		1	120	120	3000 *

Rotating point

* Value that depends on environmental conditions

Fig. 182 Lubrication point table

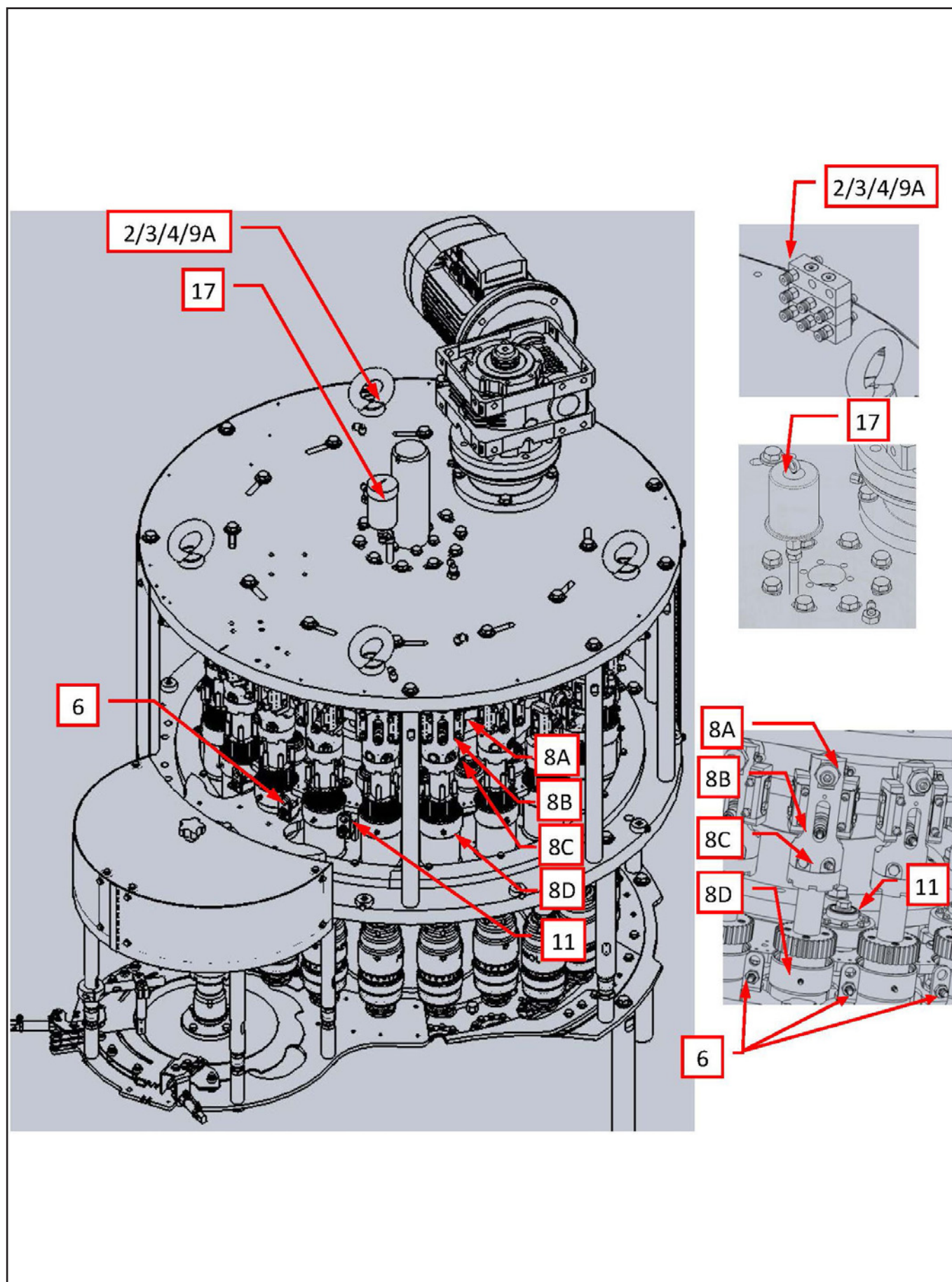


Fig. 183 Lubrication point scheme



11.6.6 LUBRICANT CARTRIDGE



At the closing machine delivery, the lubricant cartridge is not fit, thus it is not active. Before starting the closing machine, install and activate the cartridge.

11.6.6.1 LUBRICATOR DESCRIPTION

The cartridge of the lubricant is composed of:

- an adjustment screw (1).
- A body containing the lubricant (2).



The lubricant cartridge may provide lubrication intervals from 1 to 12 months. The interval is defined according to the alignment of the screw with the number on the ring nut of the cartridge.

AROL S.p.A. suggests for the lubrication of its machine to rotate the adjustment screw on number 6 of the ring nut on the cartridge.



Fig. 184

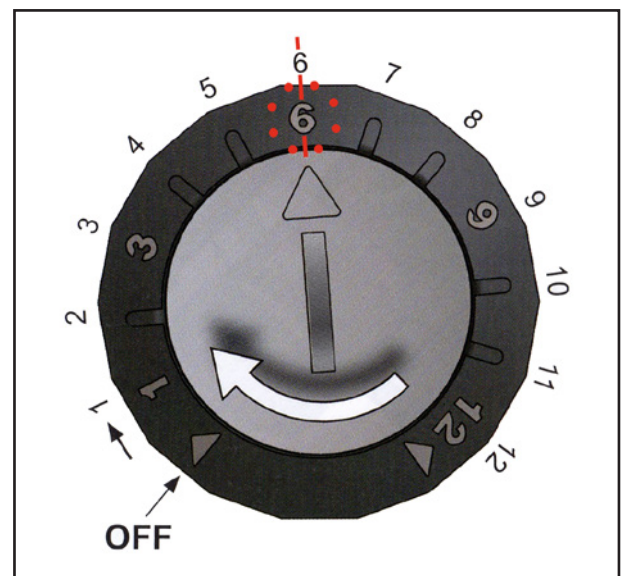


Fig. 185



During the supply period, the lubricator must not be removed to maintain the pressure inside it.



11.6.6.2 CARTRIDGE REPLACEMENT

When the cartridge must be replaced, operate as follows:

- Remove the plug from the discharge opening.
- Screw by hand the lubricating cartridge on the suitable coupling.

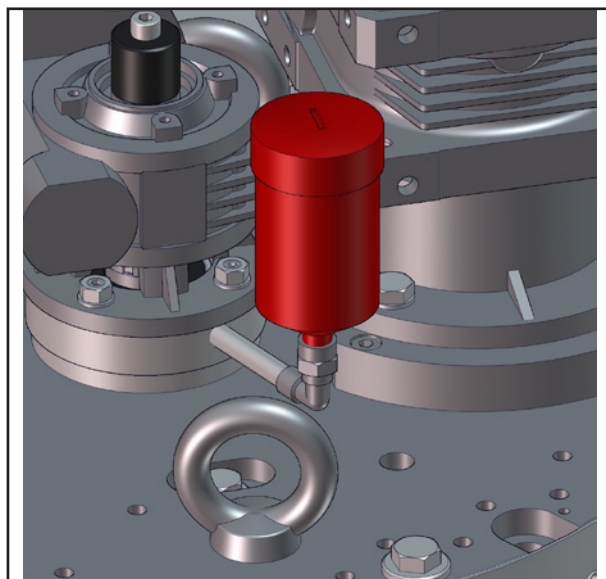


Fig. 186

- Rotate the adjustment screw on number 6 of the ring nut on the cartridge.



Fig. 187

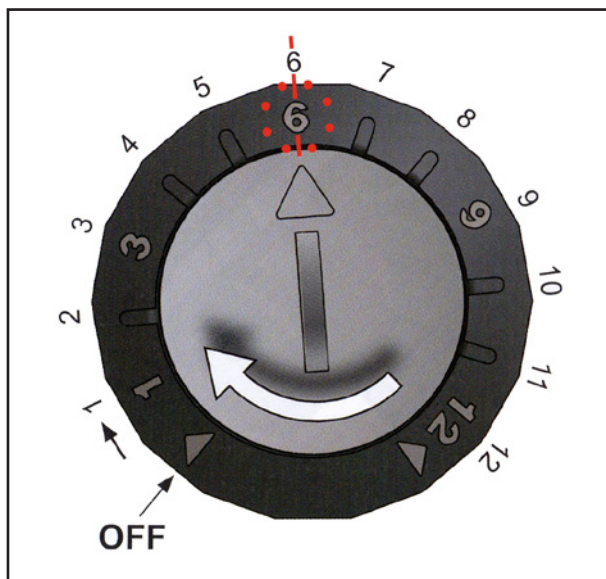


Fig. 188



11.6.7 CLOSURE HEAD LUBRICATION



Since it is objectively difficult to express an amount of lubricant to be laid with a brush, with "stroke" refers to the amount of grease that is retained, plunging up to half the bristles of a brush number 8 off about 10 mm high 18 mm bristle.

Use lubricant "**K**" type.

11.6.7.1 CLOSURE HEAD "PK 314"

- Clean the springs and apply a layer of lubricant on all the coils. Use a brush.

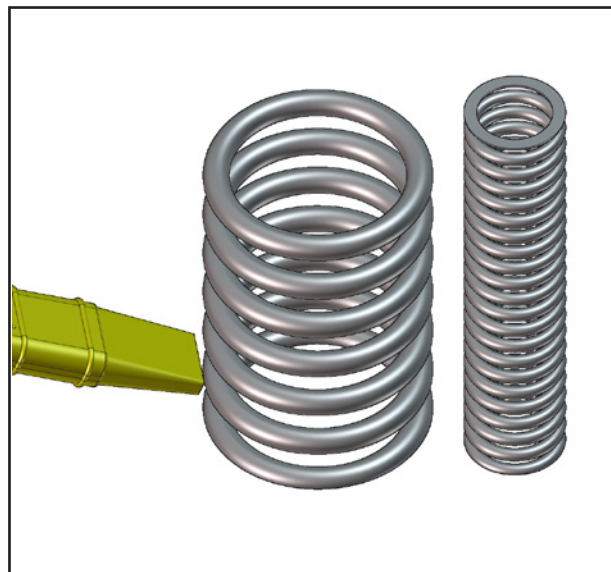


Fig. 189

- Apply a layer of lubricant on the inside of the head upper bush. Use a brush.
- Apply a layer of lubricant on the dowel bushes. Use a brush.

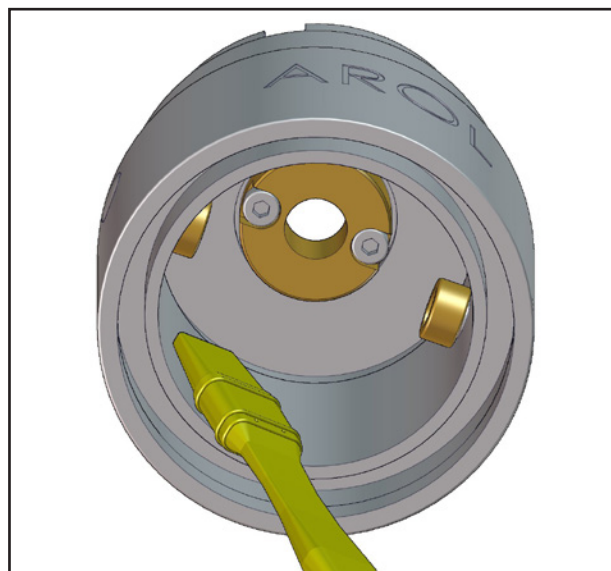


Fig. 190



- Apply a layer of lubricant on the seat of the head upper bush. Use a brush.



Fig. 191

- Apply a layer of lubricant on the inner rod. Use a brush.

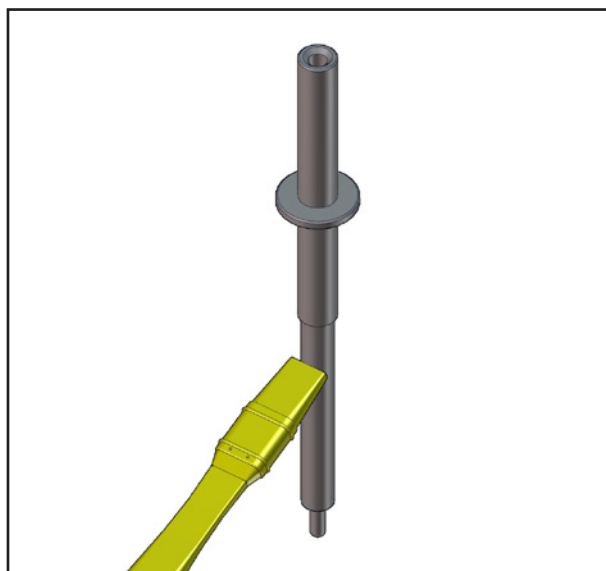


Fig. 192

- Clean the bearings and apply a layer of lubricant on all surfaces. Use a brush.

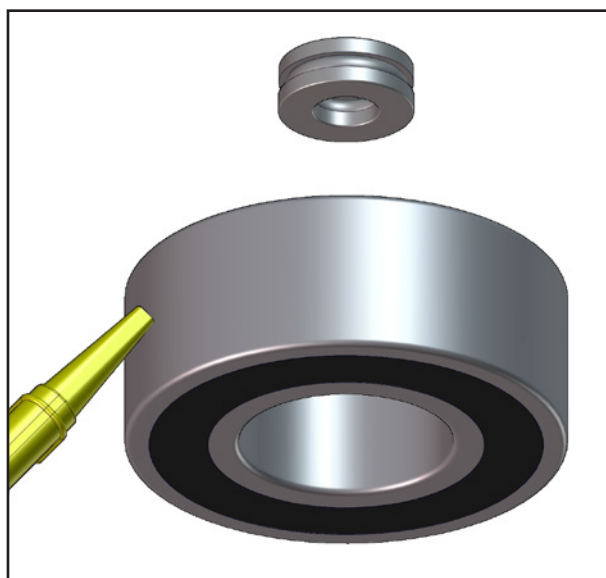


Fig. 193



- Clean the magnetic rings and apply a layer of lubricant on all surfaces. Use a brush.

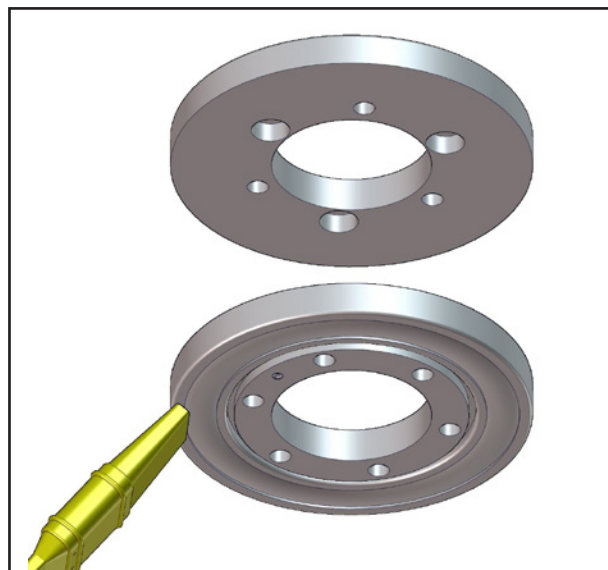


Fig. 194

- Clean inside and outside the head central body; apply a layer of lubricant inside and outside. Use a brush.



Fig. 195

- Clean the release seats of the head central body; apply a layer of lubricant inside the seats. Use a brush.

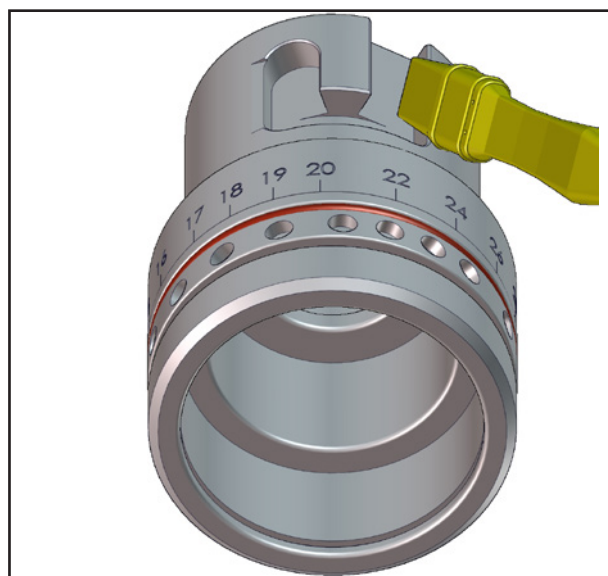


Fig. 196



- Clean the magnets holder ring-nut; apply a layer of lubricant inside the ring-nut. Use a brush.

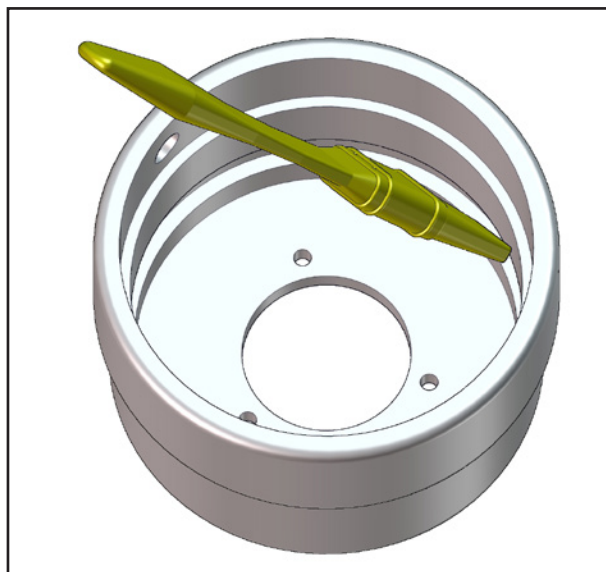


Fig. 197



11.6.8 CLOSURE CHUCKS LUBRICATION



Since it is objectively difficult to express an amount of lubricant to be laid with a brush, with "stroke" refers to the amount of grease that is retained, plunging up to half the bristles of a brush number 8 off about 10 mm high 18 mm bristle.

Use lubricant "**K**" type.

11.6.8.1 "PK 314" CLOSURE CHUCK

- Apply a layer of lubricant on the inside thread of the closure chuck. Use a brush.

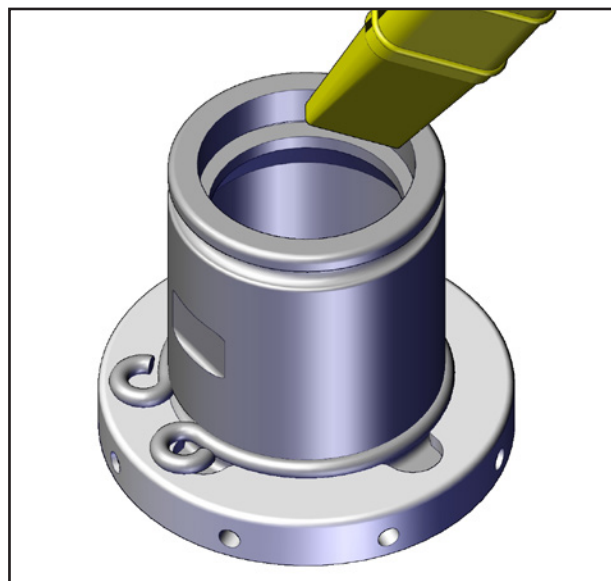


Fig. 198



11.6.9 BONFIGLIOLI REDUCER LUBRICATION

11.6.9.1 INTRODUCTION

On the reducer is fitted with the nameplate on the side shown. On this plate there are all the references and all the indispensable indications for its correct identification.

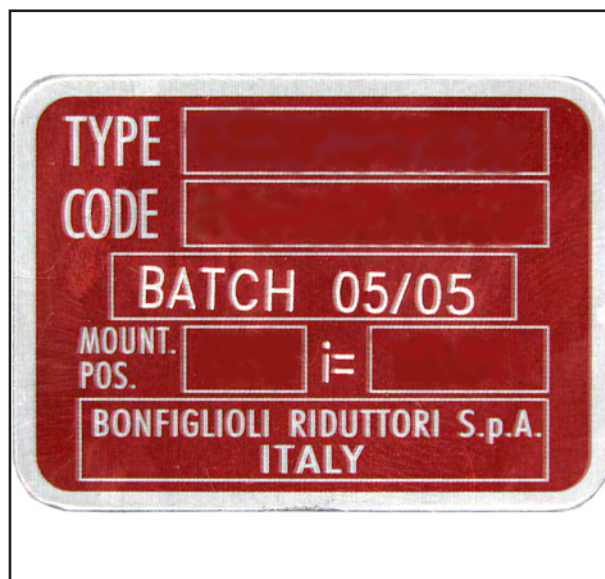


Fig. 199

11.6.9.2 LUBRICATION

VF30....VF49 AND W63....W86 REDUCERS

For reducers of this group, that are on a average at high power, the oil lubrication is adopted. These reducers have not charge, discharge and level oil plugs and so do not need any maintenance having already the right oil quantity.

VF130....VF150 AND W110 REDUCERS

For reducers of this group, that are on a average at low power, the permanent lubrication with synthetic oil is adopted. Reducers have charge, discharge and level oil plugs



Reducers are supplied with a lubrication oil load **Shell TIVELA OIL S320** type.



It is recommended, if the lubricants is not of the recommended ones, that it has an equivalent composition on synthetic nature and viscosity. Moreover it must have the suitable antifoam additives.

OIL QUANTITY TABLE (LITRES)

		B3	B6	B7	B8	V5
VF130	HS	3,90	2,50	2,50	2,30	3,30
	P(IEC)	3,00	2,50	2,50	2,30	3,30
VFR130		0,40	0,50	0,50	0,70	0,40
W110	P80...P132	1,50	1,70	1,70	1,90	1,70
	M2-M3	1,50	1,70	1,70	1,90	1,70
	$7 < i < 15$	1,50	1,70	1,70	1,90	1,70
	$20 < i < 100$	2,70	1,70	1,70	1,90	1,70



11.6.9.2.1 OIL REPLACEMENT



Do not mix oil with different trade marks and characteristics and check that the oil in use has high anti-foam characteristics and EP.

If an identical lubricant is not available, completely empty the reducer and clean it inside with a light solvent, before the next filling.

- Place a container of adequate capacity under the discharge plug.
- Remove the charge, discharge and level plug and let the oil freely flow.



To make easier the discharge operation, it is recommended to operate with hot oil.

- Wait some minutes so that all the oil is went out, then screw again the discharge plug, after the gasket has been replaced.
- Pour new oil only after the reducer is in its definitive position, until the half of the level plug is reached.
- Screw the charge plug, after the gasket has been replaced.



The exhausted oil must be disposed according to the environment protection norms.

If there are leaks, before to reset the lubricant quantity, it is necessary to find the cause of the problem before starting up again the reducer.



11.7 EXTRAORDINARY MAINTENANCE

11.7.1 SPINDLES CONTROL CAM WEAR CHECK



To check the piston control cam race wear proceed as follows:



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

- Remove the safety guard.
- Check the conditions of the spindles control cam in the cap application point. The worn part is usually the roller sliding race; if the wear is such to form a difference over 1 mm the cam must be replaced.
- Check the wear along the whole sliding race and above all in the area where the piston reaches its bottom dead center.



Fig. 200

To replace the cam, contact the *Technical Service*.

11.7.2 TIMING OF THE PHASE SENSOR OF THE TRANSFER DISK



To obtain a correct distribution of the caps must be positioned in an appropriate way the phase sensor.

Is sufficient, acting with the jog, bring the closure machine in a situation where the tip of the transfer star-wheel is located 1 mm beyond the central axis of the caps chute.

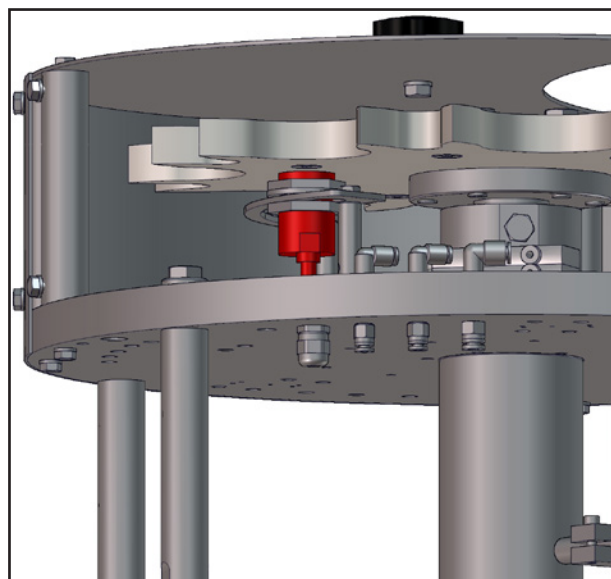


Fig. 201

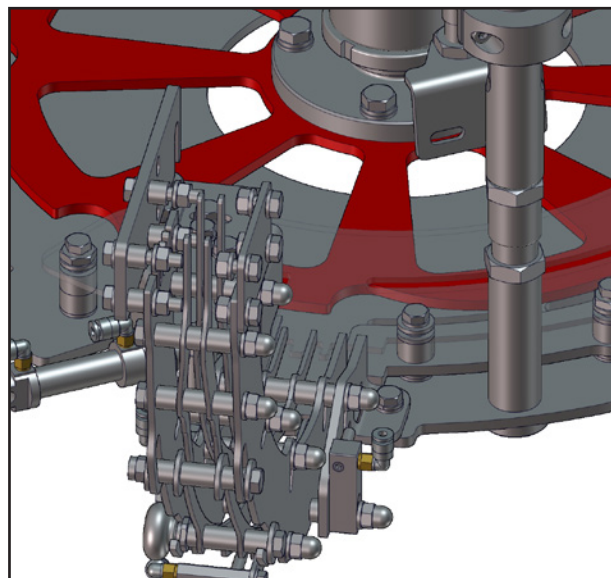


Fig. 202

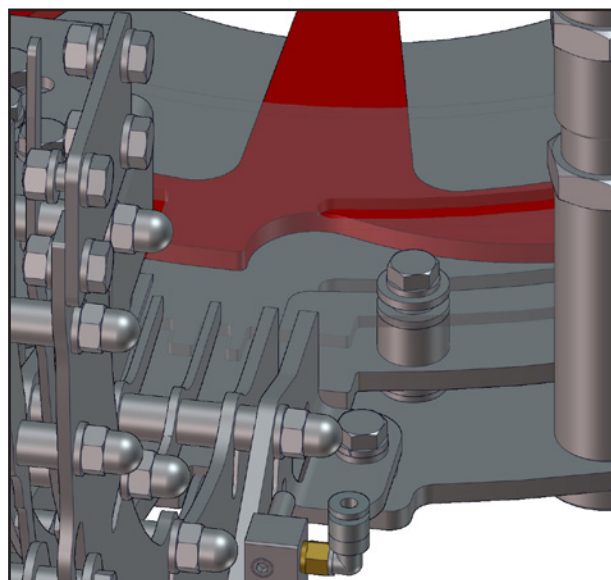


Fig. 203



Now, adjust the timing sensor so that it is at the start of the reading of the screw embedded in the gear of the transfer disk (At the reading start, the sensor LED lights up.).



Remembers that in case of continuous flow of bottles the solenoid valve must remain energized.

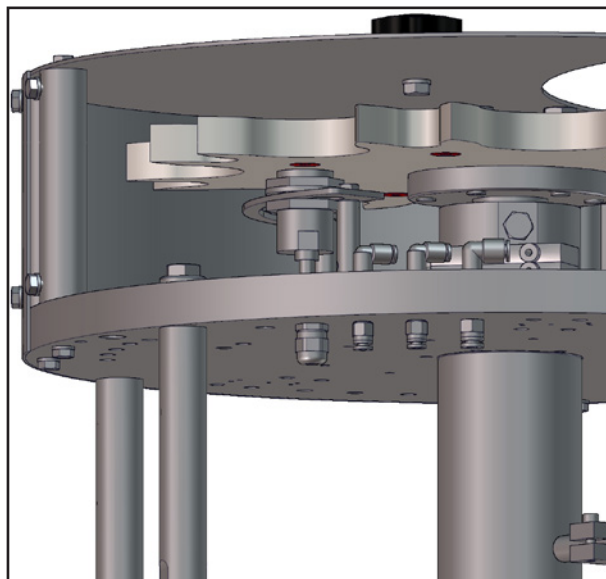


Fig. 204

11.7.3 TRANSPORT EQUIPMENT TIMING



11.7.3.1 ALIGNMENT OF THE TRANSFER STAR-WHEEL WITH THE CLOSURE HEADS

If the transfer star-wheel and the closure heads are not perfectly aligned, the complete transfer disk must be timed, operating as follows:

- Make the closing machine jog pressing the outer *JOG* button, till a cavity of the central star-wheel is aligned with the closure machine center and the closure head center.
- If the centers are not aligned:



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

- Remove the top cover of the transfer disk.

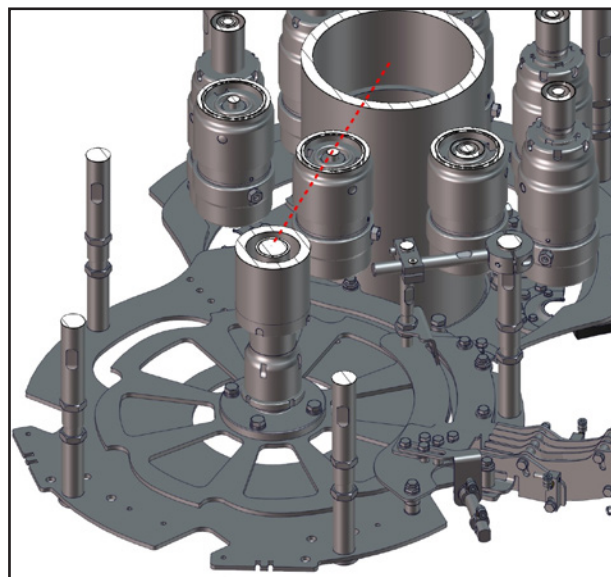


Fig. 205

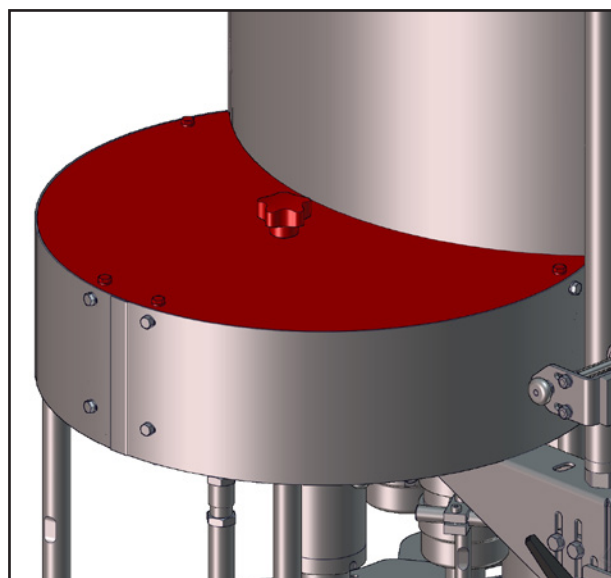


Fig. 206

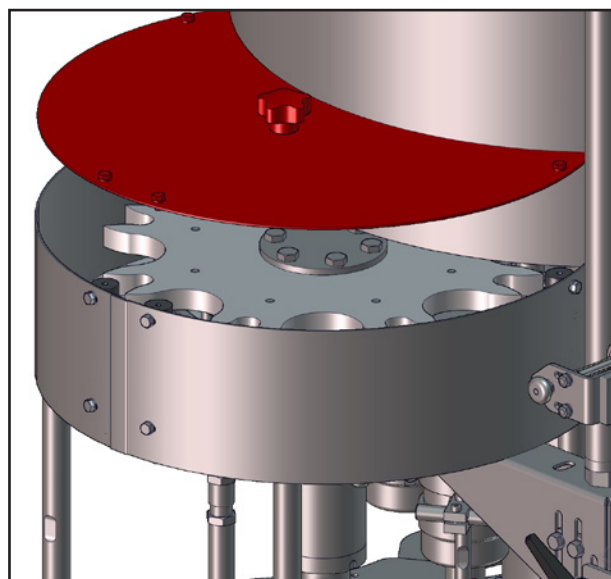


Fig. 207



- Loosen the bolts that secure the disk rotating gear.

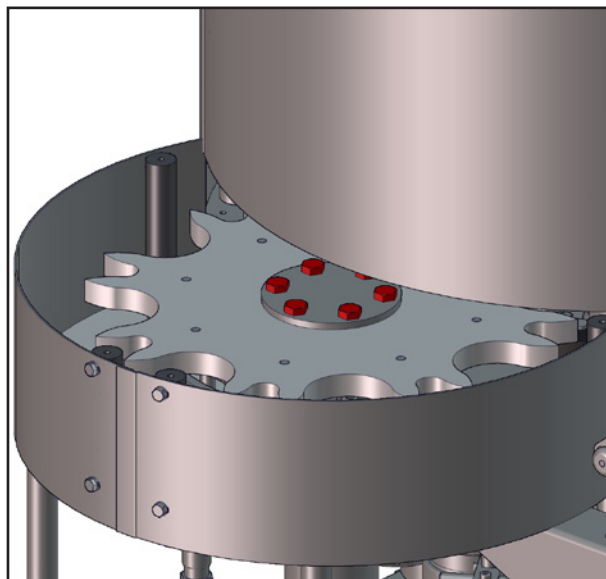


Fig. 208

- Rotate the caps transfer star-wheel till getting the required alignment.
- Strongly lock the bolts that secure the disk rotating gear.

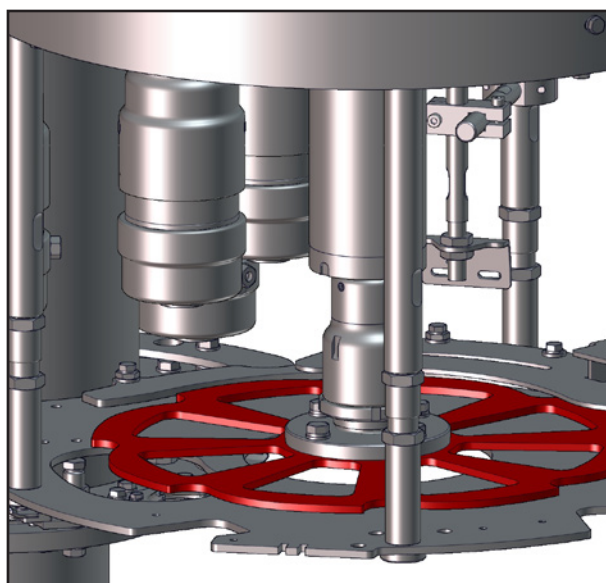


Fig. 209

- Replace the top cover of the transfer disk.

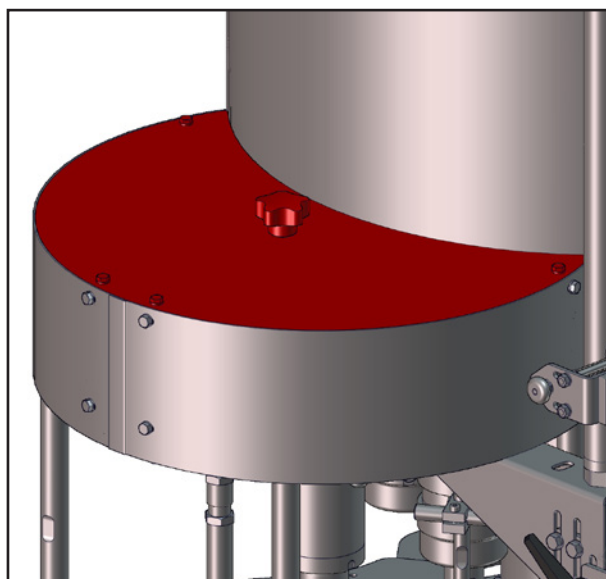


Fig. 210



12 FAULTS AND REMEDIES

The list of the faults in the following pages is useful to detect as fast as possible their reasons and possible solutions.

The research and the interventions must be carried out methodically (from the most probable to the less probable reason) and in logical sequence.

Proceed by steps: a first intervention, a test on the obtained result, then passage to the next intervention.

The main closing machine units are:

- Closure unit
- Caps/corks distribution.
- Caps/corks feeding.



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



FAULT	CAUSE	REMEDY
Lifting or lowering the closure carousel, the same does not move vertically.	Interference with the container conveying equipment, foreign matters in the lifting mechanism, other.	Identify the reason for the overload (interference with the container conveying equipment, foreign matters in the lifting mechanism, other).
	The fastening screw of the sliding bush has not been loosened.	Loosen the fastening screw.
	Min. or max. limit of closing carousel vertical range reached.	Make sure the closing machine has been preset for the container size to be processed. If so contact <i>Technical Assistance Service</i> .
The containers break in the closure step.	The closure carousel is too low.	Position the closure carousel at the correct height for the container to be processed.
	Container transfer equipment not suitable.	Replace the equipment with the one suitable for the size to be processed.



FAULT	CAUSE	REMEDY
Caps are not descending from the sorter.	The sorter exit-mouth is clogged.	Remove possible caps, cap parts or foreign matters. Check on the pressure gauges if the pressure of the air jets is correct.
	Caps are not correctly sized or are strained.	Use caps with the suitable size and even features.
The cap screws badly on the bottle.	Closure chuck not exact.	Use the closure chuck suitable for the cap to be processed.
	Cap fragments in the closure chuck.	Remove the fragments from the closure chuck inside.
	The rubber spheres seal ring on the closure chuck is worn.	Replace it with one of the correct size.
	Bottle-neck guides not suitable for the bottle size to be processed.	Replace the guides in the next with those suitable for the bottle size to be processed.
	Broken compensating spring.	Replace the spring with a spare one. For the replacement see the relative sub-chapter of the closure head.
	Container threads/cap not compatible.	Check that the bottle to be processed and the used cap correspond exactly.



FAULT	CAUSE	REMEDY
The cap is not totally screwed.	Insufficient magnetic torque.	Increase the head tightening torque. For the adjustment see the relative sub-chapter of the closure head.
	The cap slips in the chuck.	Verify the wear of the chuck and possibly replace it with a spare one. Verify if the closure chuck is suitable for the cap to be processed, possibly replace it with a suitable one.
	The bottles rotate around themselves.	Check the wear of the centering device teeth.
	Closure head springs not working.	Push the closure head body upwards. The elastic reaction of the compensating spring must be felt. For the replacement see the relative sub-chapter of the closure head.
The cap is too much screwed.	Excessive magnetic torque.	Decrease the closure head tightening torque. For the adjustment see the relative sub-chapter of the closure head.
Problems in the cap distribution.	The caps fall from the closure head.	The rubber spheres seal ring is worn or the ball seats are dirty. Replace the worn parts and clean carefully by an air jet (do not use water or liquids).
	The caps do not enter the closure heads.	Check the regular operation of the electro-pneumatic circuit.
		Check the perfect alignment between transfer star-wheel and closure heads.



13 SCRAPPING AND DISPOSING



As there are in force different norms of observation, it is necessary to follow the prescriptions fixed by laws and authorities of the states where the closing machine will be installed.

Proper precautions must be taken for out-of-order and scrapping of the closing machine in order to avoid possible dangerous situations.

In particular, follow these general instructions:

- disconnect the closing machine from the electric network and check that no residual voltage exists.
- Empty all the lubrication points of oil paying attention in not dispersing and dispose it according the laws in forces in the country where the closing machine has been used.
- Lock all the moving parts of the closing machine, making sure that there are no possibilities of movement, also due to accidental collision during the transfer operation.
- With the help of qualified and authorized personnel proceed to the dismantling of components and the differentiated collection of materials (steel, plastic, electrical components, wires, etc.) in order to dispose everything according to the indications forseen by the laws in force.



Always use security clothes during the operations that require direct intervention on the closing machine.



CUSTOMER					DATE		
Closure Machine Model					Serial number		
Container dimensions					Container Type		
Cap dimensions					Caps Type		
Head static torque					Bottles/min		
Compensating spring kg.					Gripper/chuck type		
N° Head	DETECTED TORQUE					AVERAGE	NOTES
	Lbs x inch	Lbs x inch	Lbs x inch	Lbs x inch	Lbs x inch	Lbs x inch	
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2							
3							
4							
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